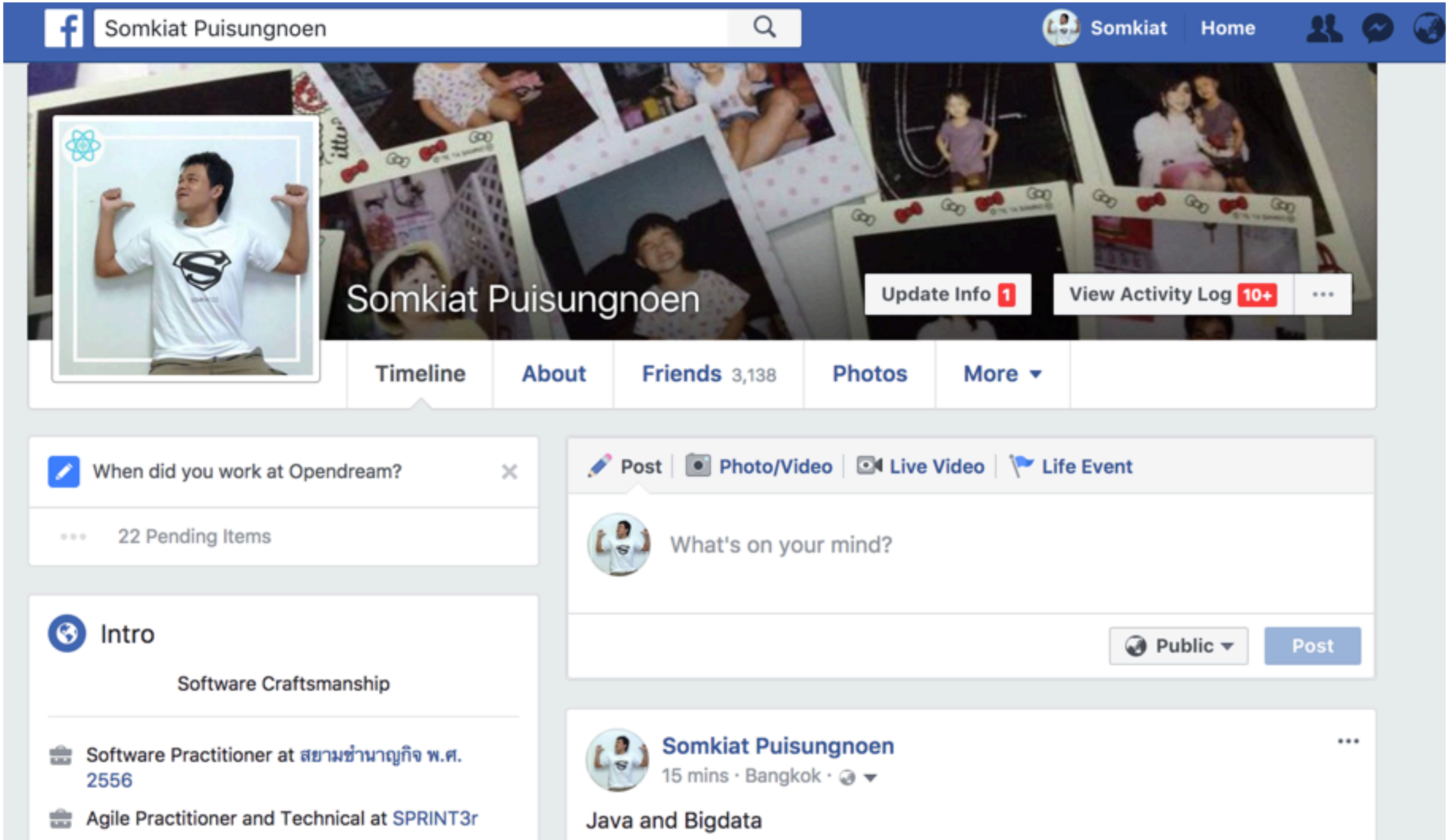
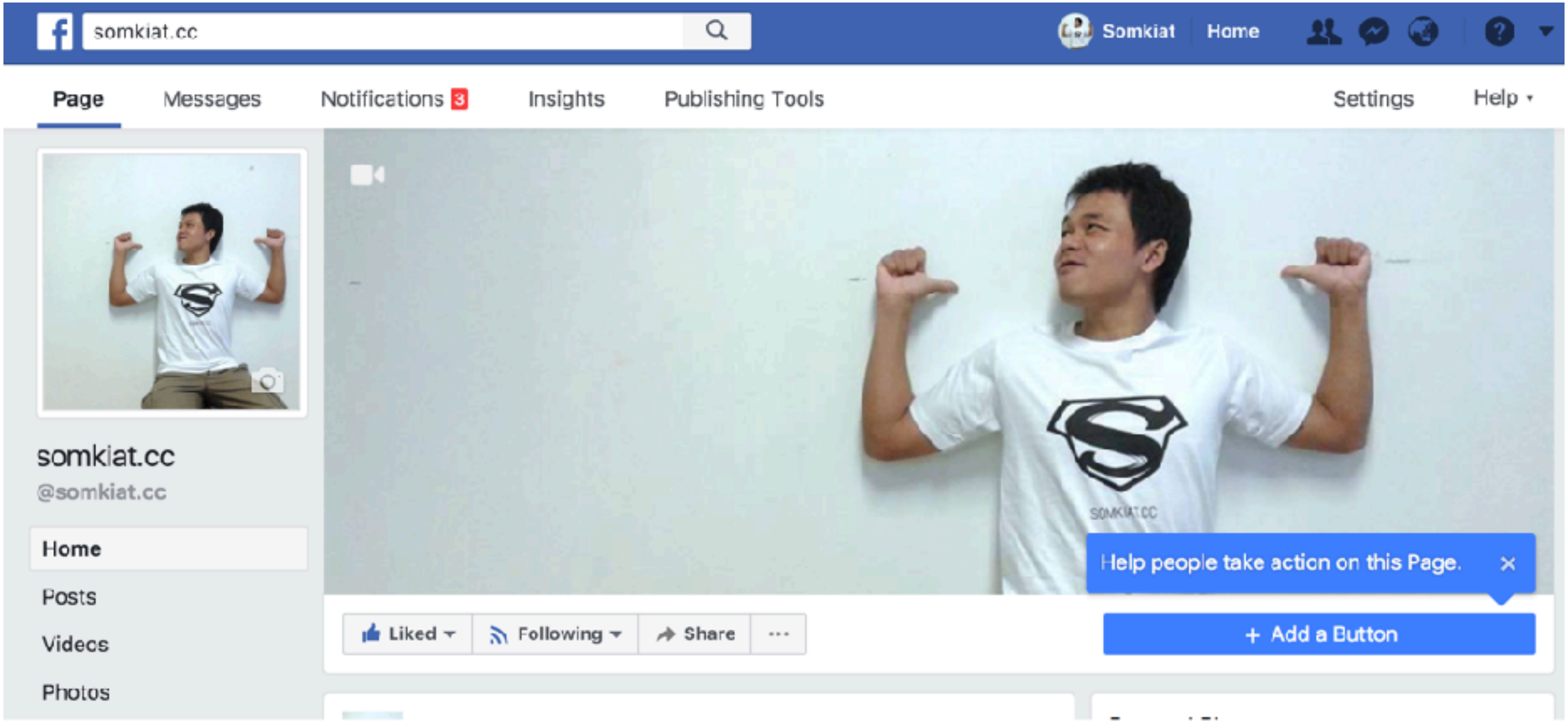
สยามชำนาญกิจ
SIAM CHAMNANKIT

DevOps series

DevOps
Modern Software Development
Docker


Somkiat Puisungnoen
Somkiat
Home
Somkiat Puisungnoen
Update Info 1
View Activity Log 10+
Timeline
About
Friends 3,138
Photos
More
When did you work at Opendream?
22 Pending Items
Intro
Software Craftsmanship
Software Practitioner at สยามชำนาญกิจ พ.ศ. 2556
Agile Practitioner and Technical at SPRINT3r
Post
Photo/Video
Live Video
Life Event
What's on your mind?
Public
Post
Somkiat Puisungnoen
15 mins · Bangkok ·
Java and Bigdata


somkiat.cc
Somkiat
Home
Page
Messages
Notifications 3
Insights
Publishing Tools
Settings
Help
somkiat.cc
@somkiat.cc
Home
Posts
Videos
Photos
Help people take action on this Page.
Liked
Following
Share
+ Add a Button

DevSecOps

Design | Develop | Test | Deploy

Monitoring and Observability

Continuous Integration and Delivery

Section 1

1. Introduction to DevOps
2. DevOps workflow
3. DevOps tools
4. DevOps team topologies
5. Version Control System with Git
6. Manage containers with Docker
7. Docker image, container, Dockerfile, compose

Section 2

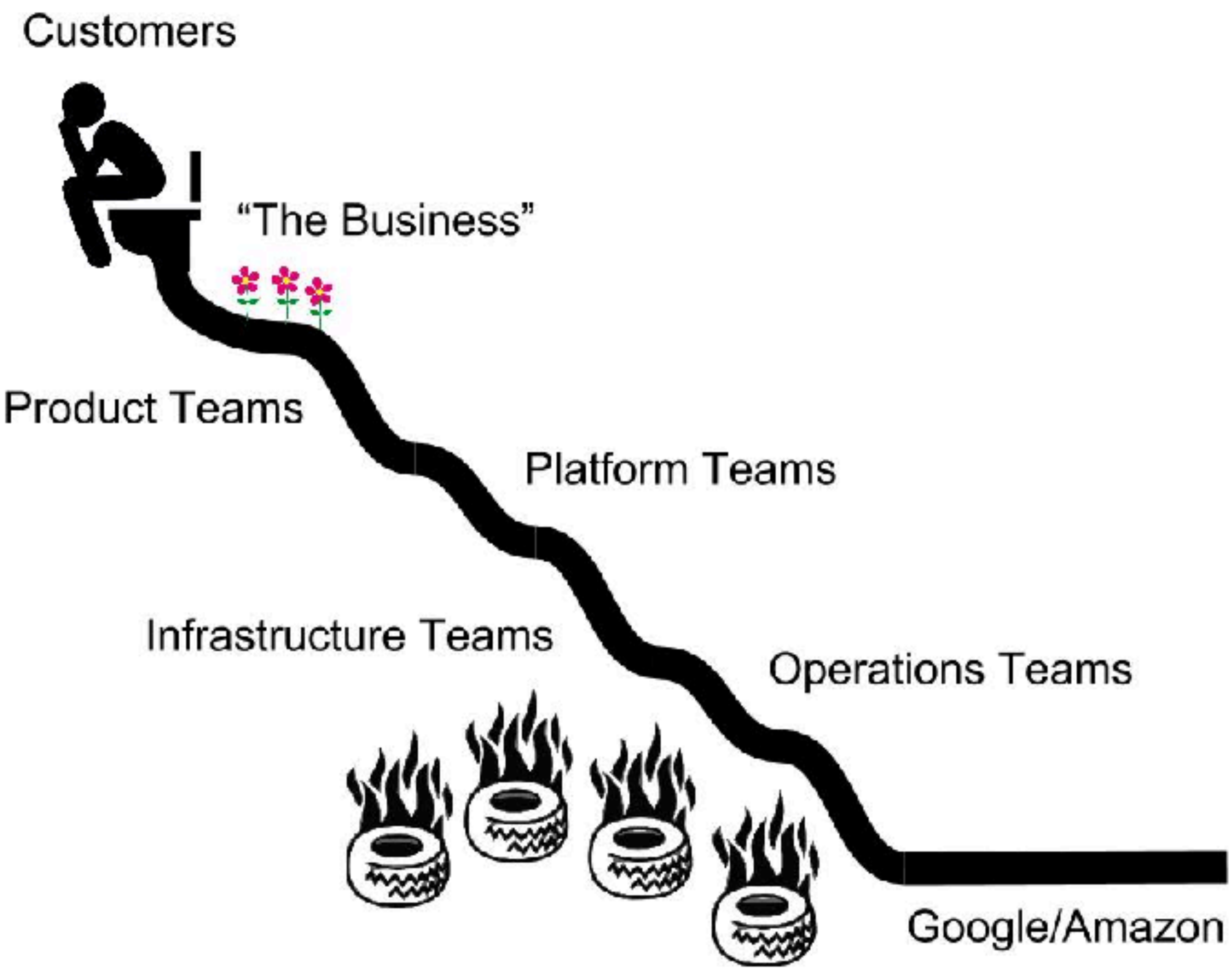
1. Continuous Integration and Delivery (CI/CD)
2. CI/CD principles
3. Design your pipeline
4. Working with Jenkins
5. Create your pipeline
6. Pipeline as a Code
7. Scalable Jenkins with Master/slave

https://github.com/up1/course-devops-workshop

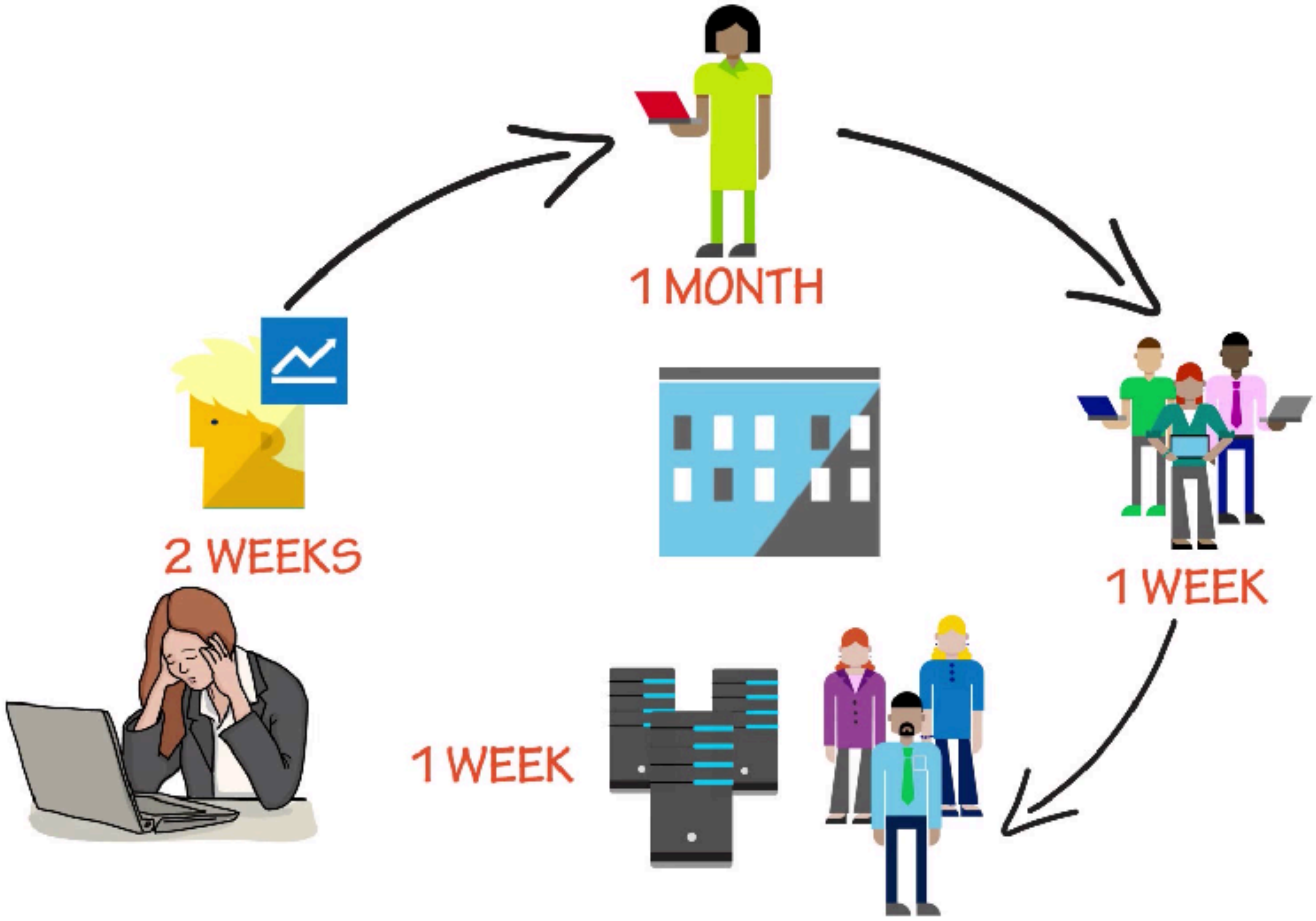
Software Delivery

Quantity vs Quality ?

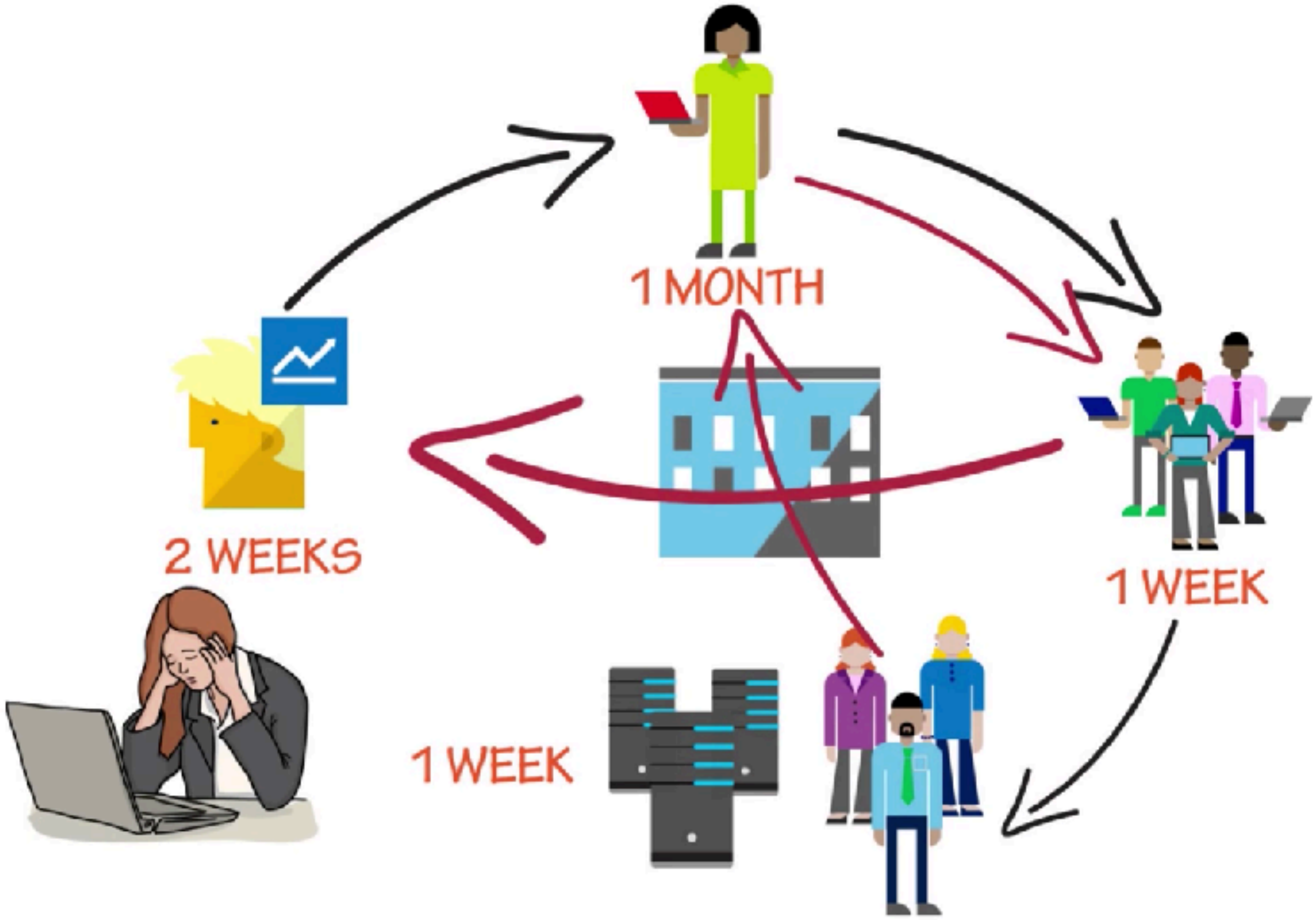
SDLC


Customers
"The Business"
Product Teams
Platform Teams
Infrastructure Teams
Operations Teams
Google/Amazon

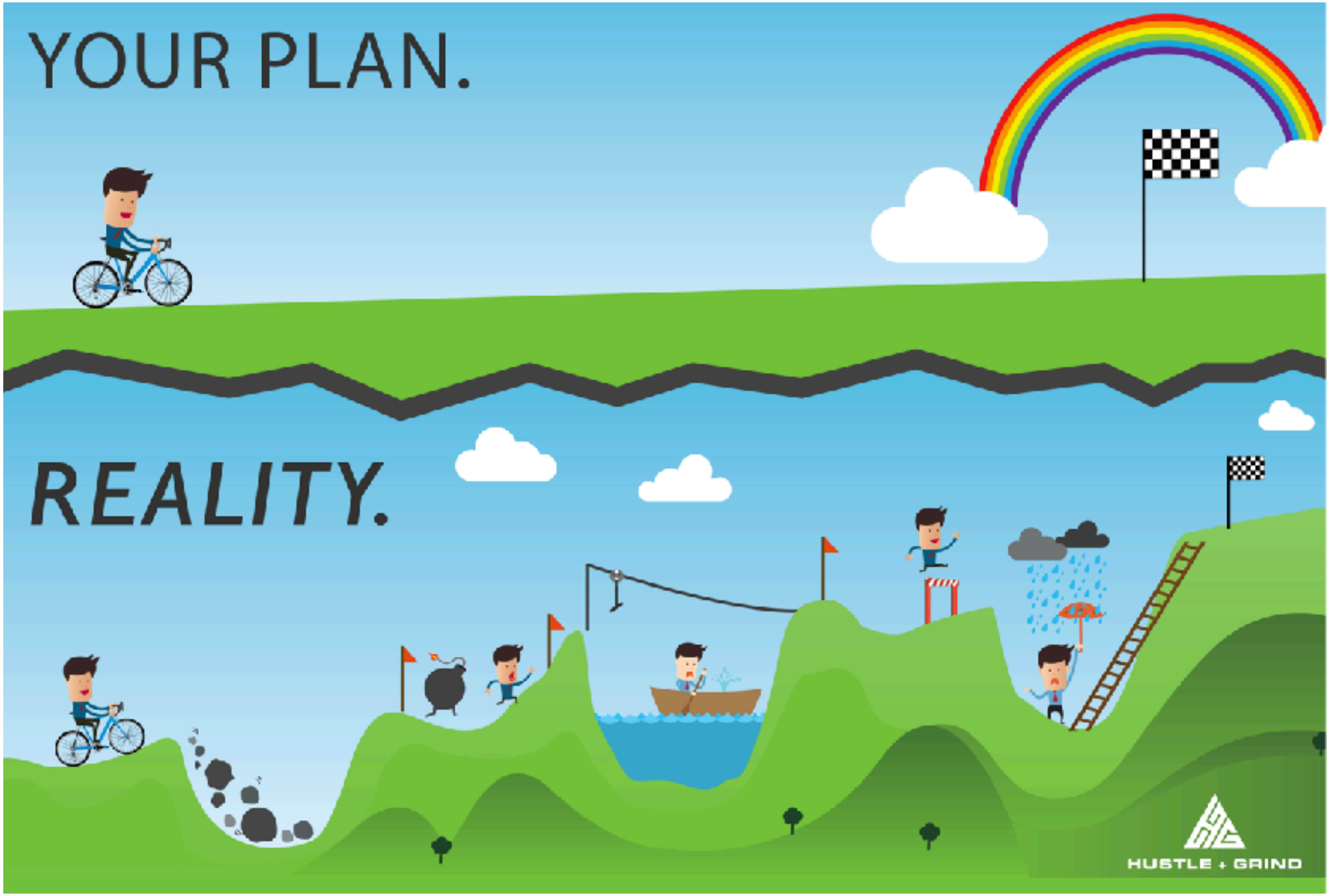
SDLC



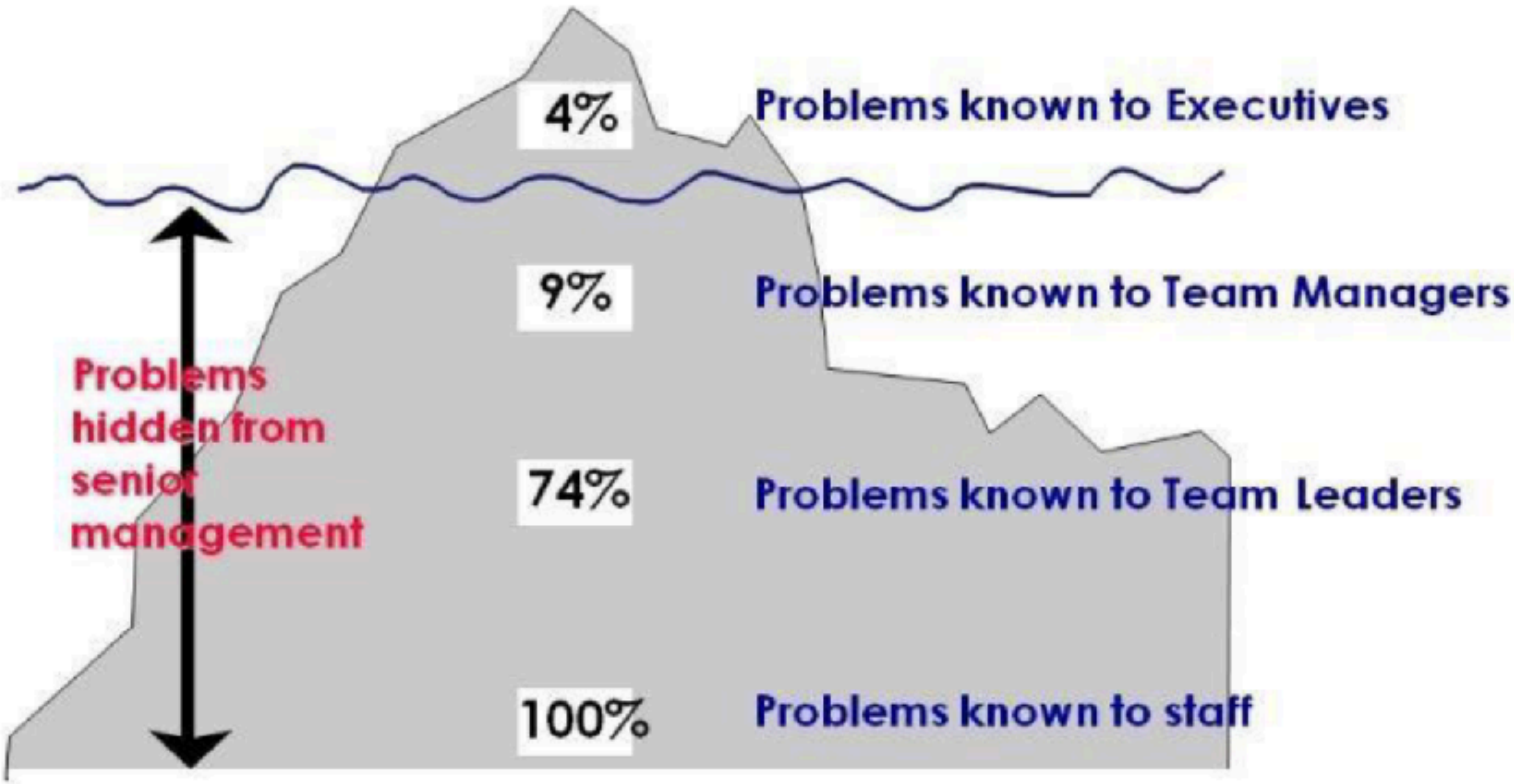
SDLC



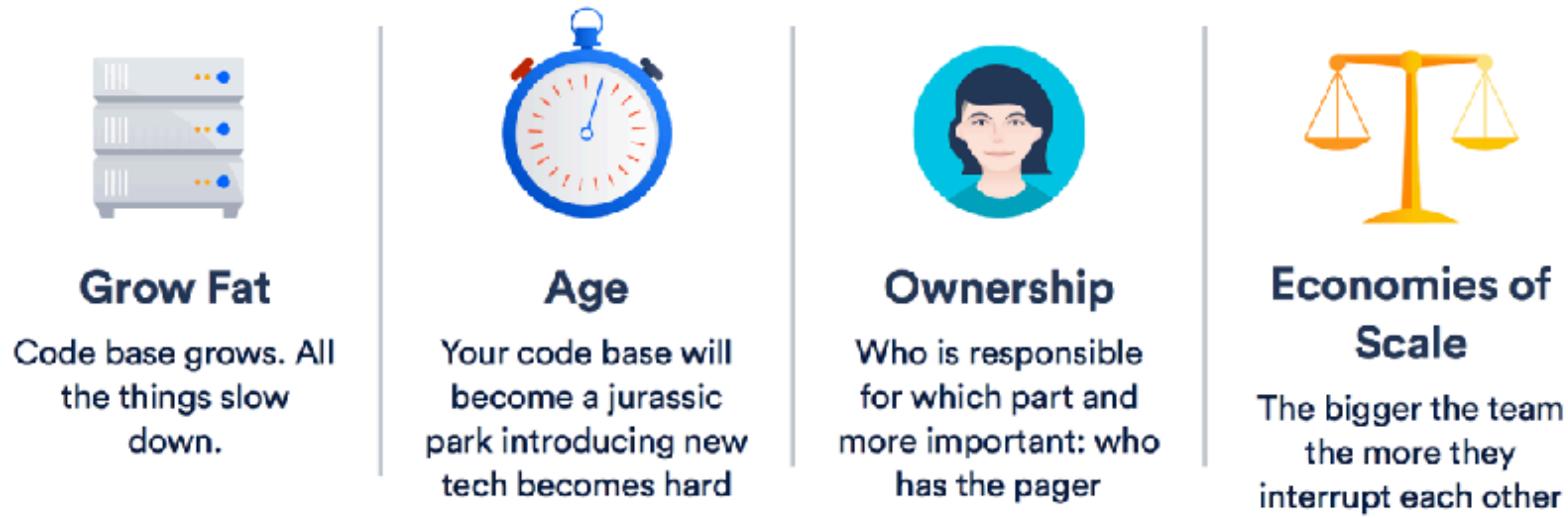
Plan vs Reality



The iceberg of ignorance



Consequence of inefficient

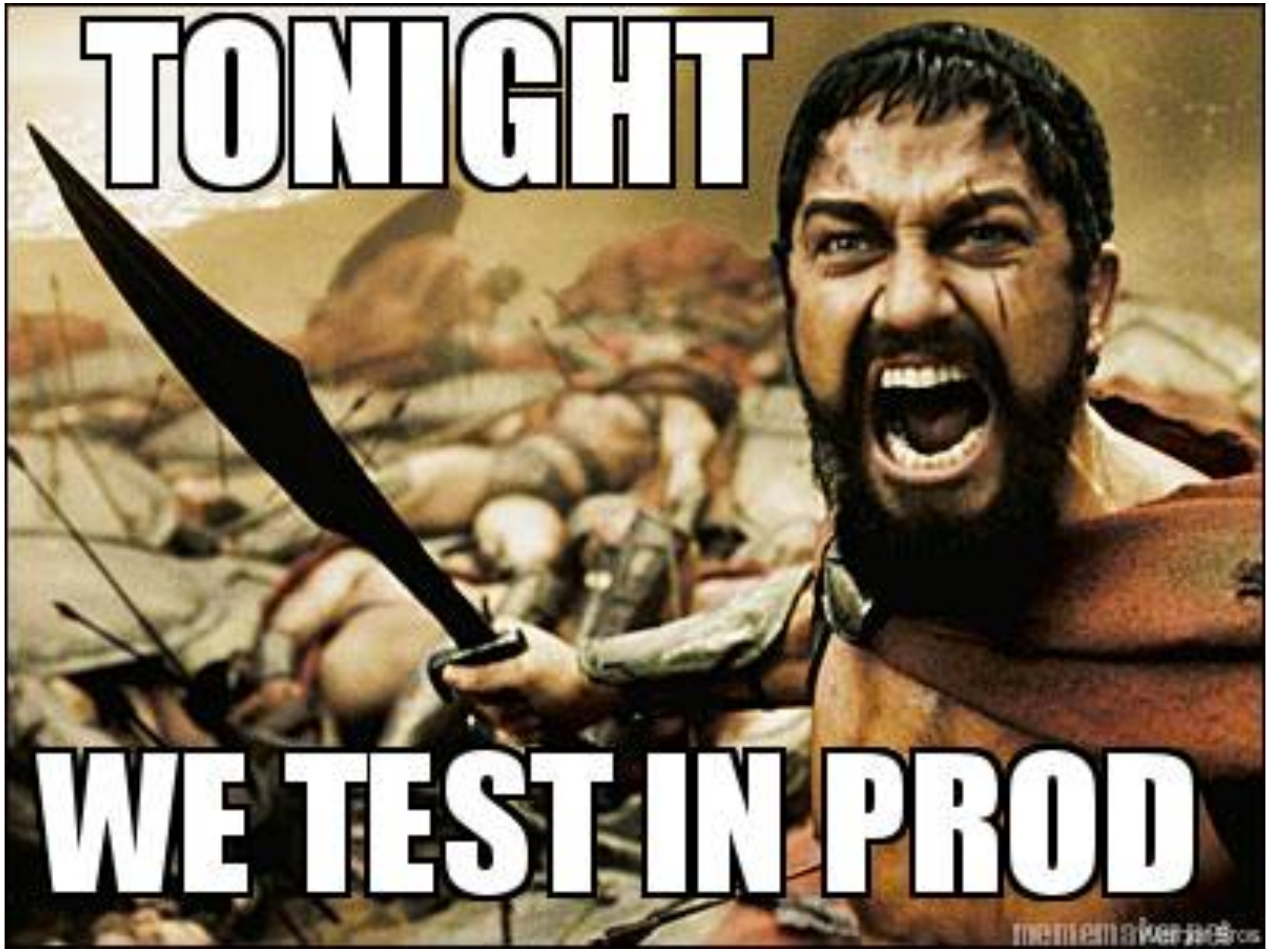


The Blame Game !!

Consequence of inefficient

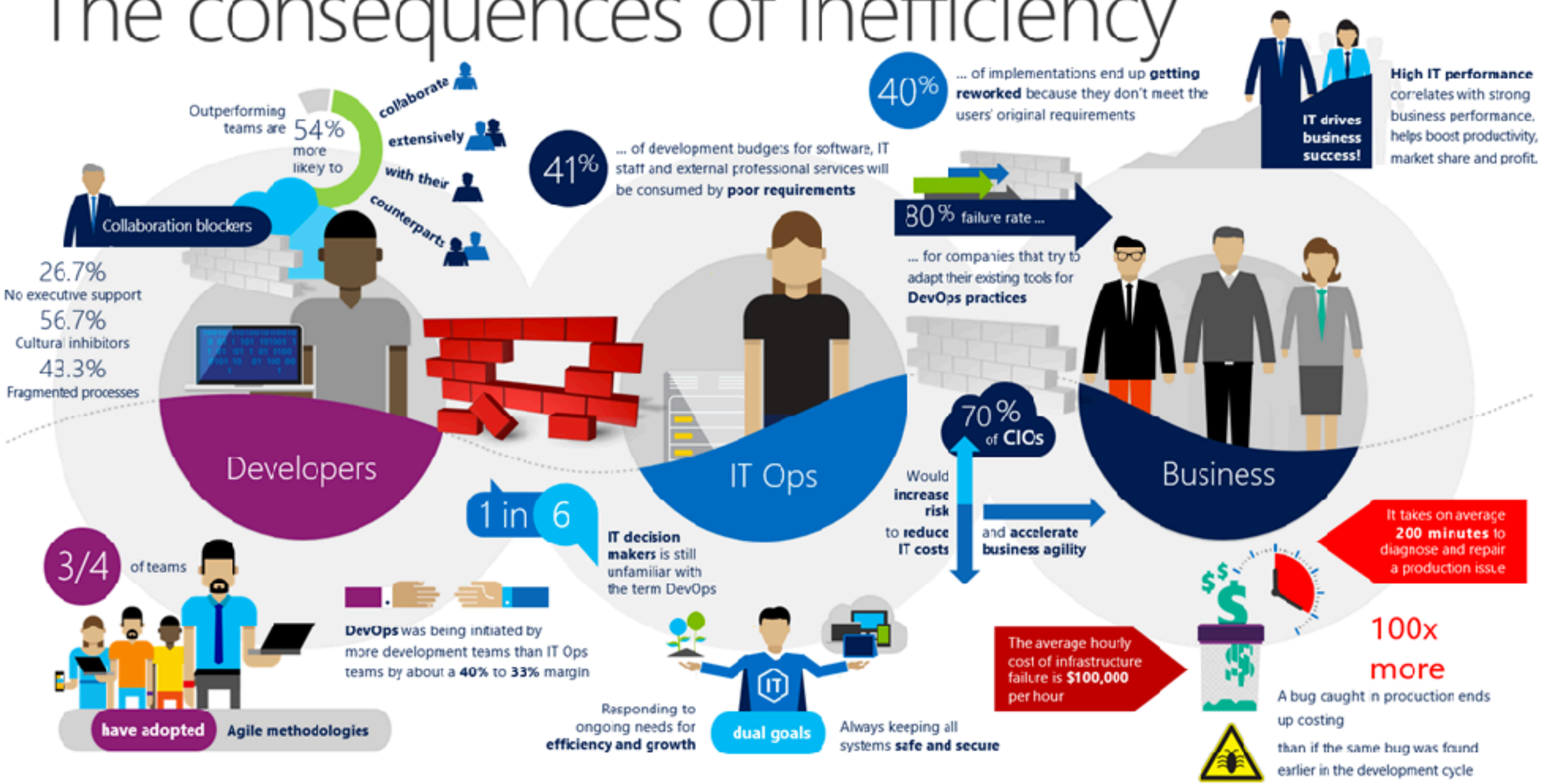
Consequence of inefficient

Consequence of inefficient

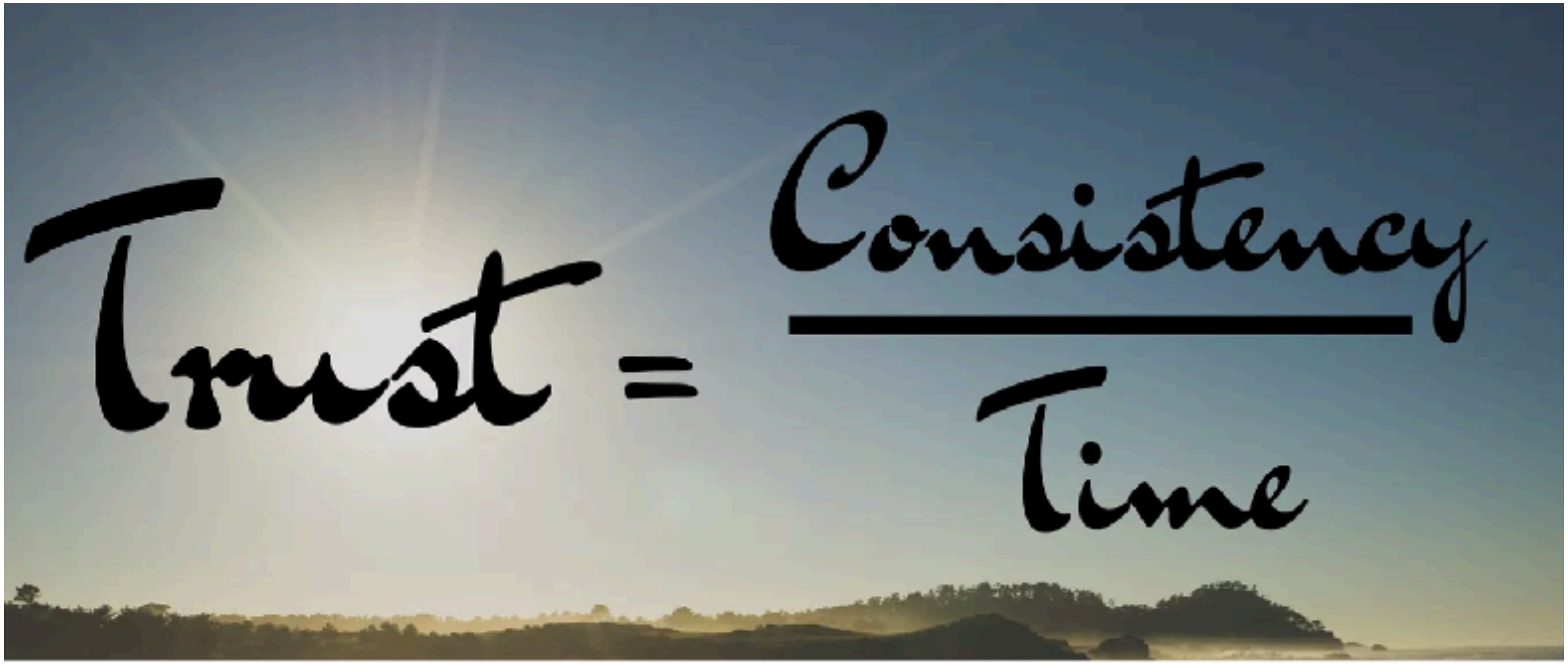


Consequence of inefficient

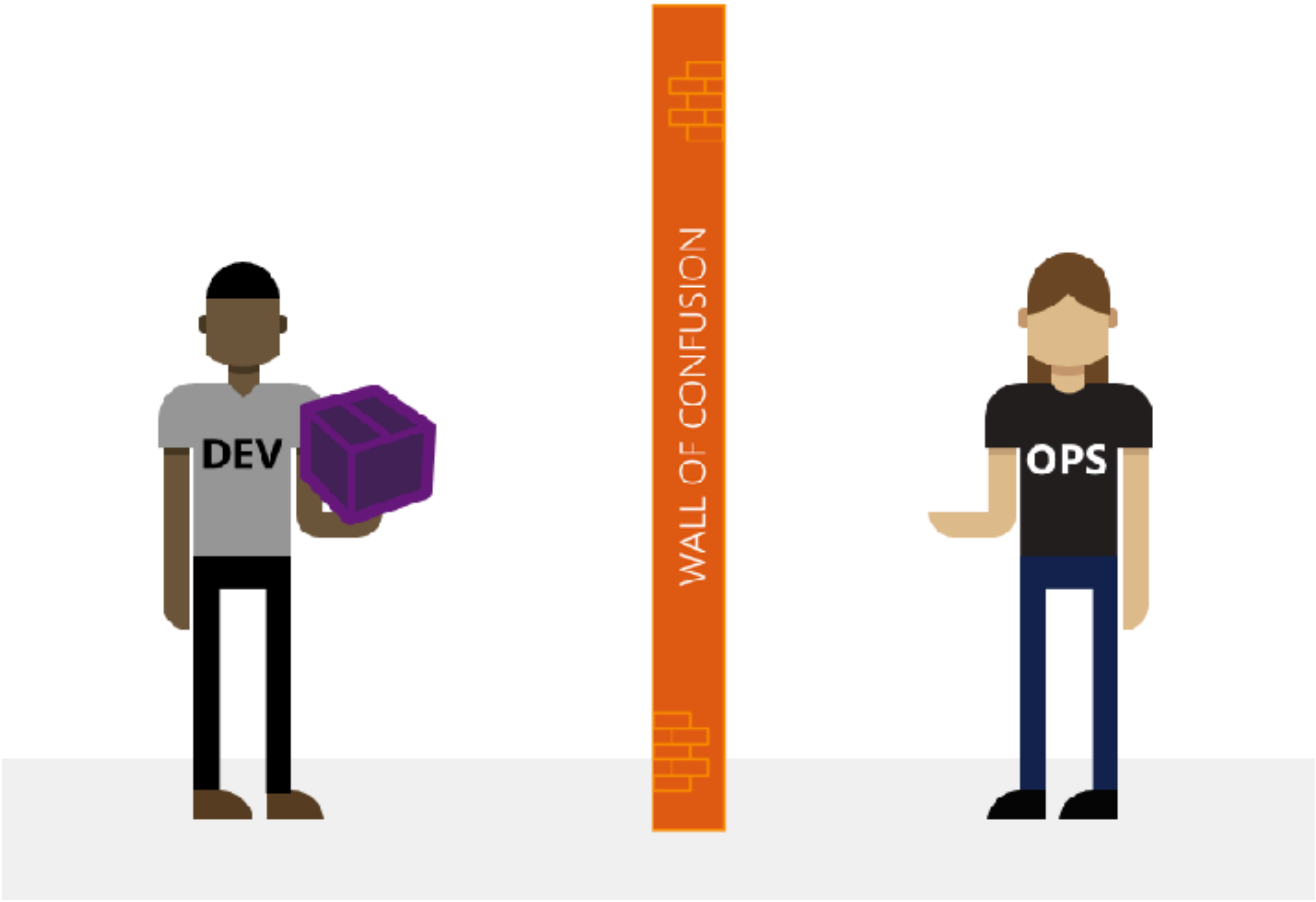
The consequences of inefficiency

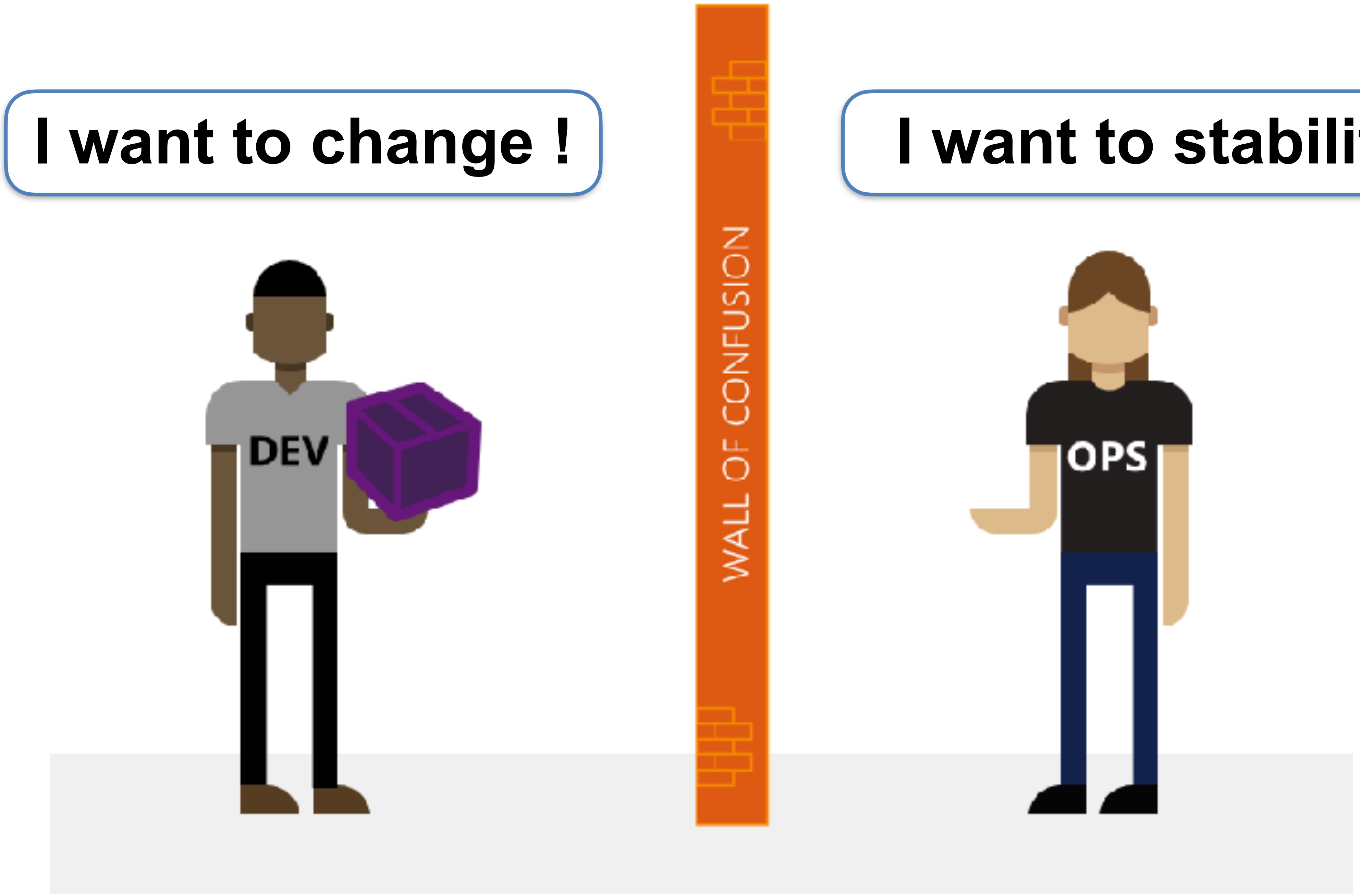


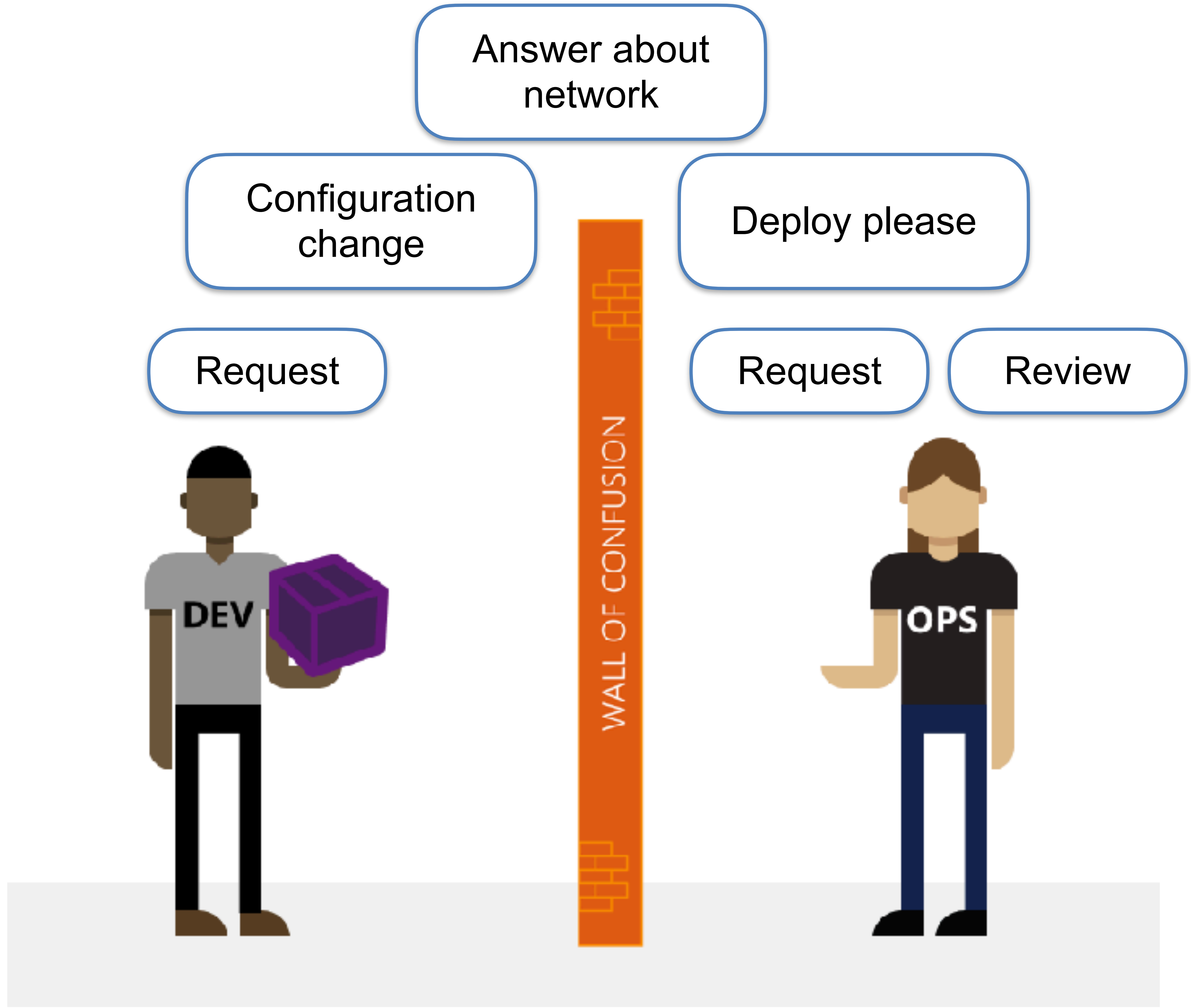
https://channel9.msdn.com/Series/DevOps-Fundamentals


Trust = Consistency
Time

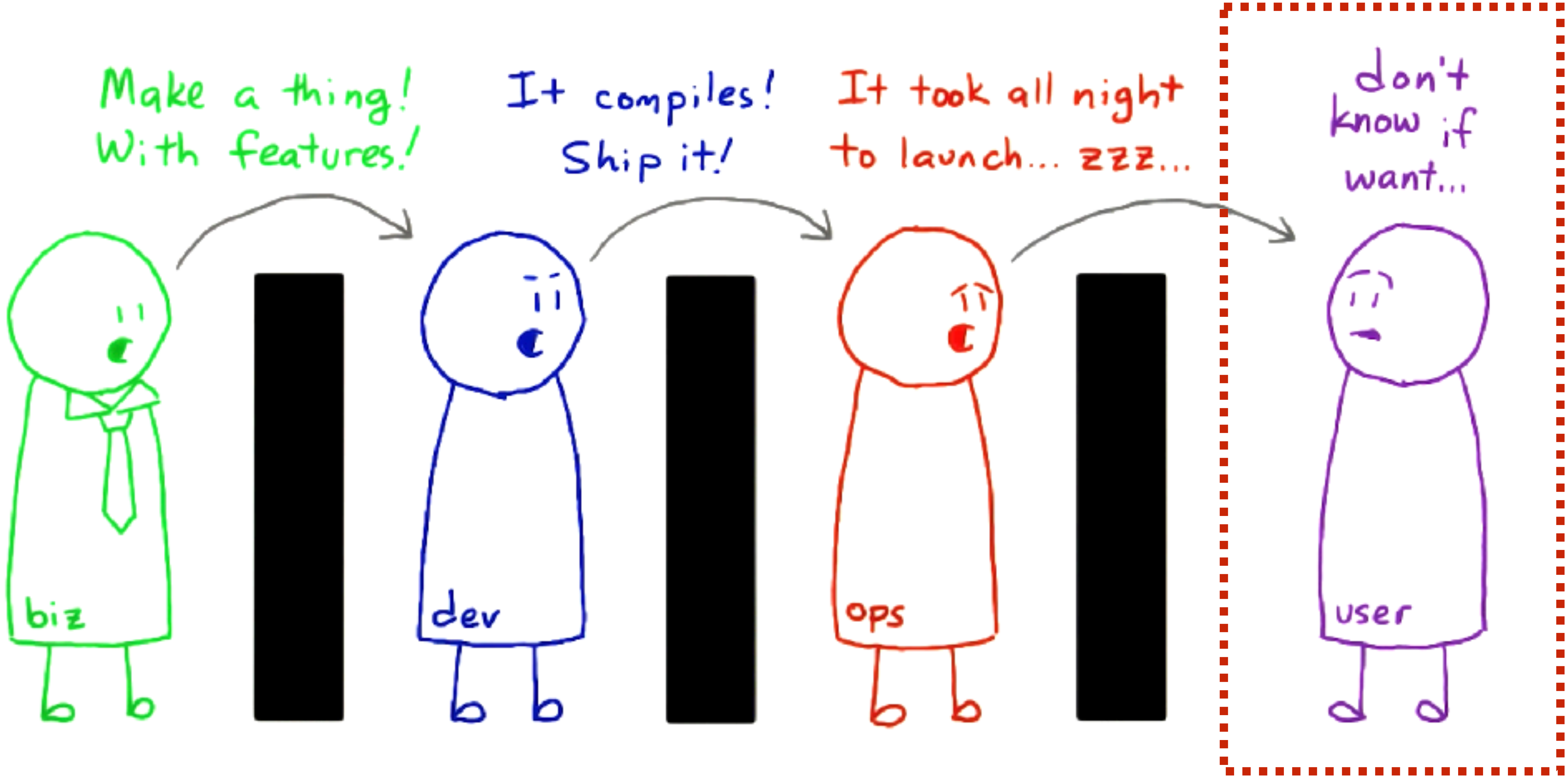
Development vs Operations


DEV
WALL OF CONFUSION
OPS


I want to change !
I want to stability !
DEV
WALL OF CONFUSION
OPS


Answer about network
Configuration change
Deploy please
Request
Request
Review
DEV
WALL OF CONFUSION
OPS

Result ?



Make a thing! With features!
It compiles! Ship it!
It took all night to launch... zzz...
don't know if want...
biz
dev
ops
user

Problems ?

Deadline Driven Development

Delayed project/product deliver

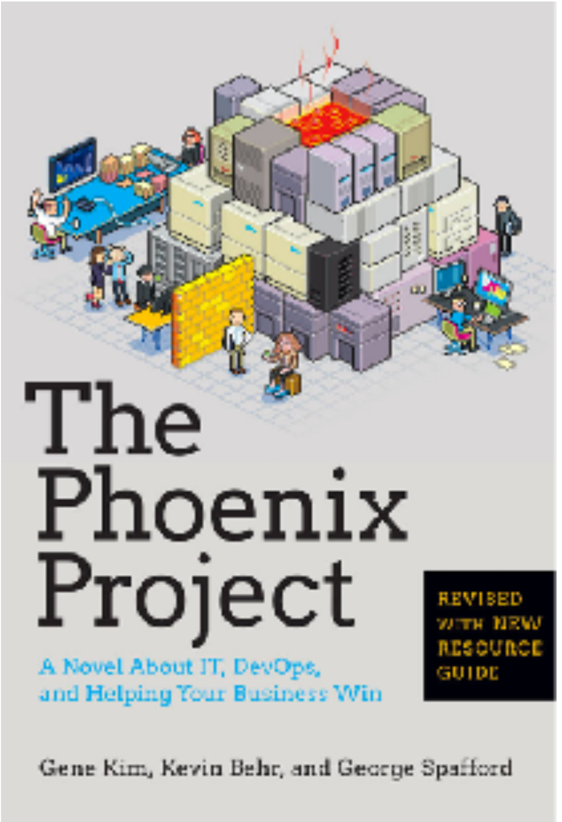
Bad quality

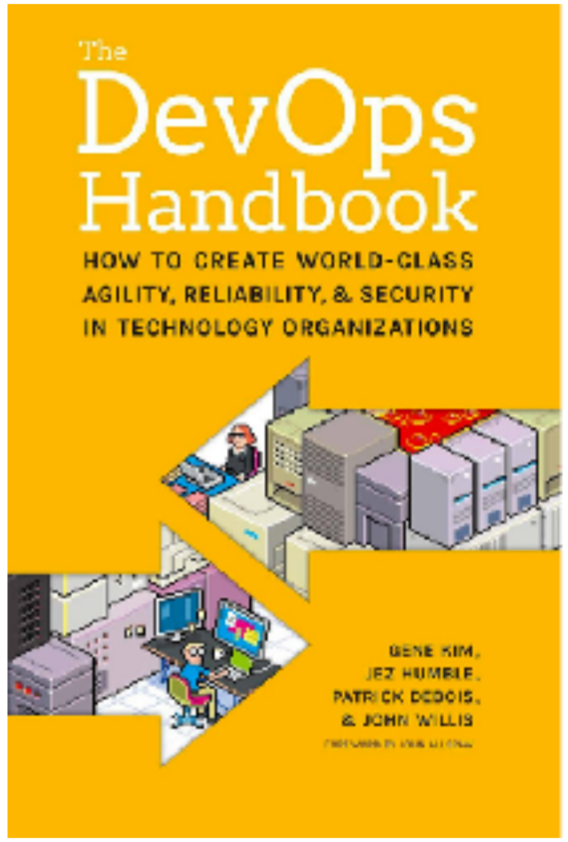
Low customer satisfaction

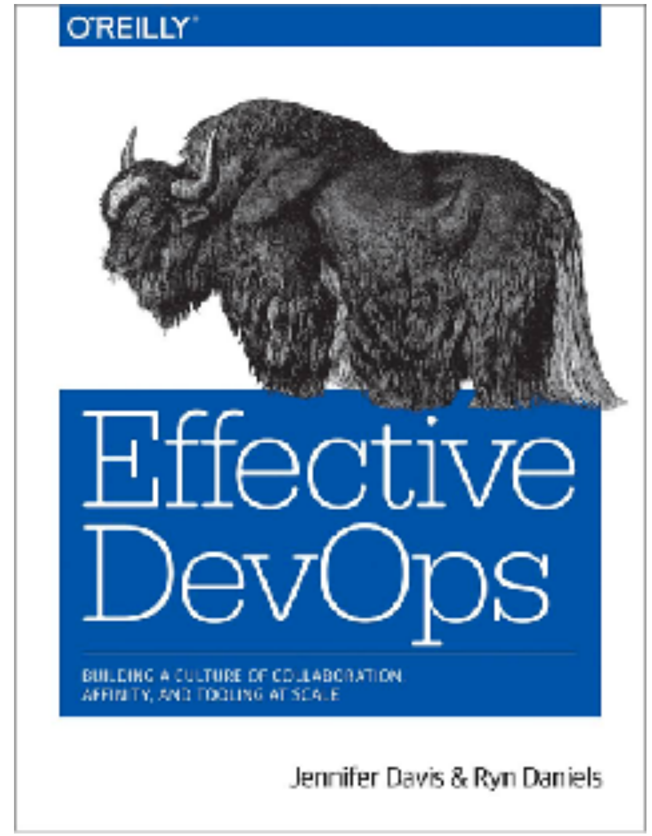
Rise of DevOps

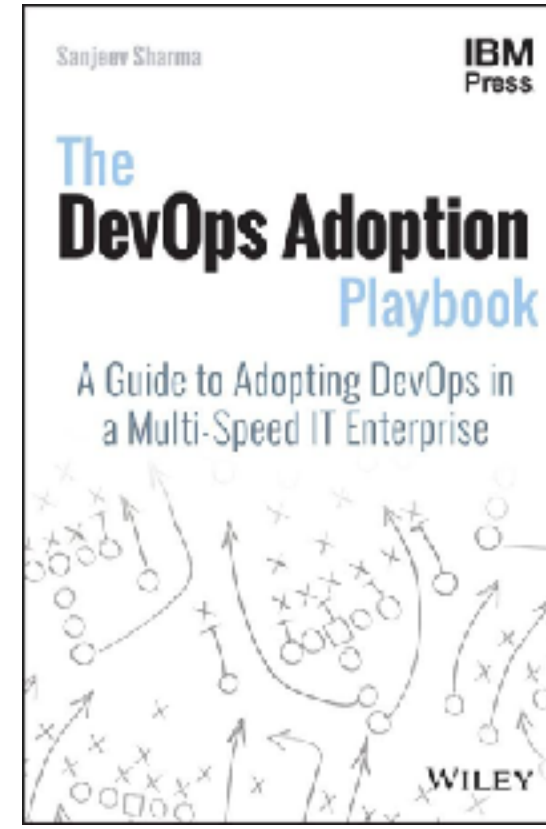

GROWTH
TOGETHER
TEAMWORK
HELP!
SUCCESS
Vision
PARTNER
30%
10%
Ideas!
Brainstorm

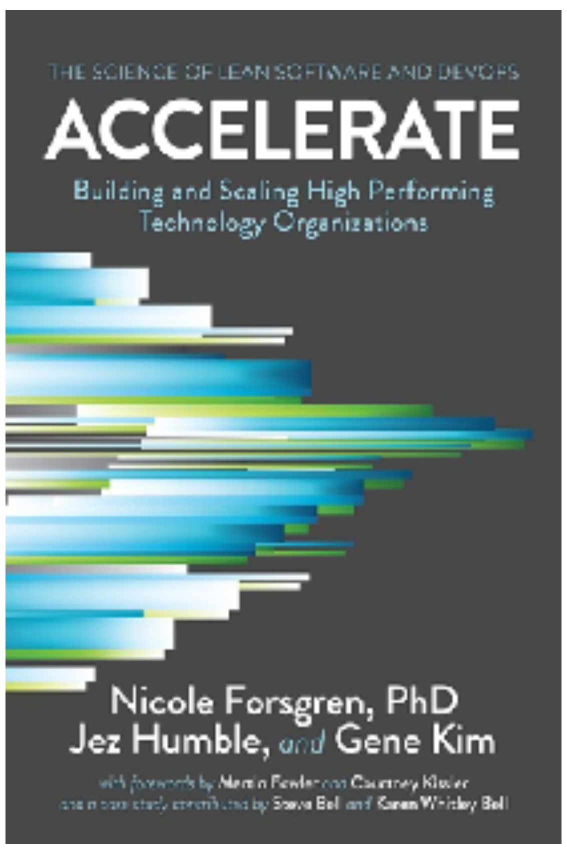
DevOps foundation

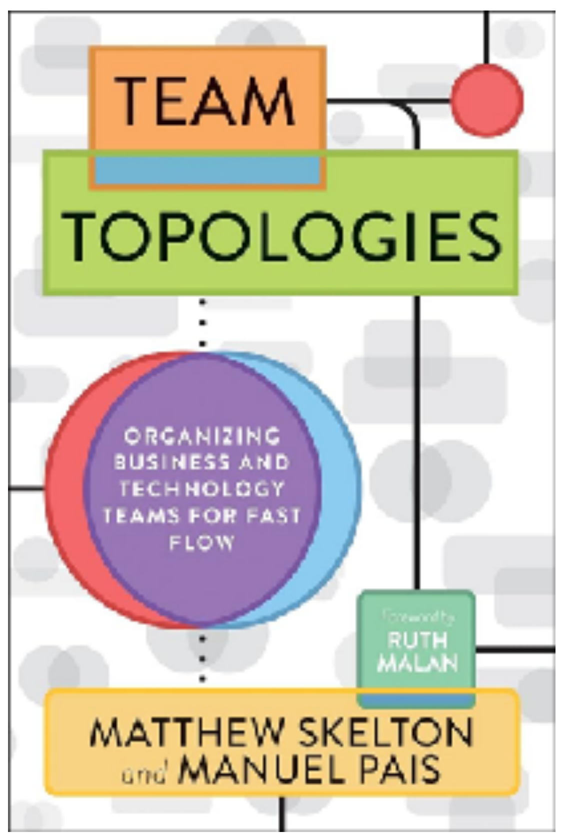

The Phoenix Project
REVISED WITH NEW RESOURCE GUIDE
A Novel About IT, DevOps, and Helping Your Business Win
Gene Kim, Kevin Behr, and George Spafford

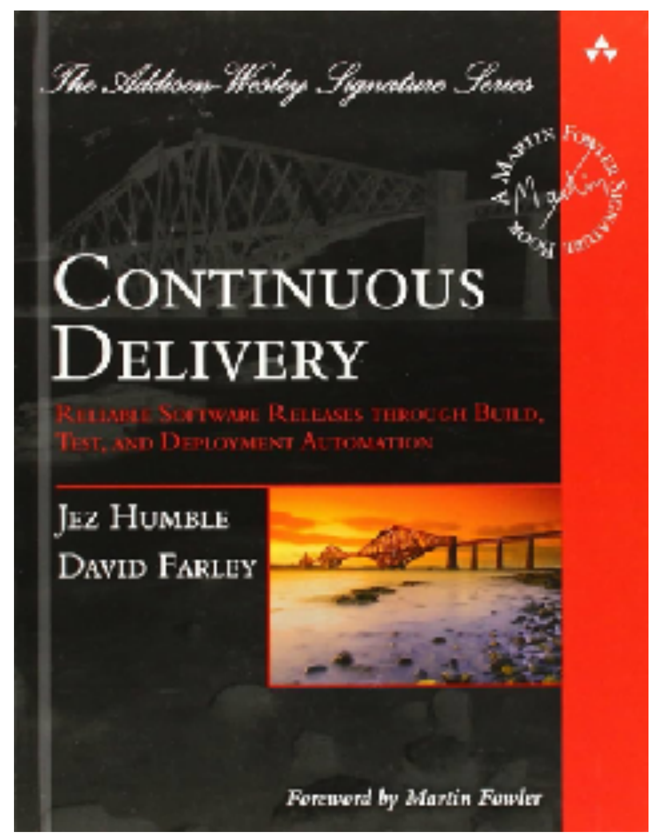

The DevOps Handbook
HOW TO CREATE WORLD-CLASS AGILITY, RELIABILITY, & SECURITY IN TECHNOLOGY ORGANIZATIONS
GENE KIM, JEZ HUMBLE, PATRICK DEBOIS, & JOHN WILLIS

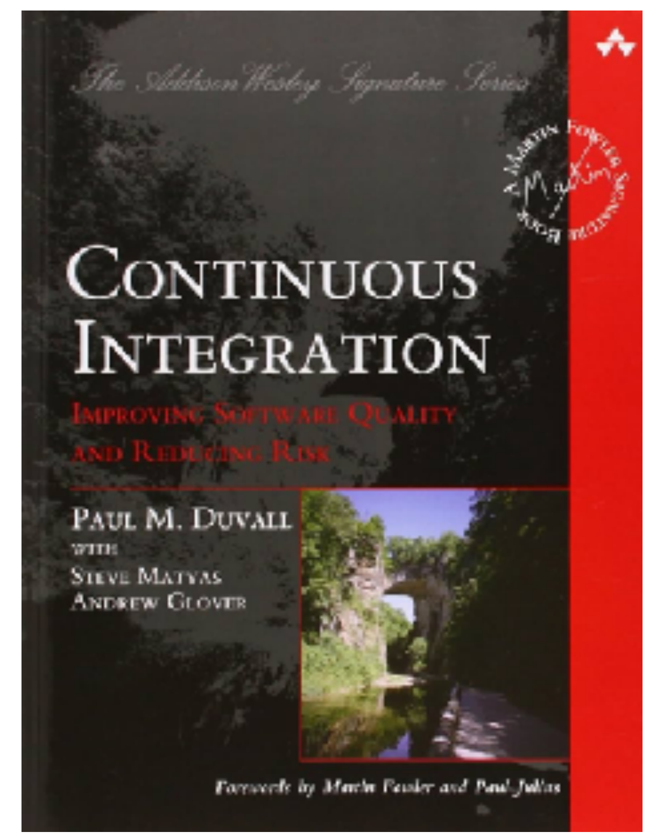

O'REILLY
Effective DevOps
BUILDING A CULTURE OF COLLABORATION, AFFINITY, AND TOOLING AT SCALE
Jennifer Davis & Ryn Daniels


Sanjeev Sharma
IBM Press
The DevOps Adoption Playbook
A Guide to Adopting DevOps in a Multi-Speed IT Enterprise
WILEY

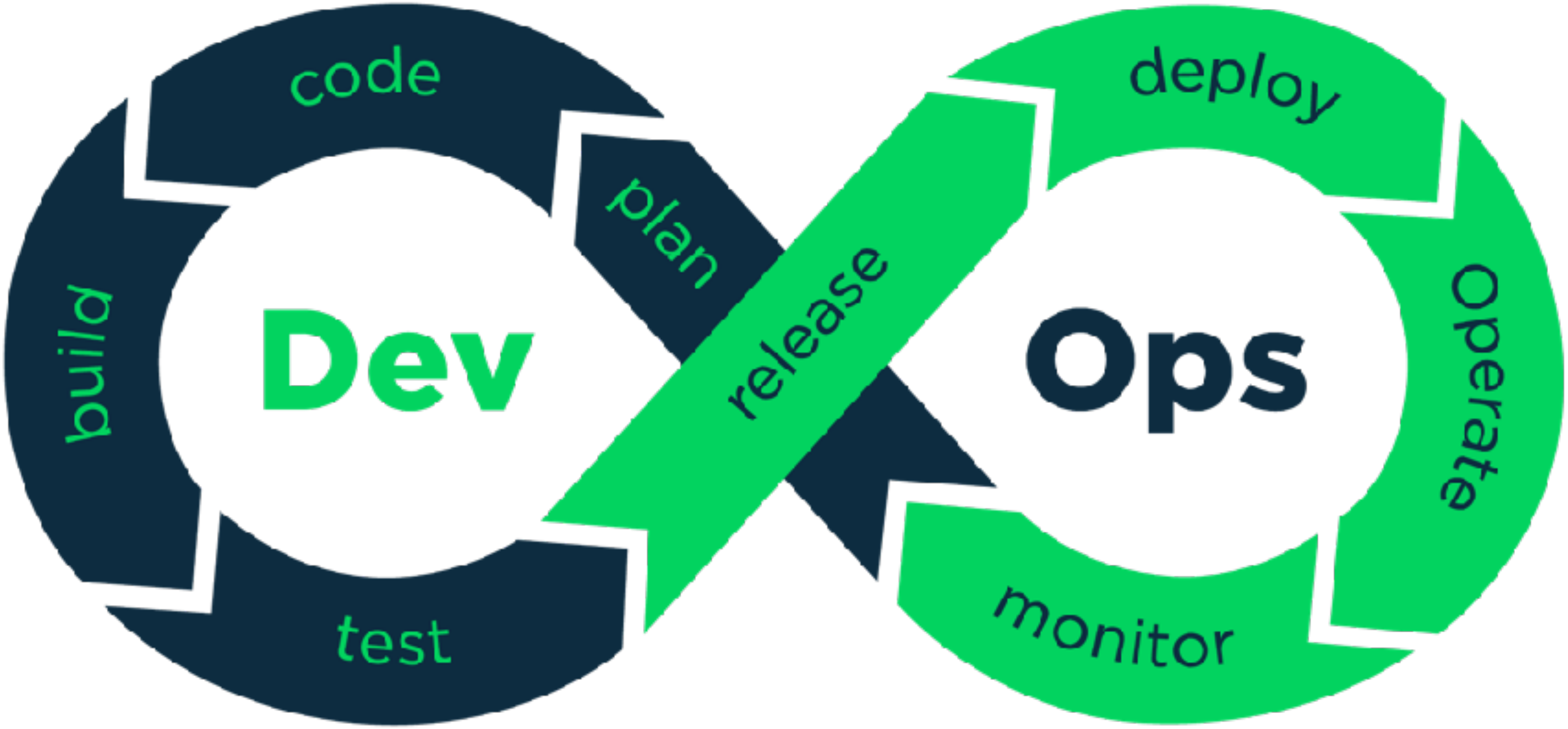

THE SCIENCE OF LEAN SOFTWARE AND DEVOPS
ACCELERATE
Building and Scaling High Performing Technology Organizations
Nicole Forsgren, PhD
Jez Humble, and Gene Kim


TEAM TOPOLOGIES
ORGANIZING BUSINESS AND TECHNOLOGY TEAMS FOR FAST FLOW
RUTH MALAN
MATTHEW SKELTON and MANUEL PAIS


The Addison-Wesley Signature Series
CONTINUOUS DELIVERY
RELIABLE SOFTWARE RELEASES THROUGH BUILD, TEST, AND DEPLOYMENT AUTOMATION
JEZ HUMBLE
DAVID FARLEY
Foreword by Martin Fowler

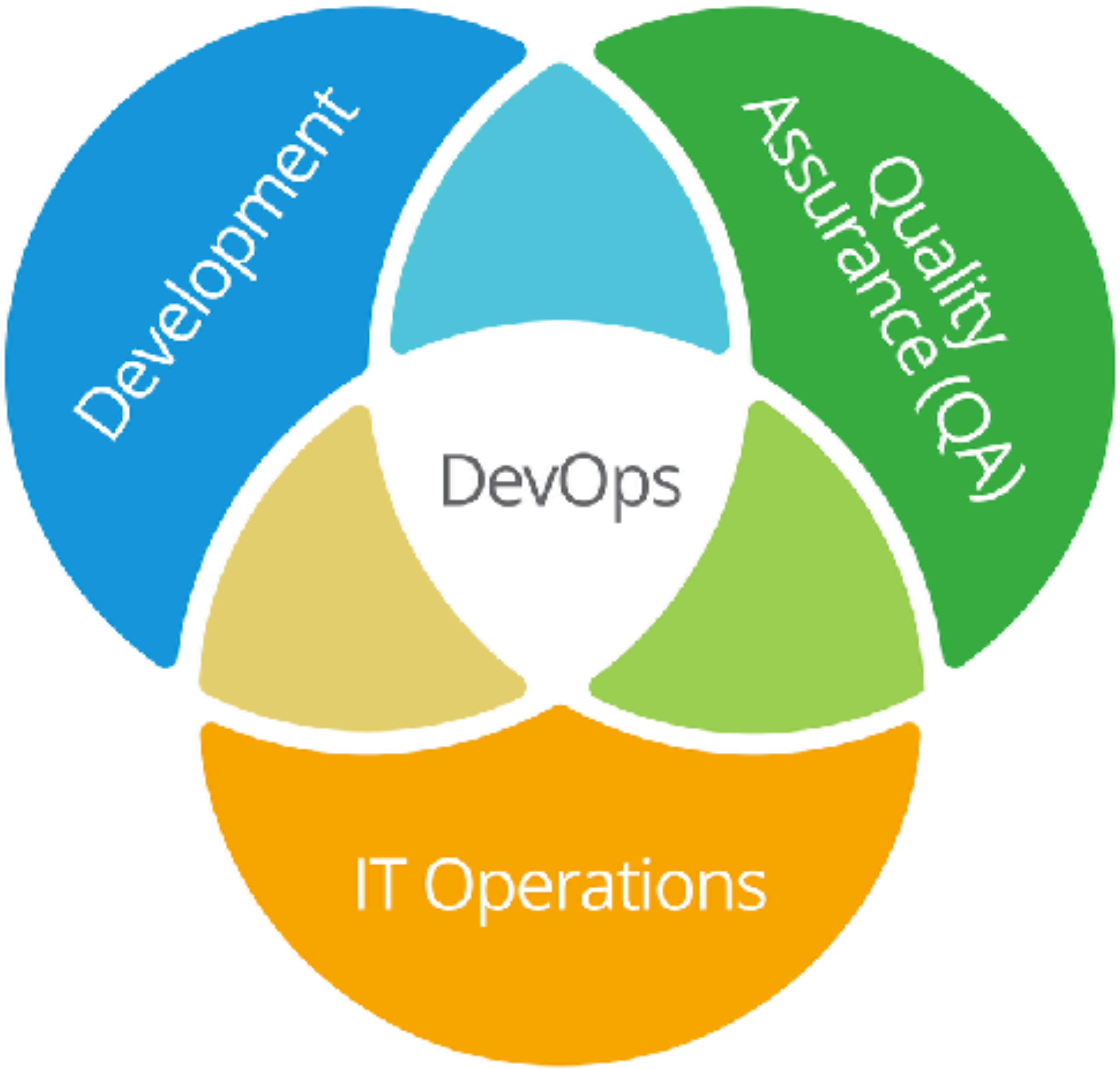

The Addison-Wesley Signature Series
CONTINUOUS INTEGRATION
IMPROVING SOFTWARE QUALITY AND REDUCING RISK
PAUL M. DUVALL
WITH
STEVE MATYAS
ANDREW GLOVER
Forewords by Martin Fowler and Paul Julius

DevOps



code
plan
build
Dev
release
test
deploy
Ops
Operate
monitor

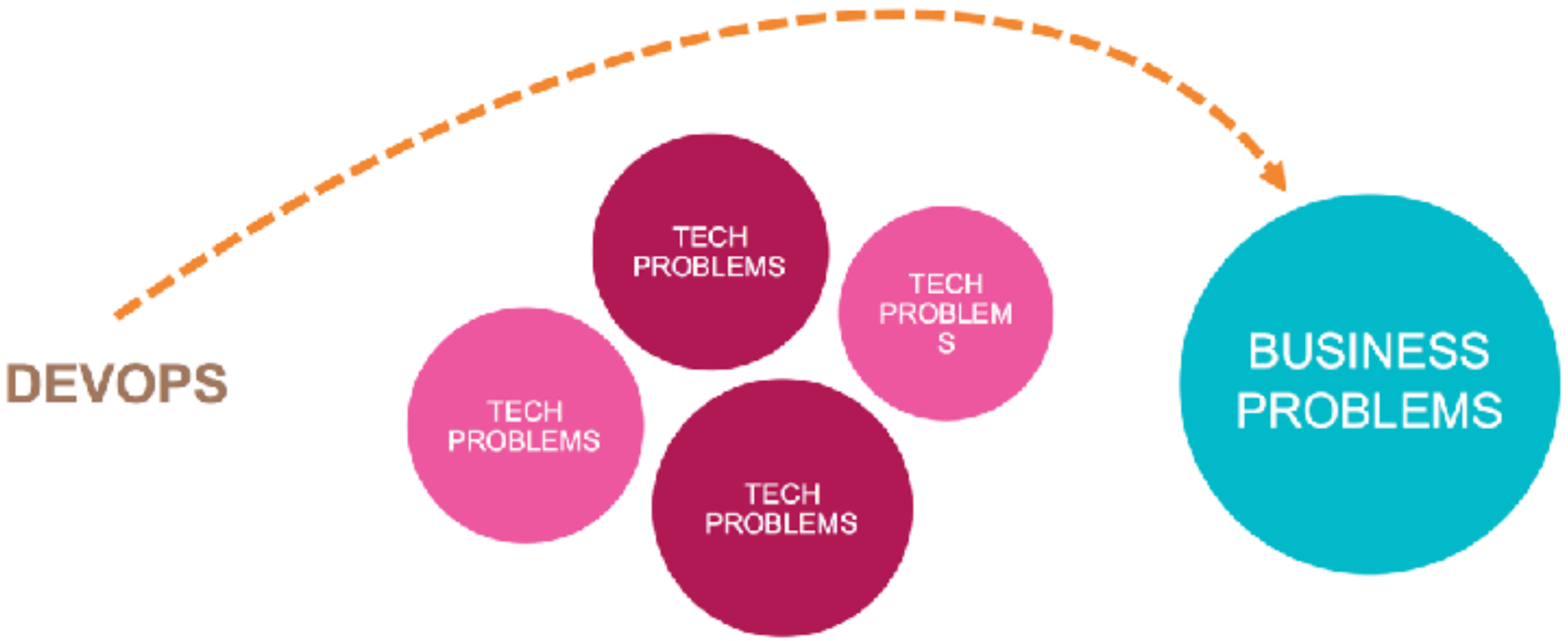
DevOps



Devops isn't any single person's job.
It's everyone's job.

DevOps

Solve **business** problem first



State of DevOps Report

High-performing teams deploy more frequently and have much faster lead times.

They make changes with fewer failures, and recover faster from failures.

200x more frequent deployments

2,555x shorter lead times

3x lower change failure rate

24x faster recovery from failures

https://puppet.com/resources/whitepaper/state-of-devops-report

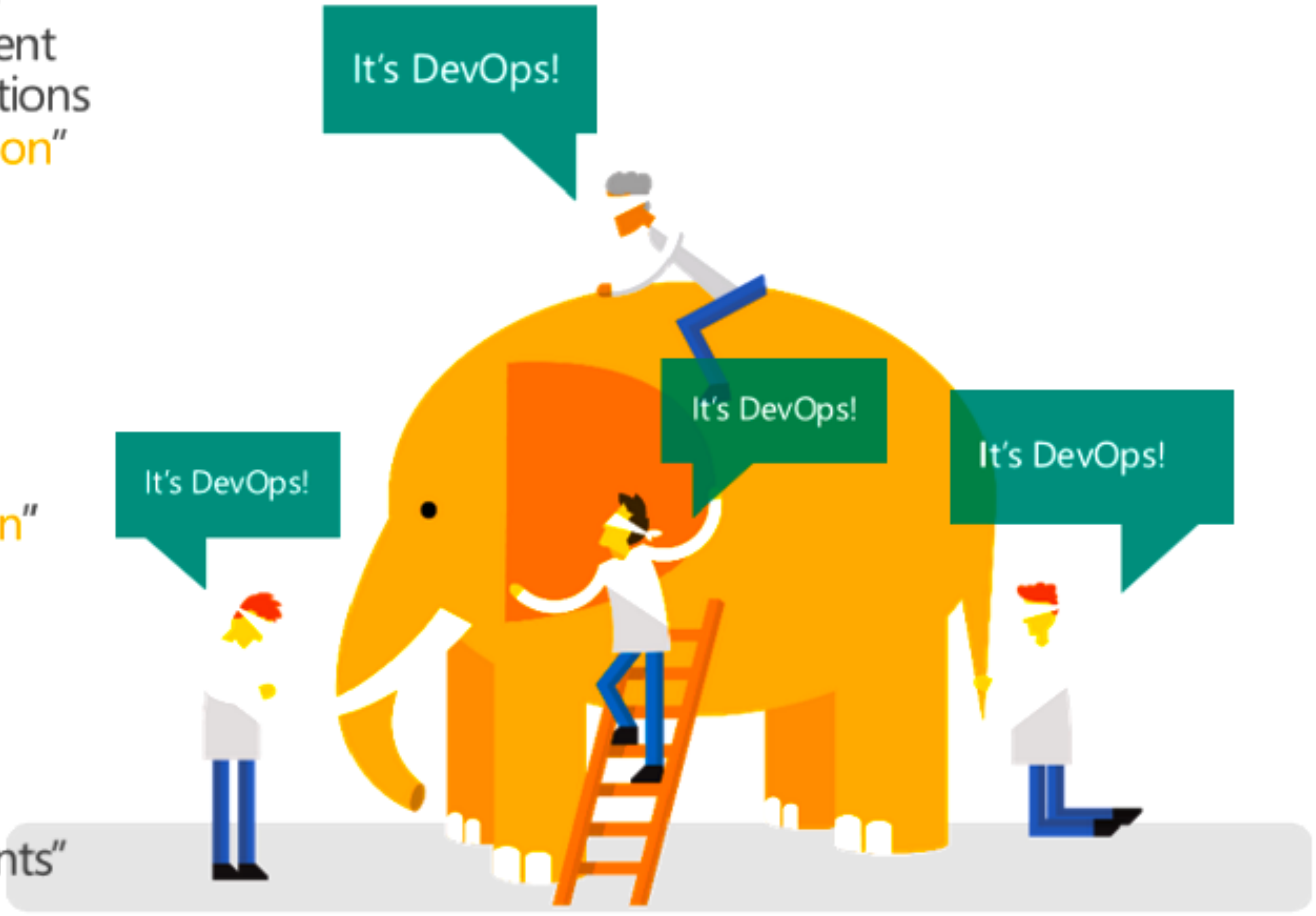
What is DevOps ?

What is DevOps ?

"**DevOps** is development and operations collaboration"

"**DevOps** is using automation"

"**DevOps** is small deployments"



"**DevOps** is treating your infrastructure as code"

"**DevOps** is feature switches"

"Kanban for Ops?"

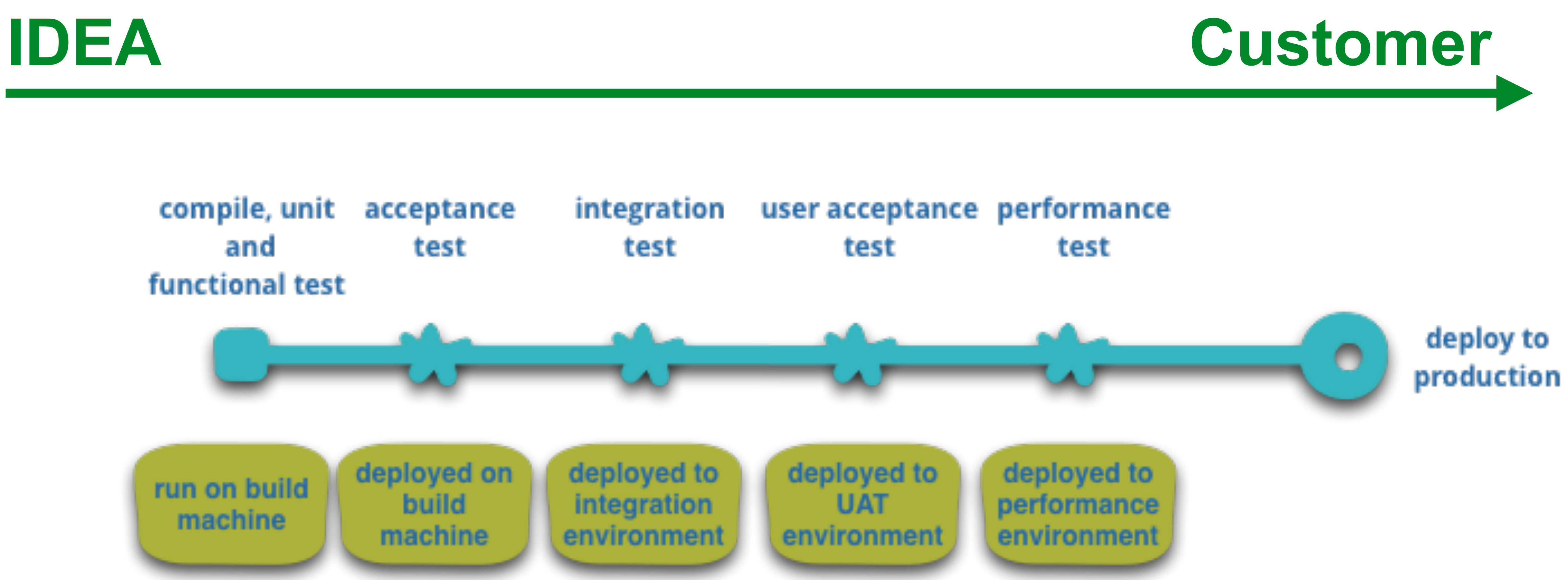
What is DevOps ?

DevOps is a **set of practices** intended to **reduce the time** between committing a change to a system and the change being placed into normal production, while ensuring **high quality**

https://en.wikipedia.org/wiki/DevOps#Definitions_and_history

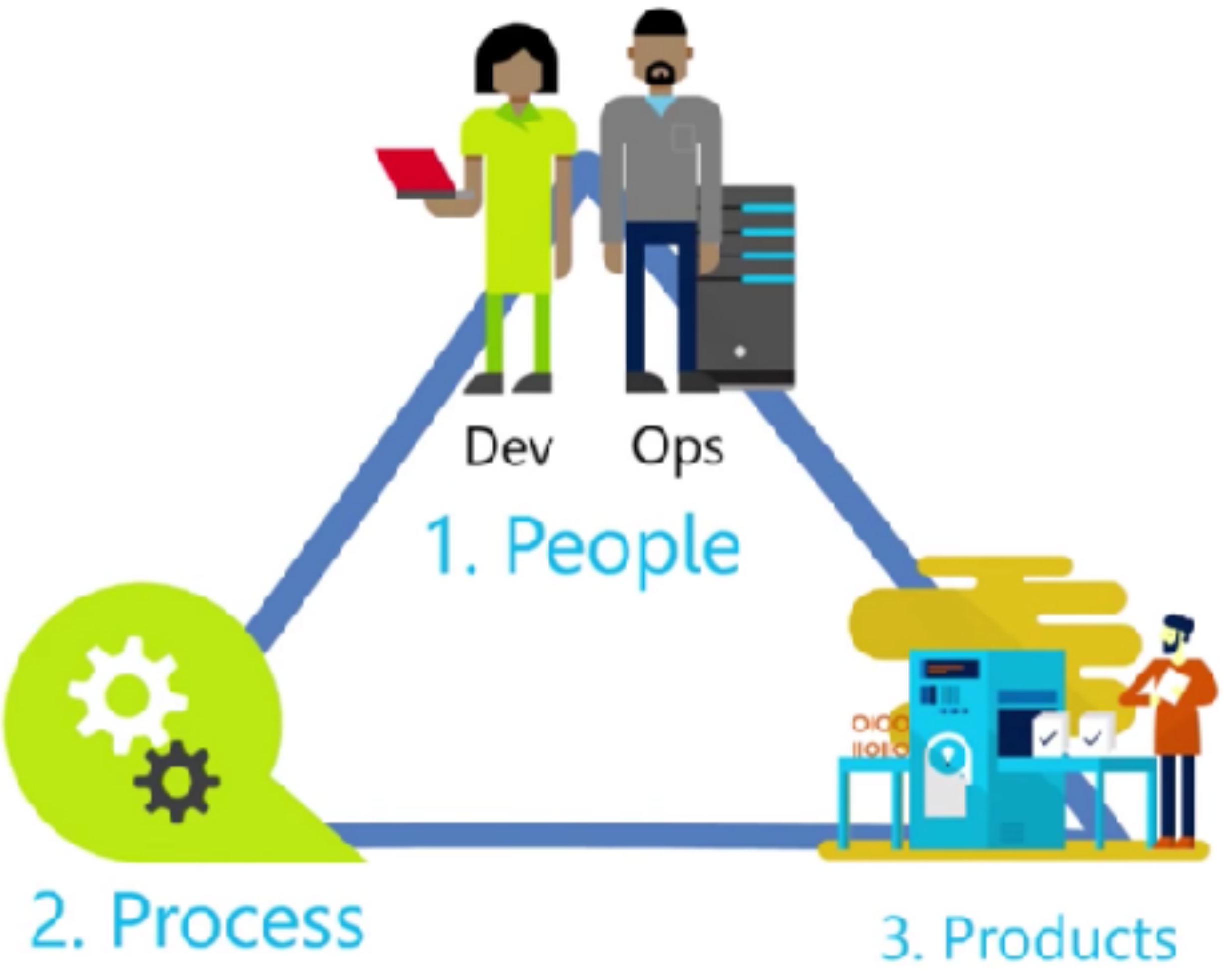
Goals of DevOps

Improve the delivery of value
for Customer and Business



Goals of DevOps

Enable the **continuous delivery** of value to customer and business



Continuous Delivery

The more often to deploy

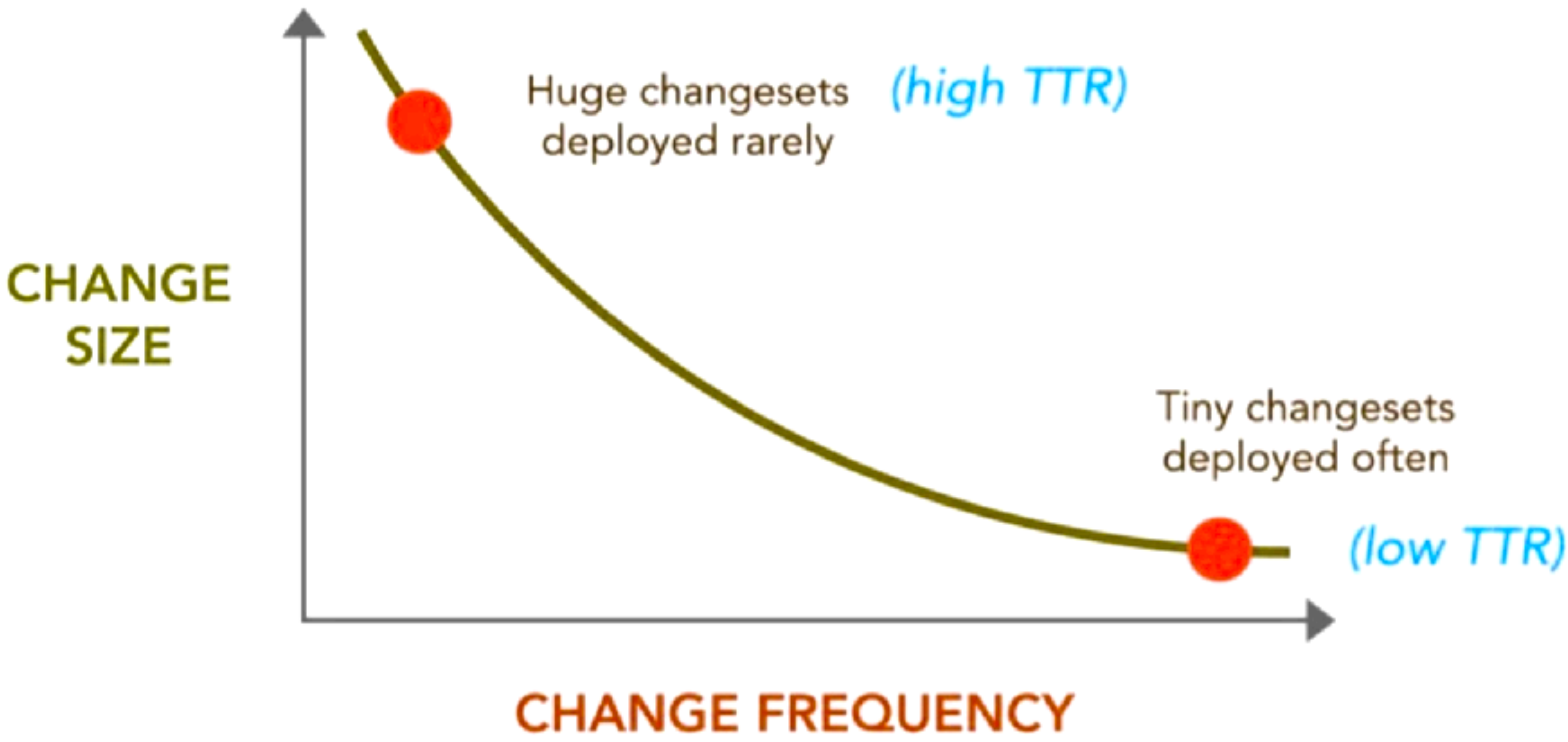
Small change

Automate it

Reduce **TTR** (Time to Repair/Recovery)

Learn and repeat

Continuous Delivery



DevOps is all about Human problems

DevOps Metrics

Lead time for changes

Change failure rate

Deployment frequency

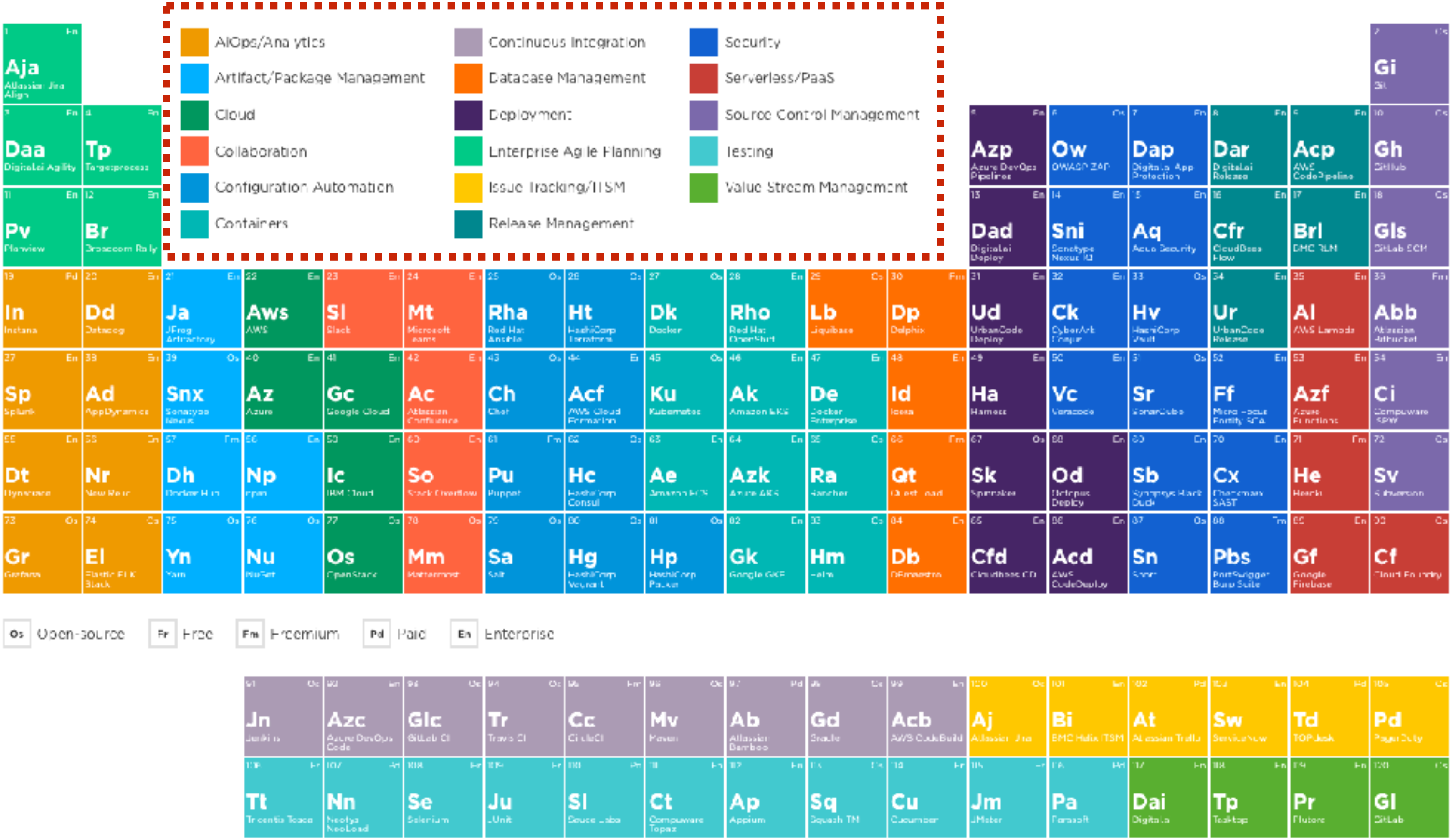
Mean time to recovery (MTTR)

Software Delivery Performance

Aspect of Software delivery performance	Elite	High	Medium	Low
Change lead time For the primary application or service you work on, what is your lead time for changes (i.e., how long does it take to go from code committed to code successfully running in production)?	Less than one day	Between one day and one week	Between one week and one month	Between one week and one month
Deployment frequency For the primary application or service you work on, how often does your organization deploy code to production or release it to end users?	On-demand (multiple deploys per day)	Between once per day and once per week	Between once per week and once per month	Between once per week and once per month
Change failure rate For the primary application or service you work on, what percentage of changes to production or released to users result in degraded service (e.g., lead to service impairment or service outage) and subsequently require remediation (e.g., require a hotfix, rollback, fix forward, patch)?	5%	10%	15%	64%
Failed deployment recovery time For the primary application or service you work on, how long does it generally take to restore service after a change to production or release to users results in degraded service (for example, lead to service impairment or service outage) and subsequently require remediation (for example, require a hotfix, rollback, fix forward, or patch)?	Less than one hour	Less than one day	Between one day and one week	Between one month and six months
Percentage of respondents	18%	31%	33%	17%

https://cloud.google.com/blog/products/devops-sre/announcing-the-2023-state-of-devops-report

DevOps Tools



https://digital.ai/periodic-table-of-devops-tools

AiOps/Analytics	Continuous Integration
Artifact/Package Management	Database Management
Cloud	Deployment
Collaboration	Enterprise Agile Planning
Configuration Automation	Issue Tracking/ITSM
Containers	Release Management
Security	Serverless/PaaS
Source Control Management	Testing
Value Stream Management	

https://digital.ai/periodic-table-of-devops-tools

Workshop
Design your delivery process

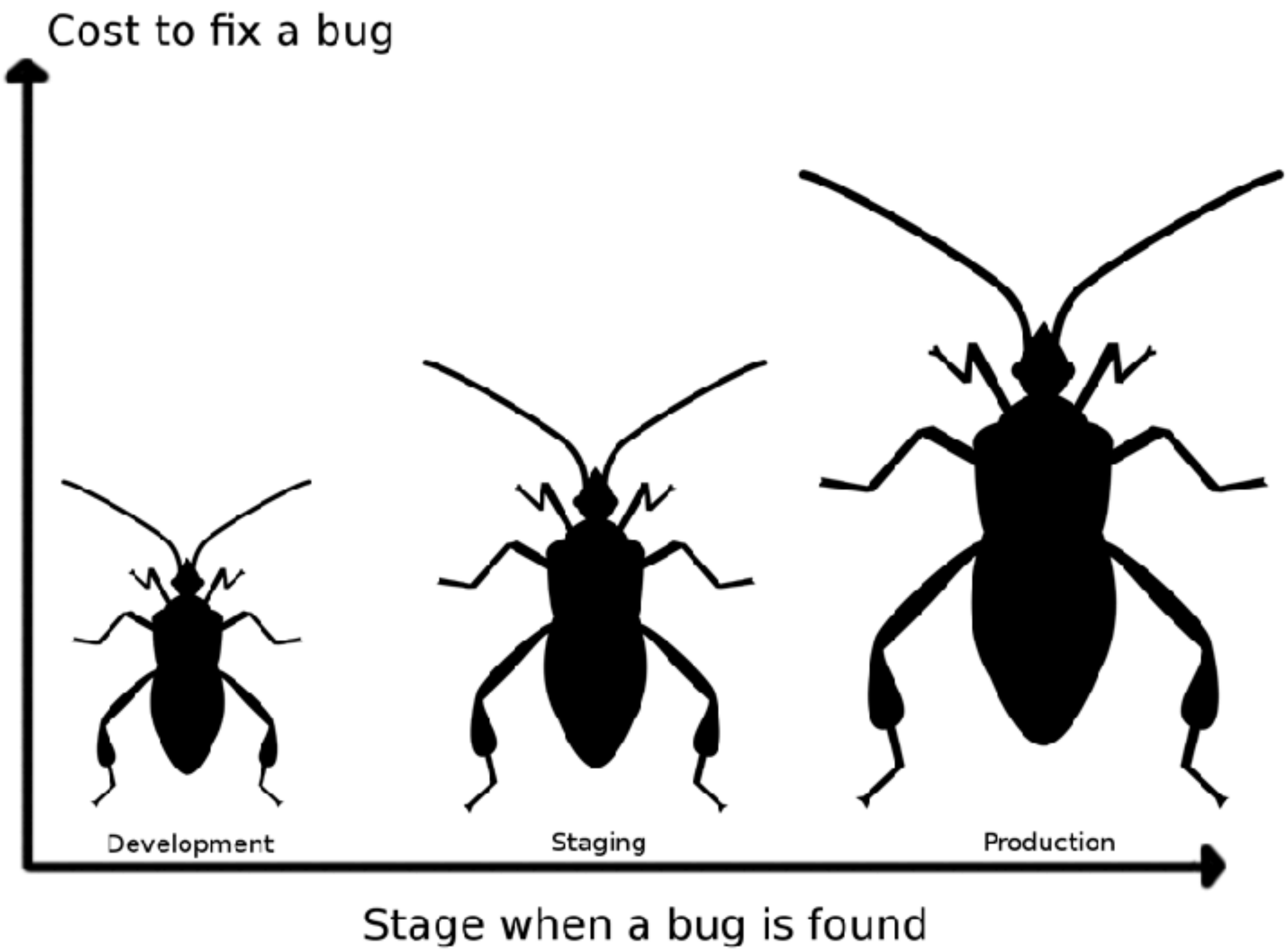
Continuous Integration
Continuous Delivery

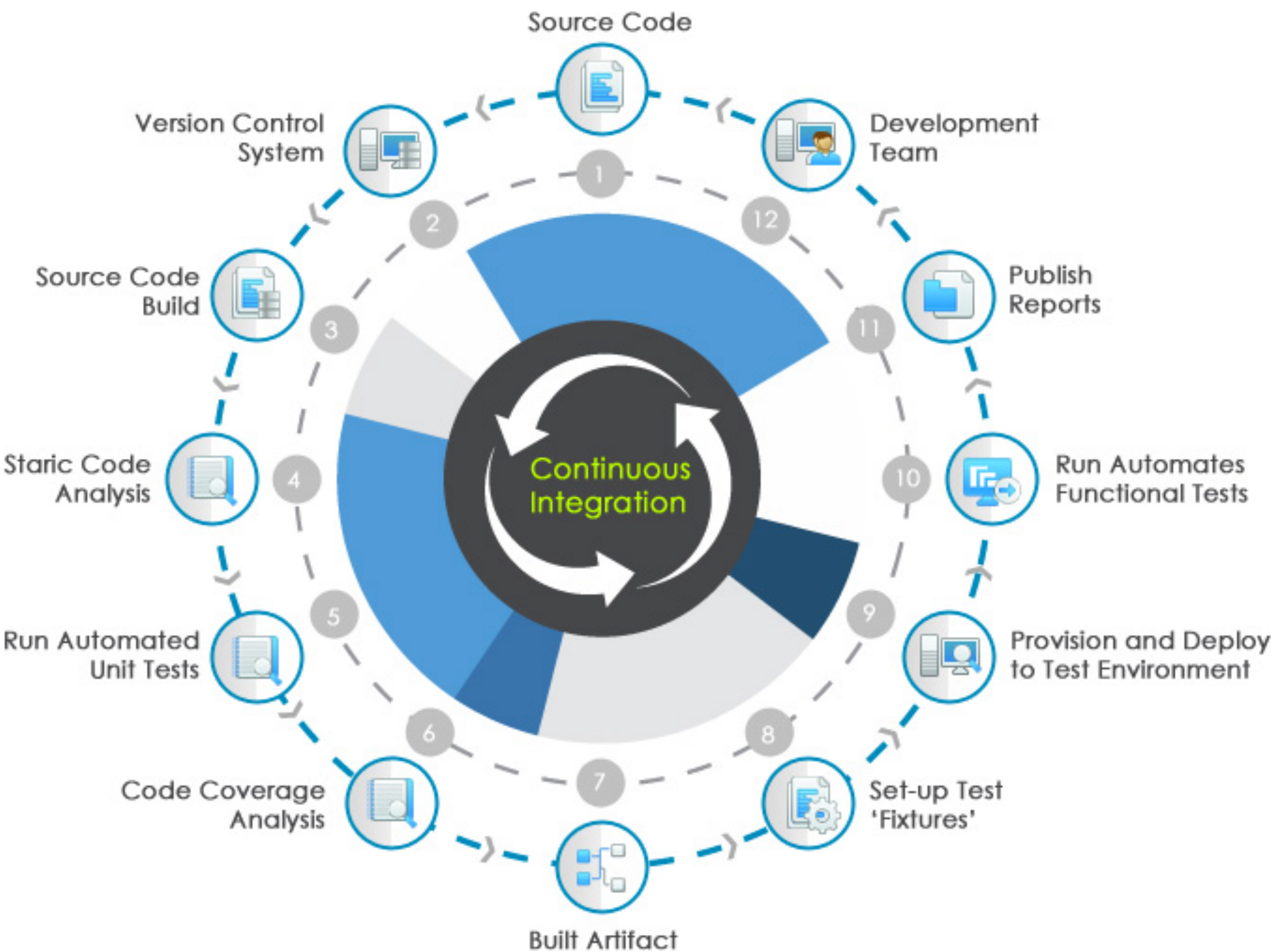
Why CI/CD ?

The cost of integration

1. Merging the code
2. Duplicate changes
3. Test again again !!
4. Fixing bugs
5. Impact on stability

The cost of integration




Source Code
Development Team
Publish Reports
Run Automates Functional Tests
Provision and Deploy to Test Environment
Set-up Test 'Fixtures'
Built Artifact
Code Coverage Analysis
Run Automated Unit Tests
Staric Code Analysis
Source Code Build
Version Control System
Continuous Integration
1
2
3
4
5
6
7
8
9
10
11
12

Jenkins

Bamboo

TeamCity

go™

Visual Studio
Team Foundation Server

Hudson

Travis

wercker

circleci

CI is about what people do

not about what tools they use

Continuous Integration

Discipline to **integrate frequently**

Continuous Integration

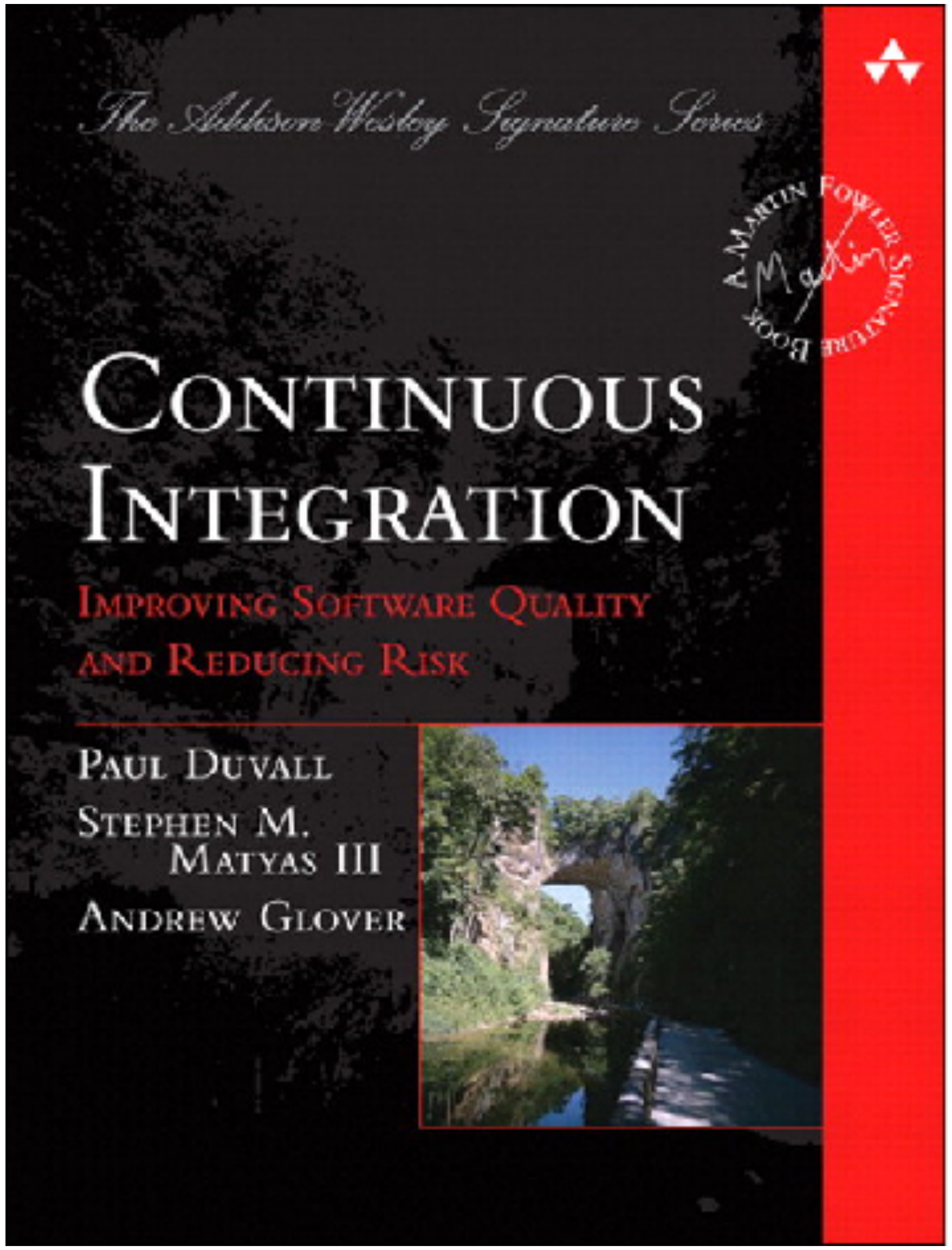
Strive to make **small change**

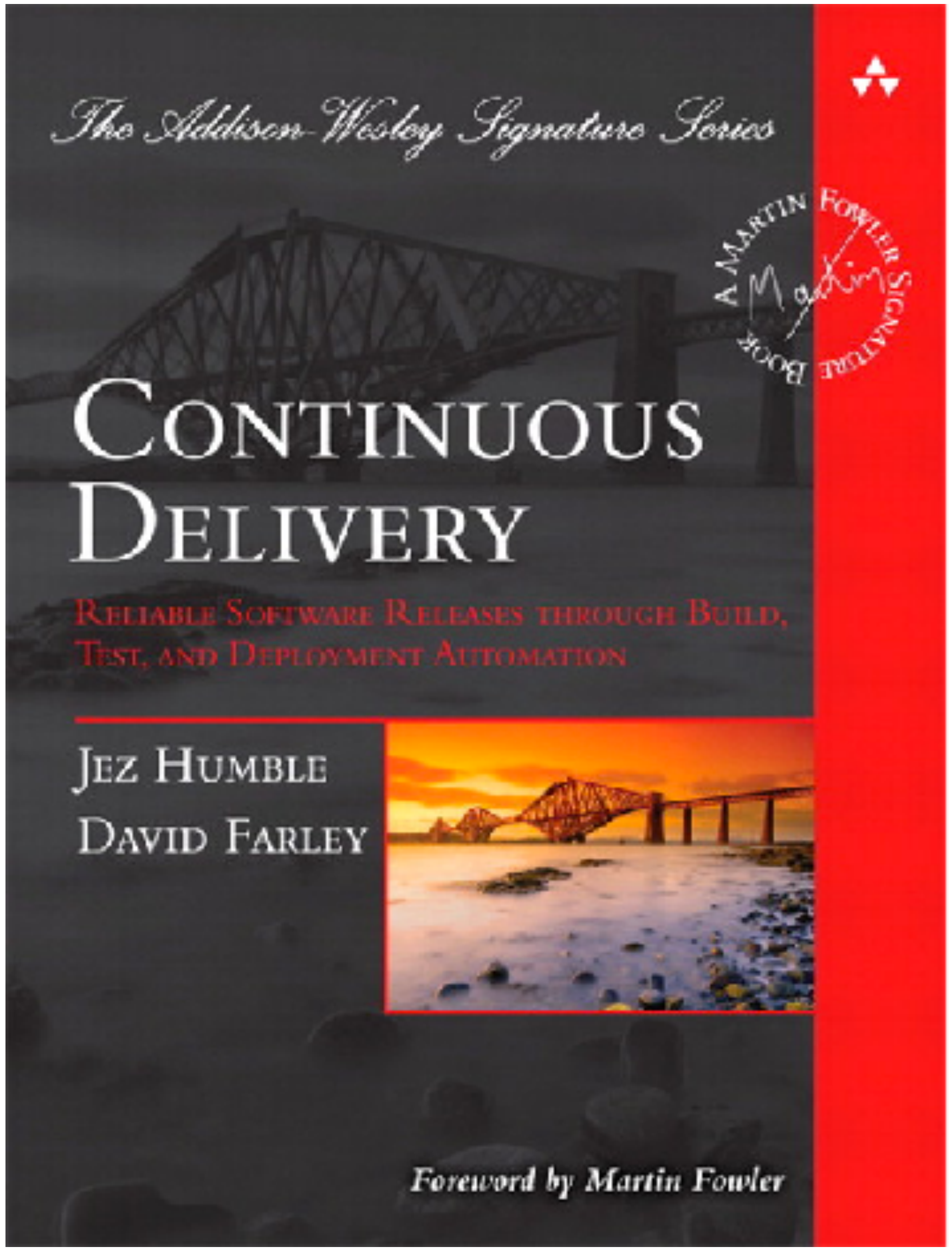
Continuous Integration

Strive for **fast feedback**

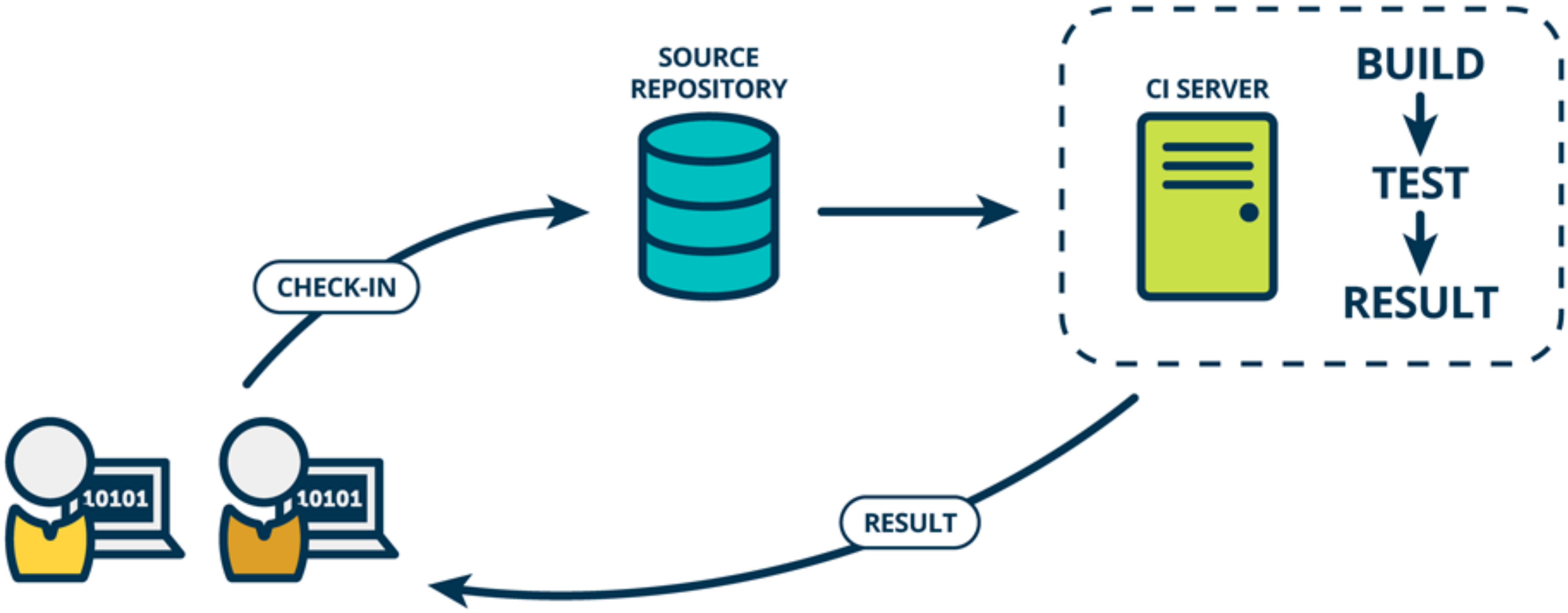
Practices
of
Continuous Integration

Improve quality and reduce risk

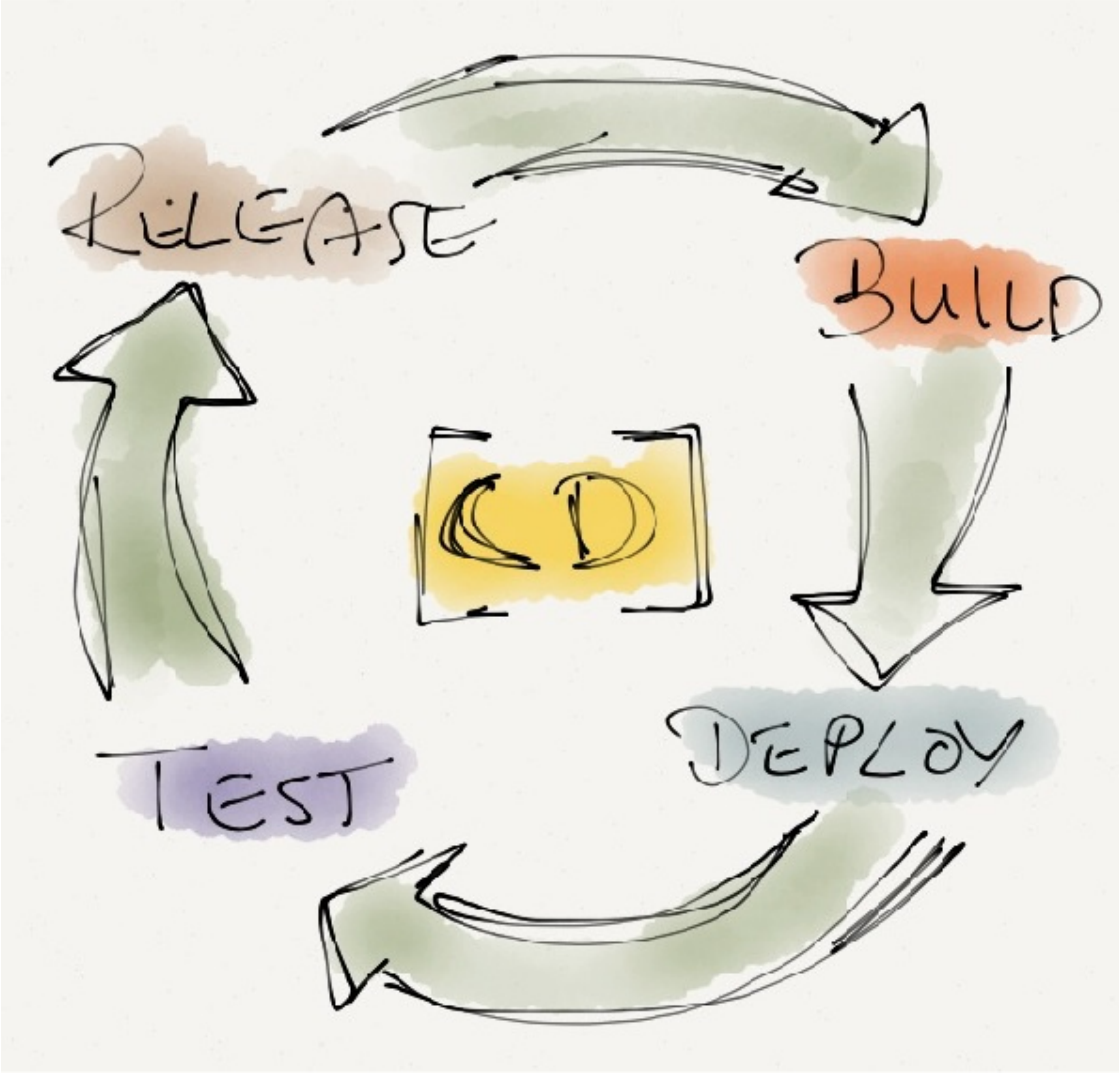




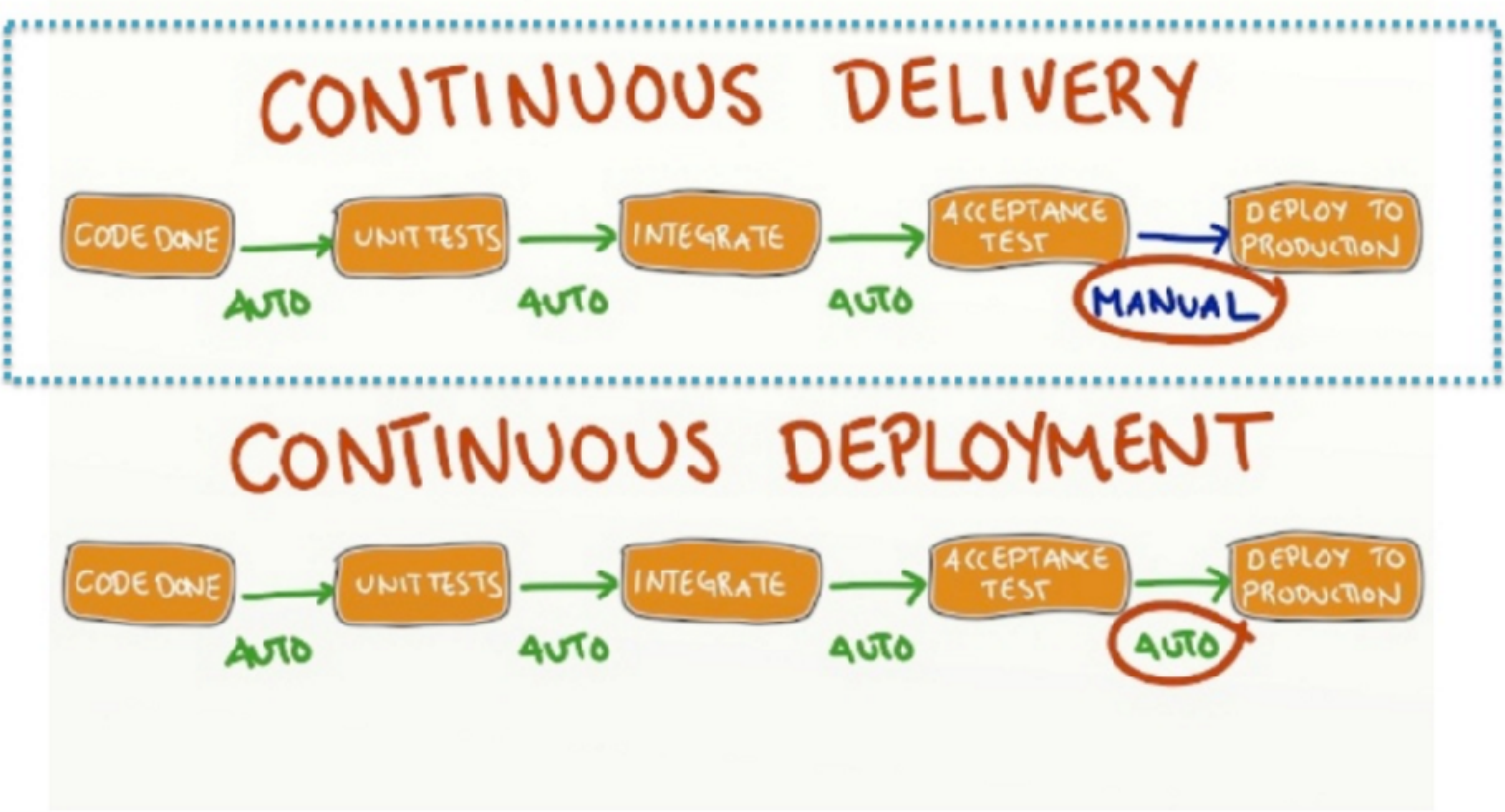
Continuous Integration



CD ?

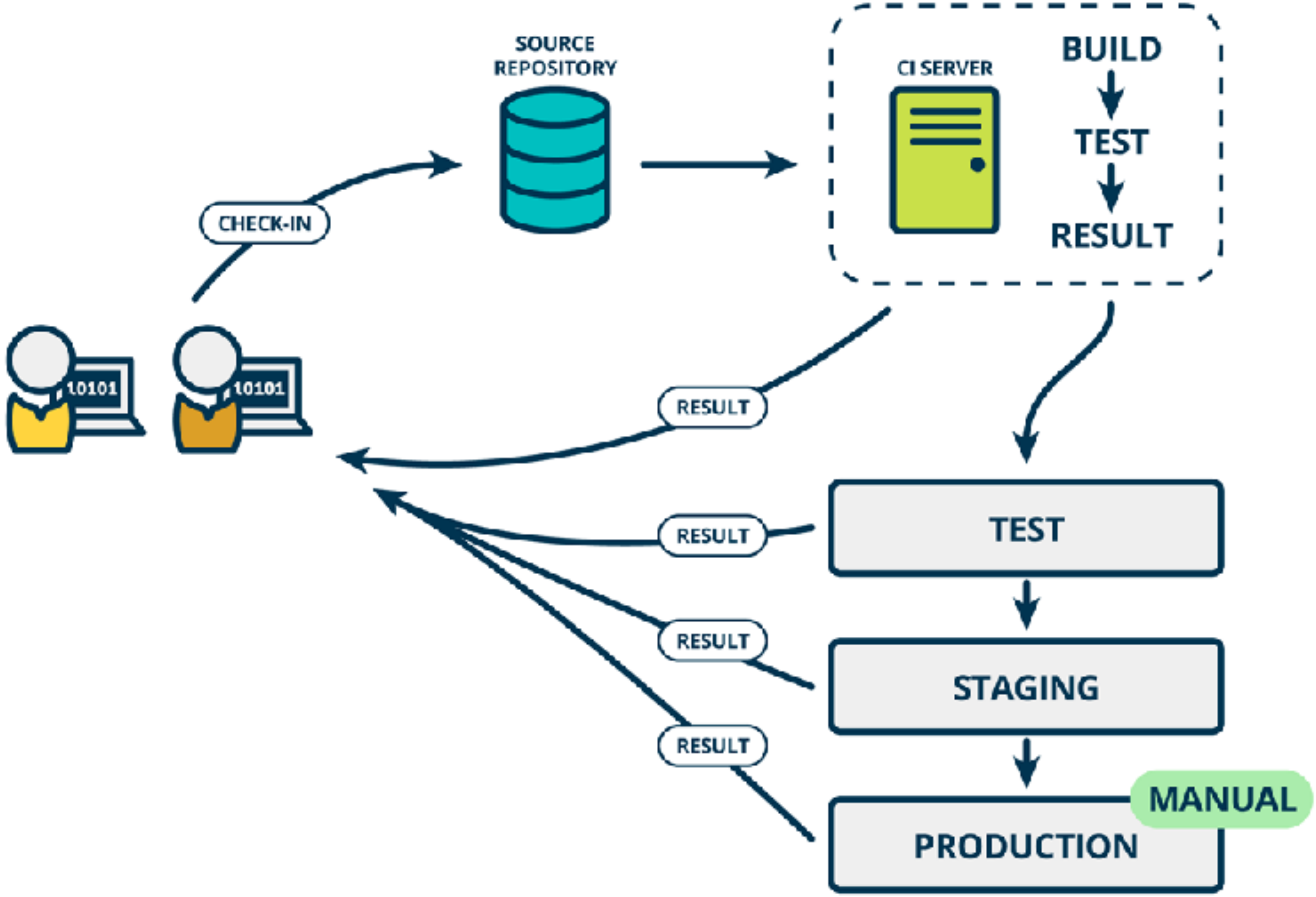


CD ?

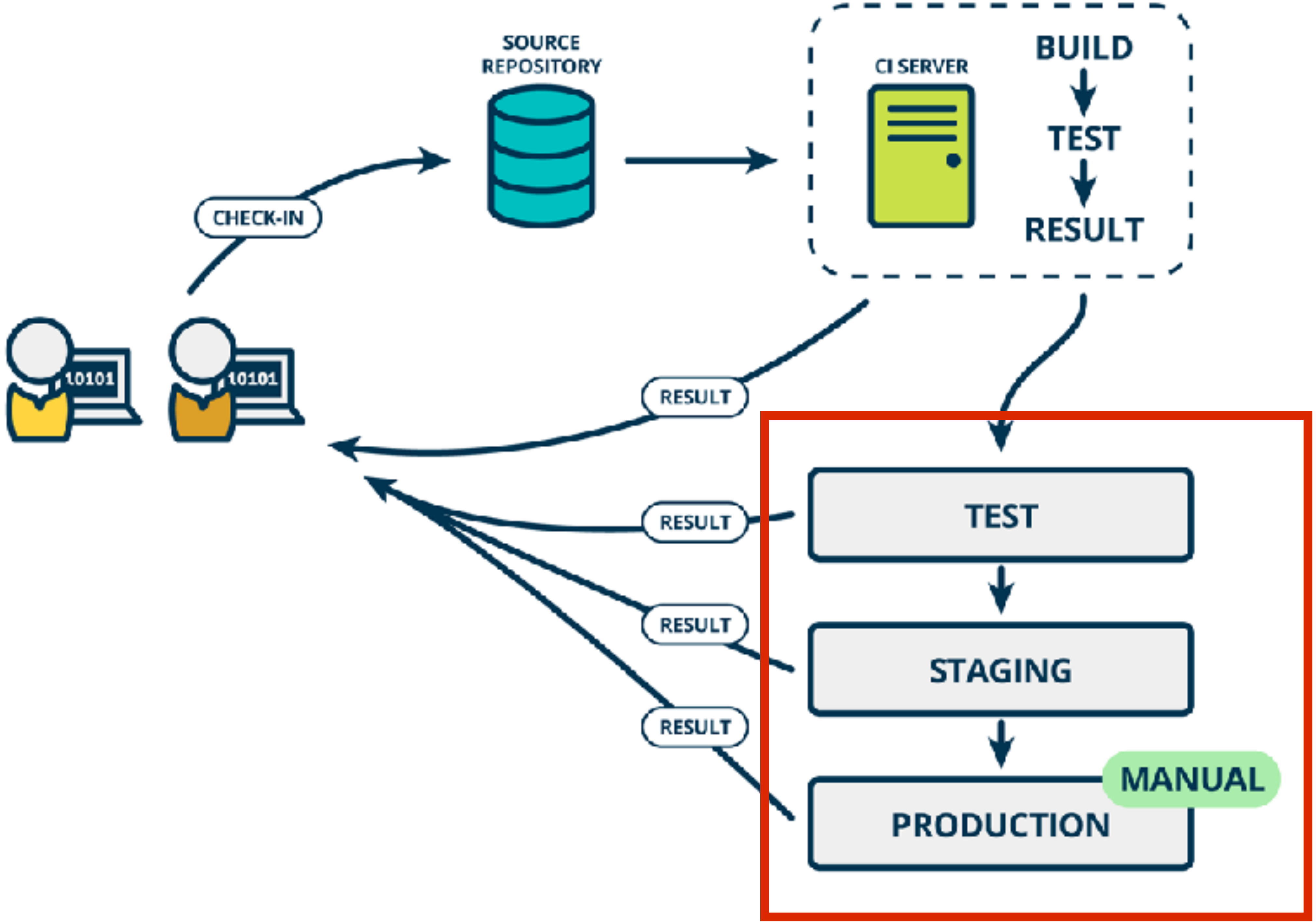


http://blog.crisp.se/2013/02/05/yassalsundman/continuous-delivery-vs-continuous-deployment

Continuous Delivery



Rise of DevOps



Continuous Integration

is a Software development **practices**

Practice 1

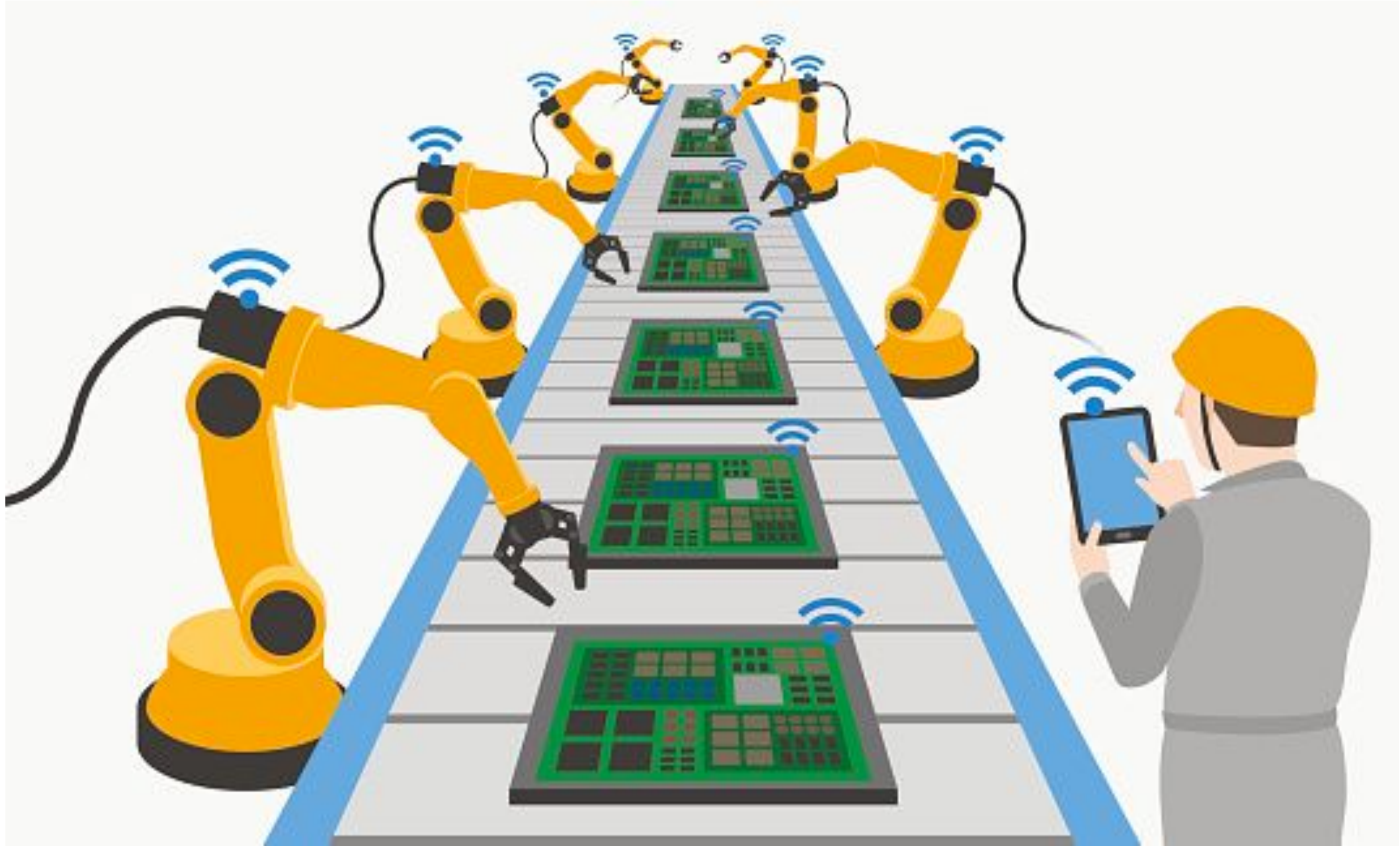
Maintain a single source repository

In general, you should store in source control everything you need to build anything

Practice 2

Automated the build

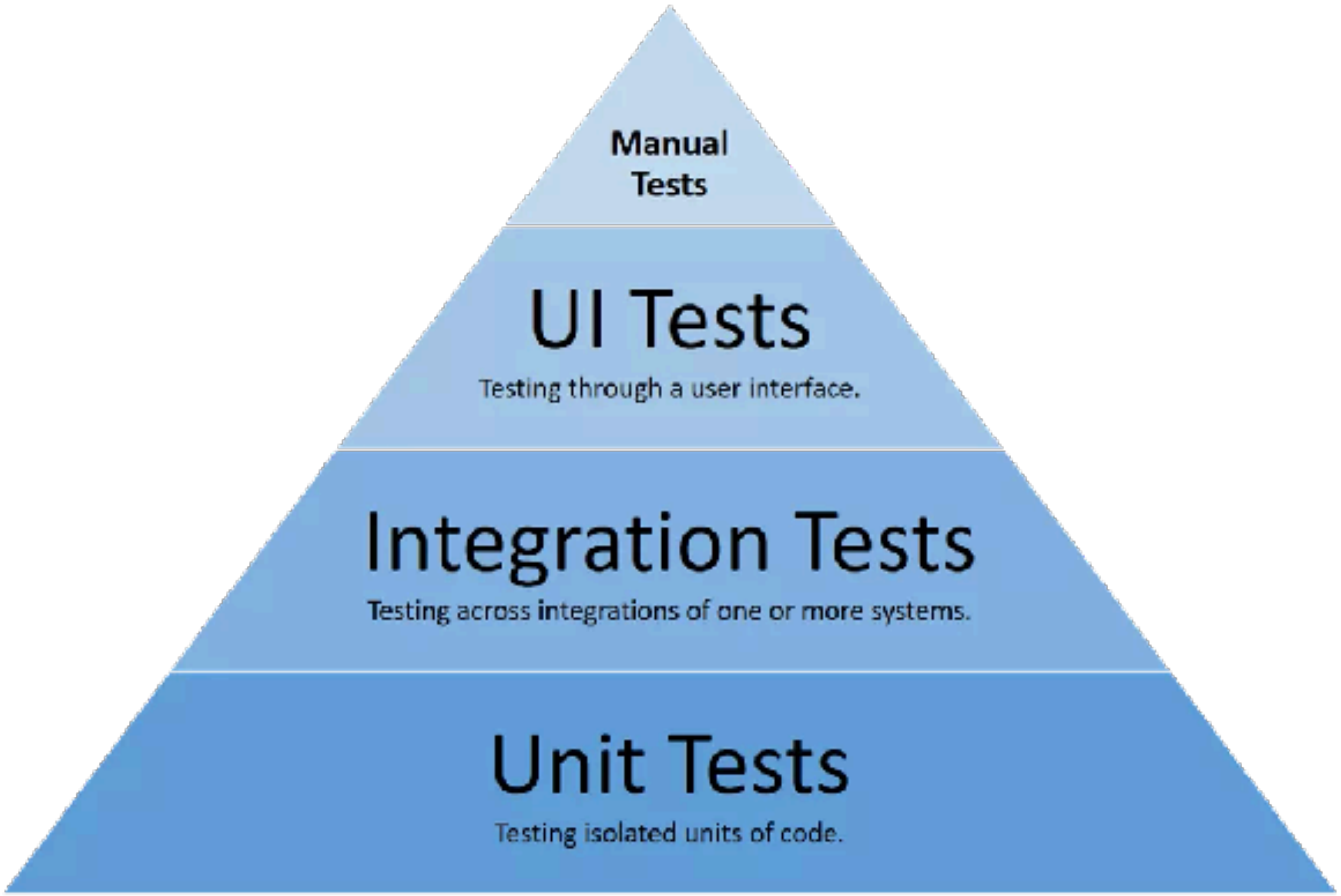
Automated environment for builds

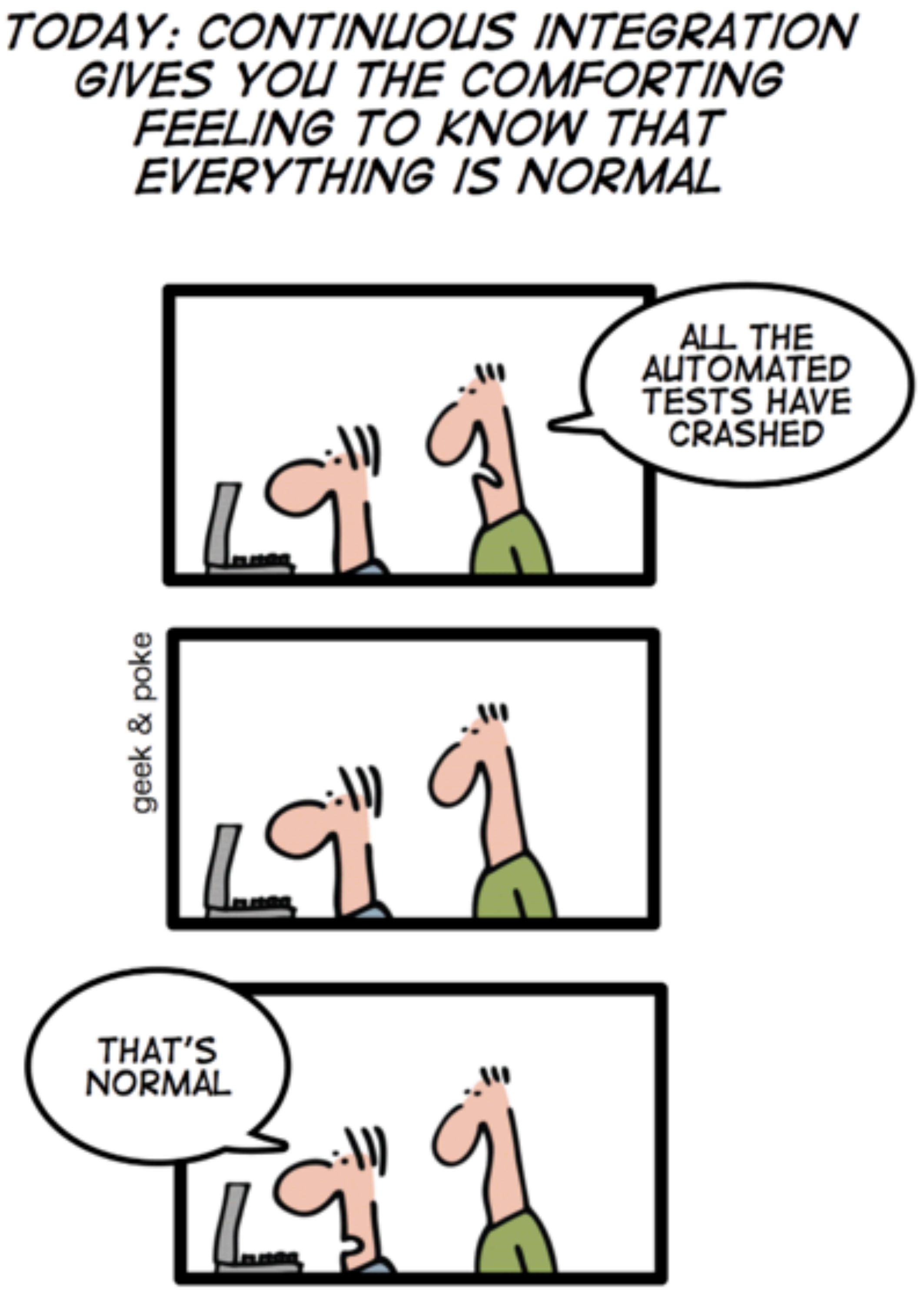


Practice 3

Make your build **self-testing**

Build process => compile, linking and **testing**





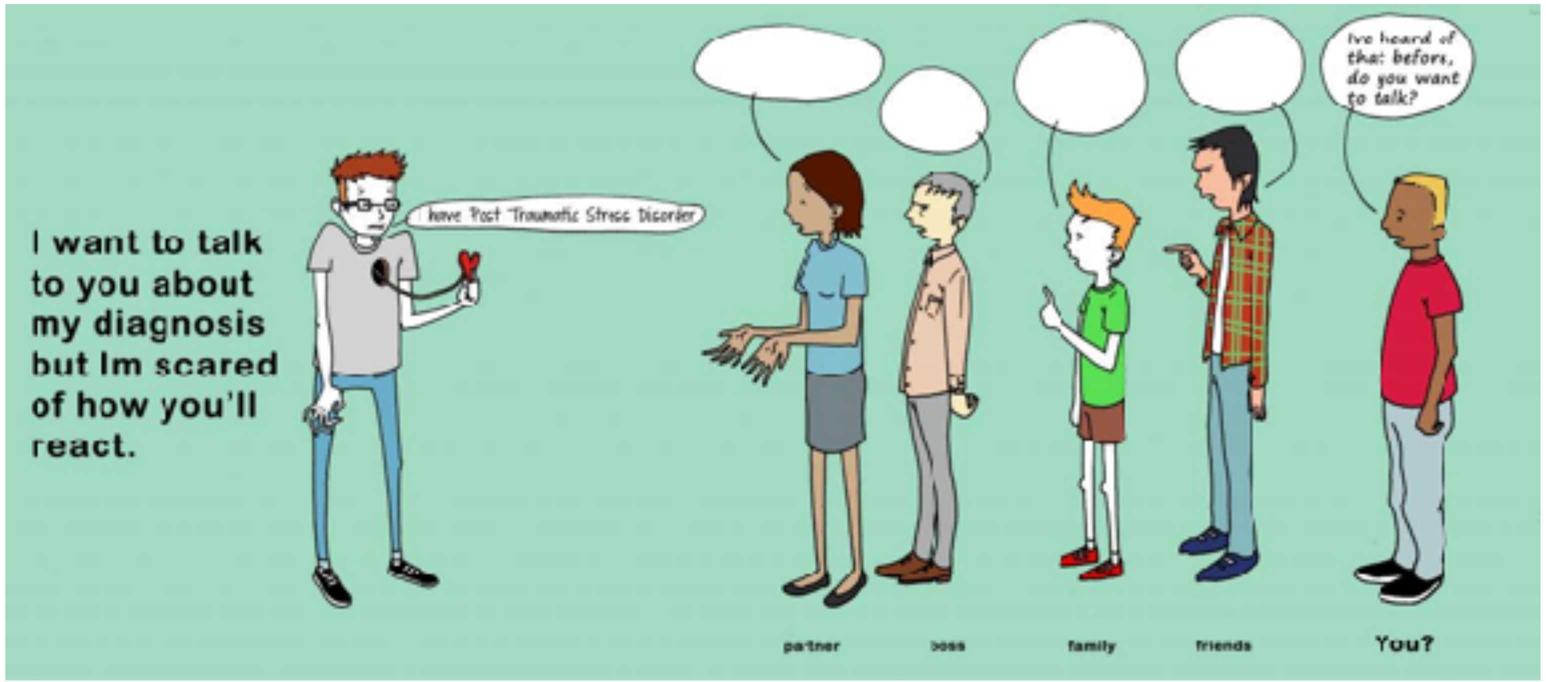
http://geekandpoke.typepad.com/

Practice 4

Everyone **commits** to the mainline everyday

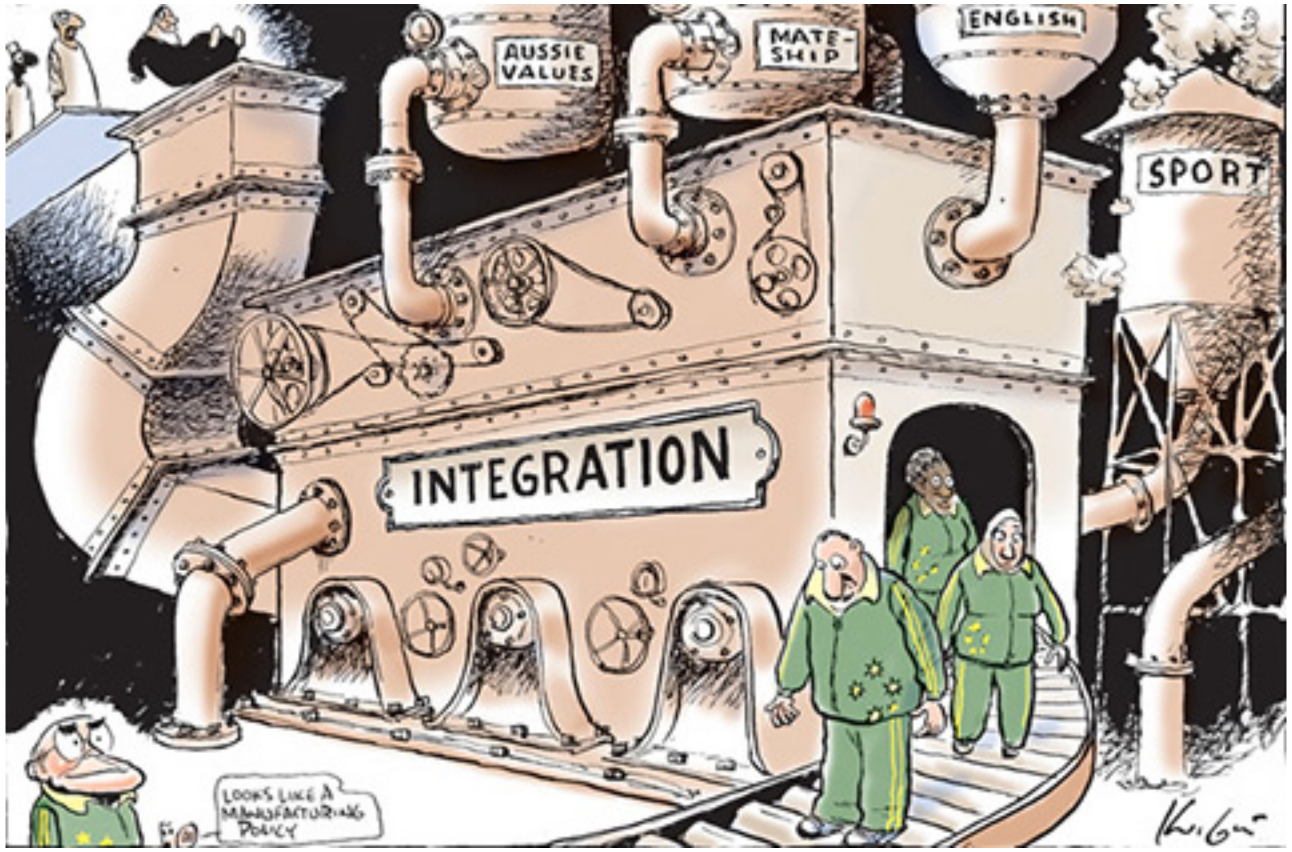
Integration is about **communication**

Integration allows developers to **tell** other developers



Practice 5

Every commits should build the mainline on an **Integration machine**



Nightly build is not enough for Continuous Integration

Practice 6

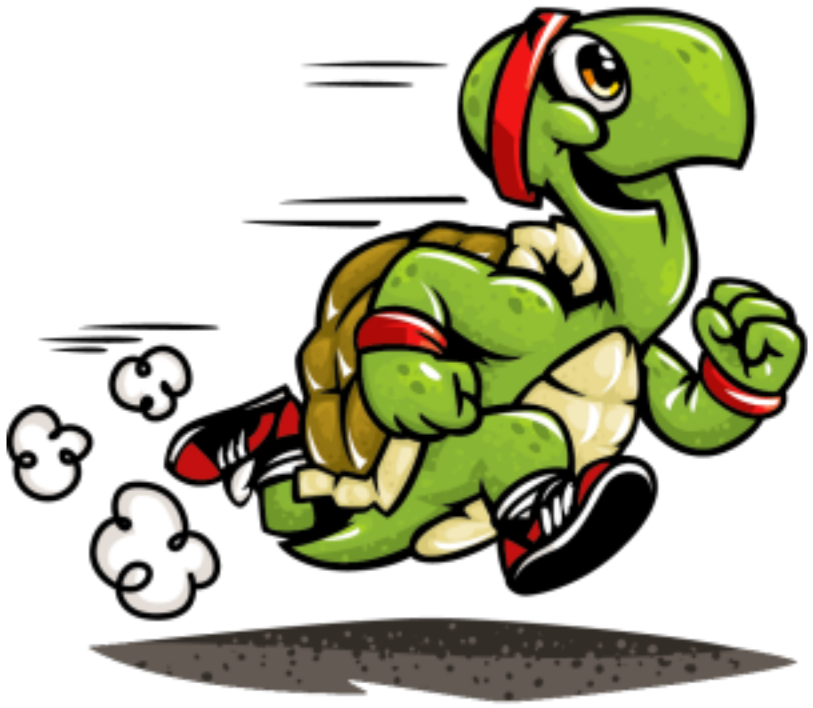
Fix broken builds **immediately**

"Nobody has a higher priority task than fixing the build"

Practice 7

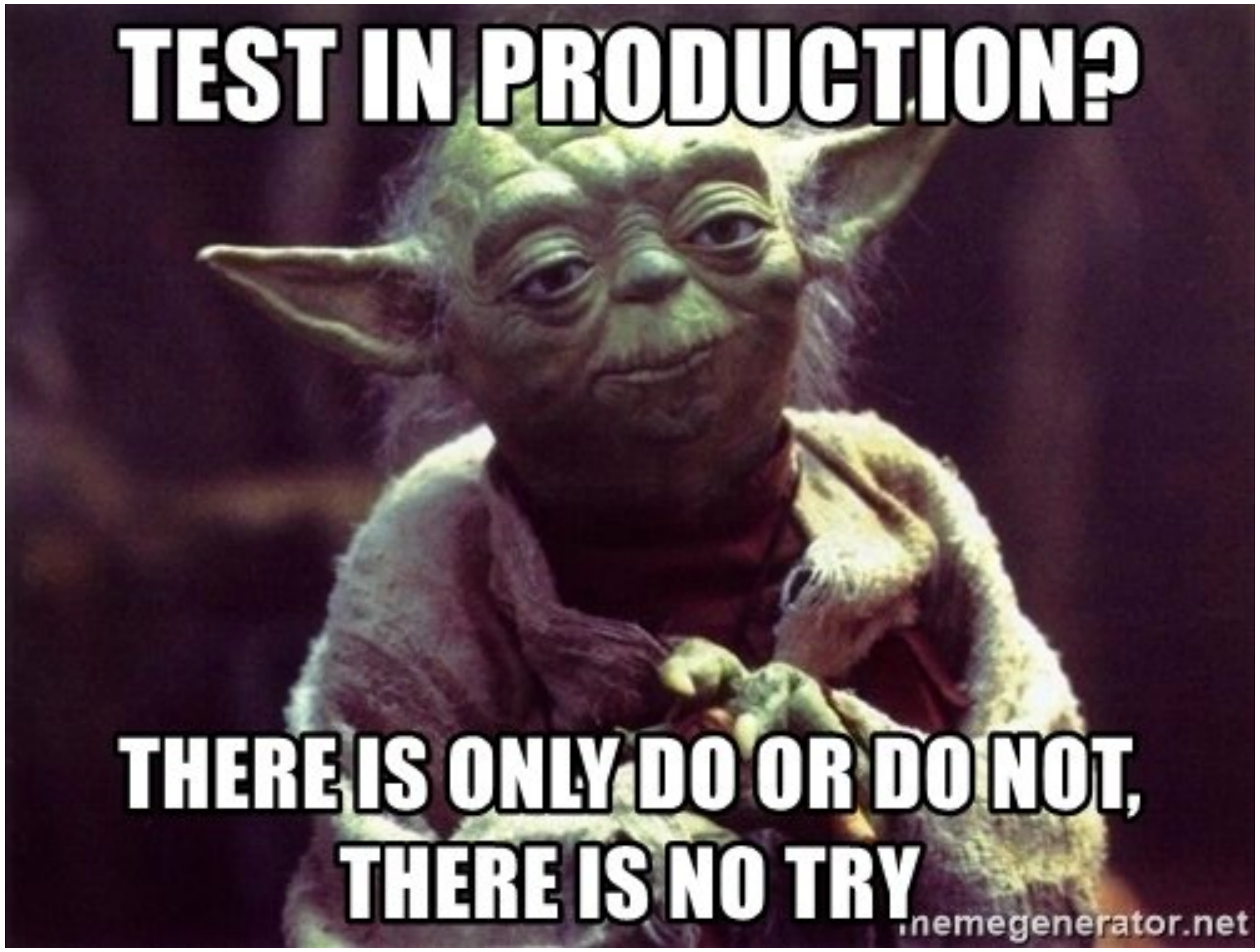
Keep the build **fast**

Continuous Integration is to provide rapid feedback



Practice 8

Test in clone of the **Production** environment



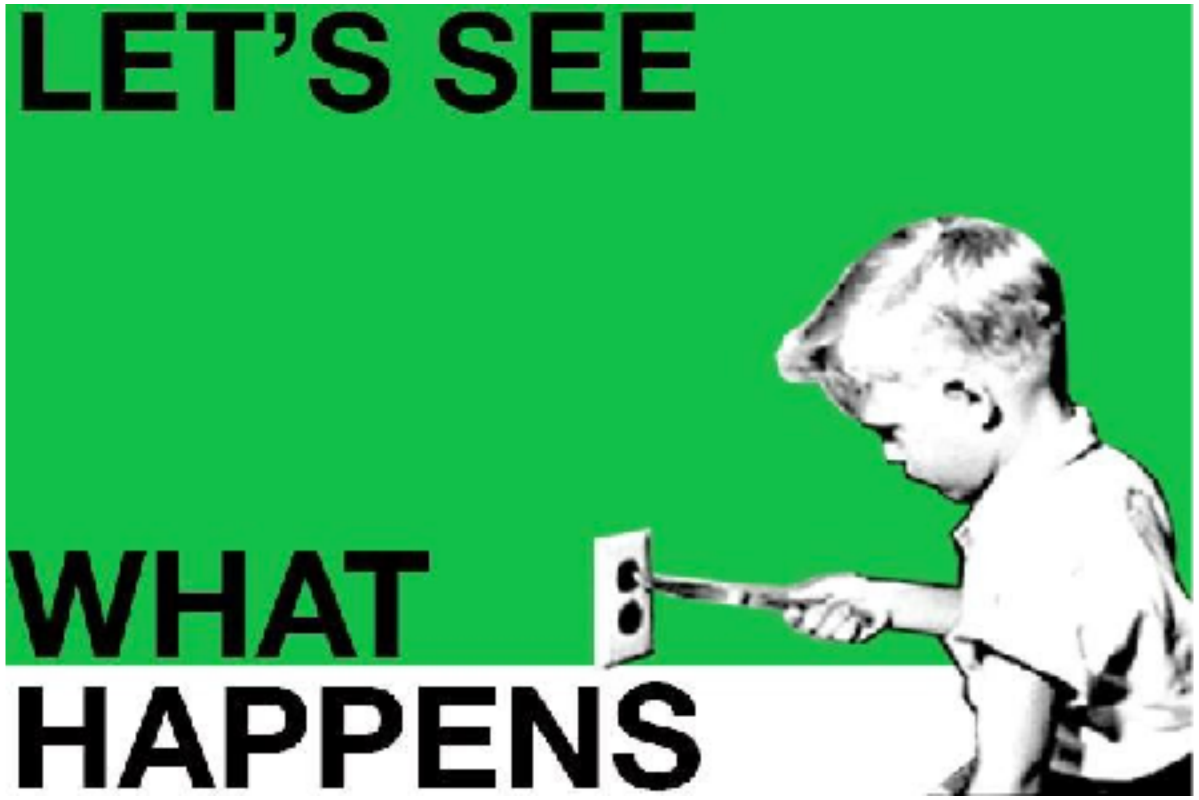
Practice 9

Make it easy for anyone to get
the latest executable

Make sure well known place where people can find

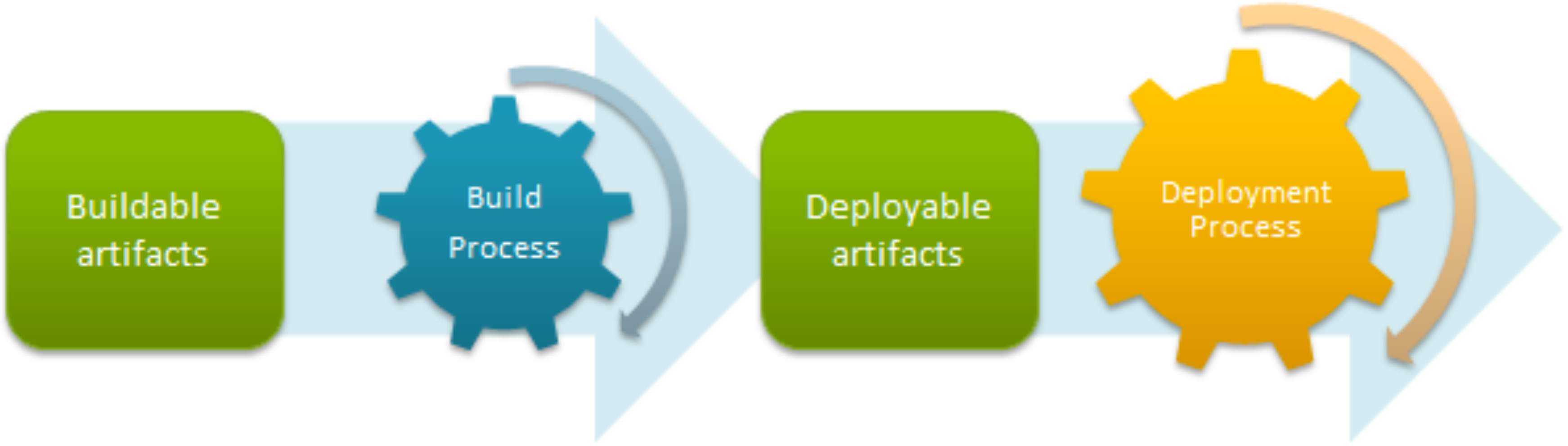
Practice 10

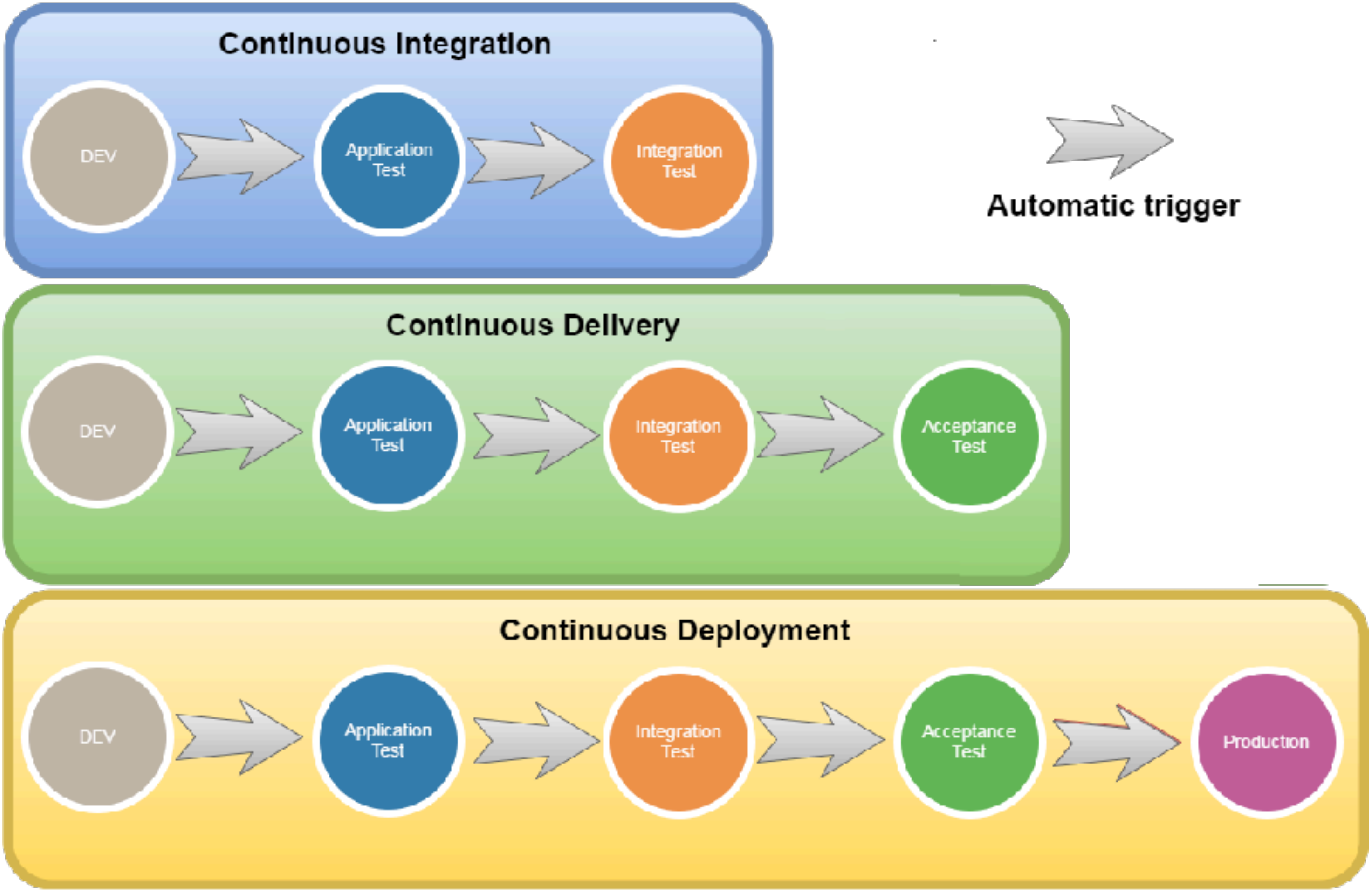
Everyone can see what's happening
Easier to see the state of the system and changes
Show the good information



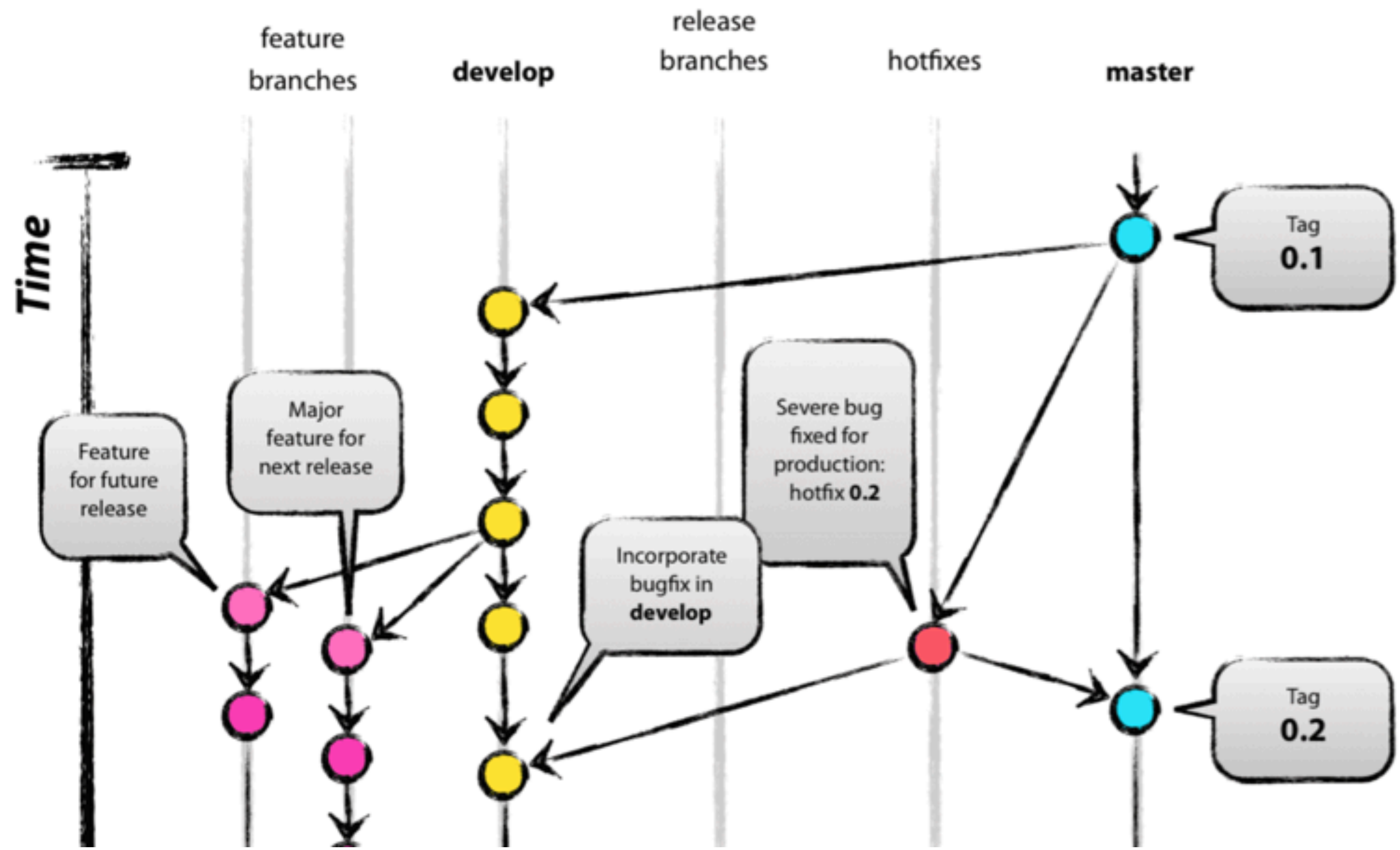
Practice 11

Automated deployment




Continuous Integration
DEV
Application Test
Integration Test
Automatic trigger
Continuous Delivery
DEV
Application Test
Integration Test
Acceptance Test
Continuous Deployment
DEV
Application Test
Integration Test
Acceptance Test
Production

Git Branch Model



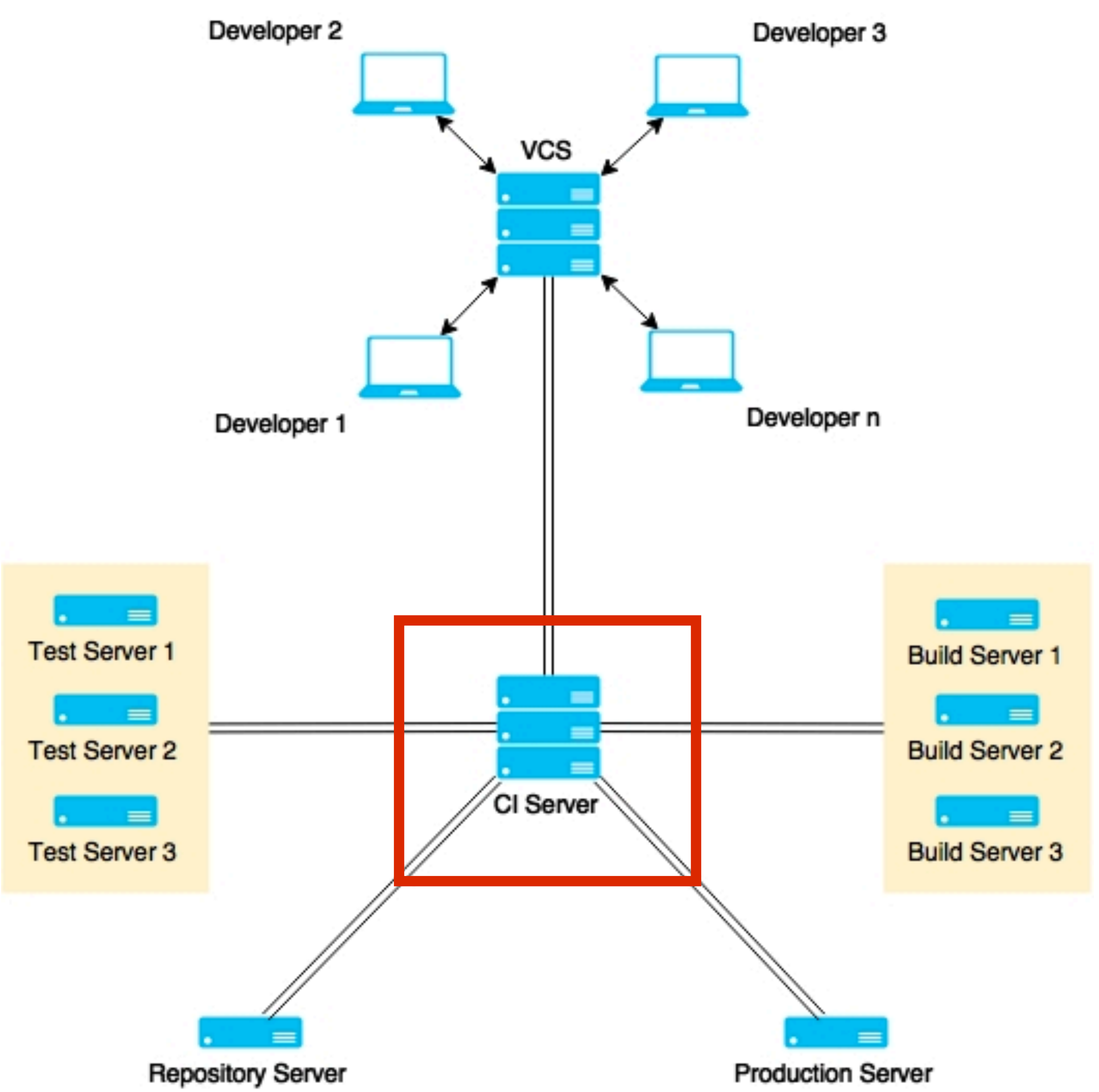
https://nvie.com/posts/a-successful-git-branching-model/

Validate, Validate and Validate

Suggestion

Keeping branches short-lived, merge conflicts are keep to as few as possible

3. Use good CI tool



CI/CD Tools

Jenkins

Bamboo

TeamCity

go™

Visual Studio
Team Foundation Server

Hudson

Travis

wercker

circleci

More …

Use static code analysis

Automated testing

Automated deployment

People discipline/habit

“Behind every successful agile project, there is a **Continuous Integration Server**”

Let’s start with CI/CD

Application and framework to manage and monitor the executable of **repeated tasks**

Jenkins

https://jenkins.io/

Centralize Continuous Integration Server

Jenkins

https://jenkins.io/

Why Jenkins ?

Easy !!

Extensible

Scalable

Opensource

Large community

Lot of plugins

Who use Jenkins ?

We thank the following organizations for their major commitments to support the Jenkins project.

Microsoft cloudbees OSL OPEN SOURCE LAB

rackspace redhat

We thank the following organizations for their support of the Jenkins project through free and/or open source licensing programs.

Atlassian Datadog JFrog Mac Cloud PagerDuty XMission

https://jenkins.io/

Commit message

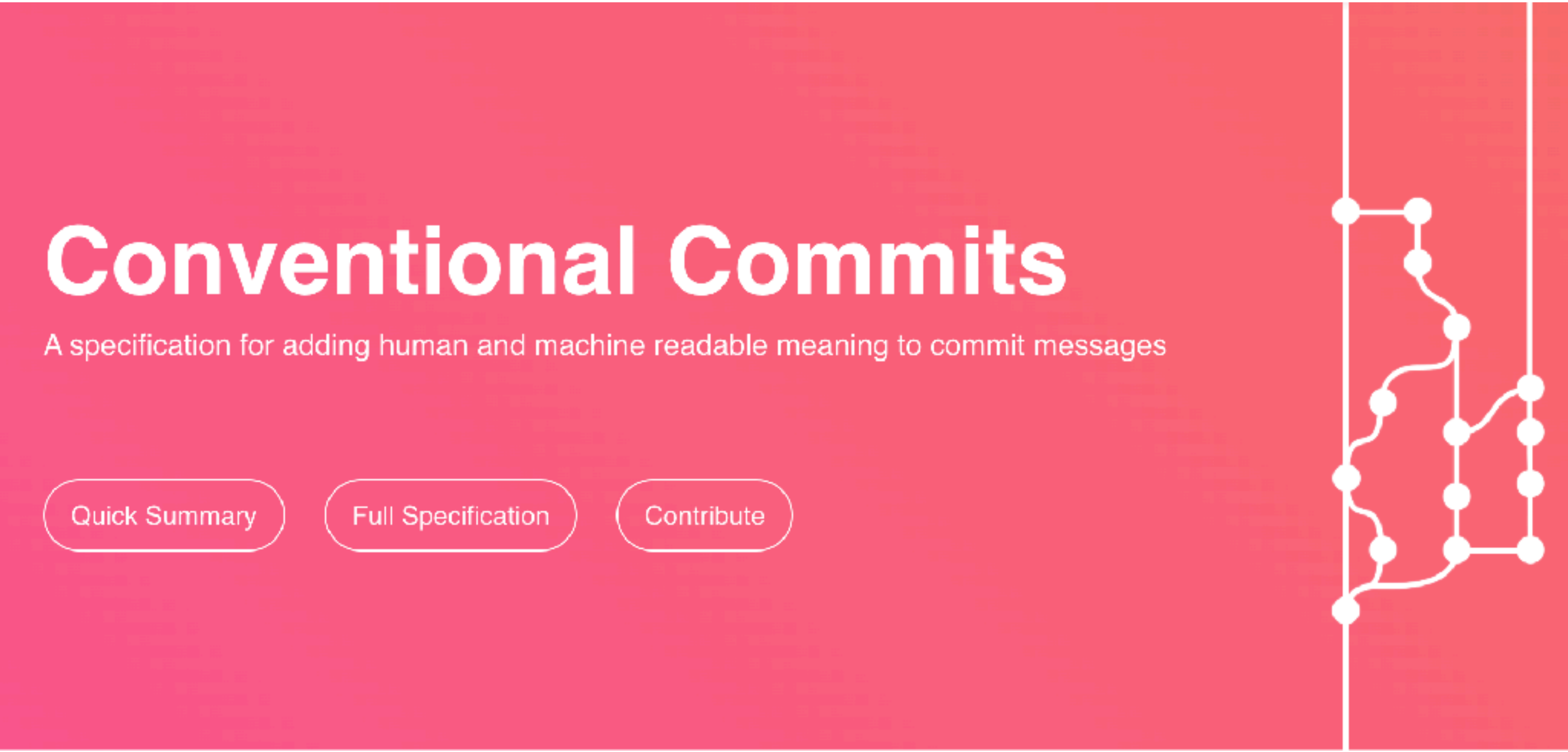
Update

TODO

fixed bug

add feature

Commit Message Format ?



https://www.conventionalcommits.org/

“Added a user object to the database.
currently only has a name and email.
no authentication yet ”

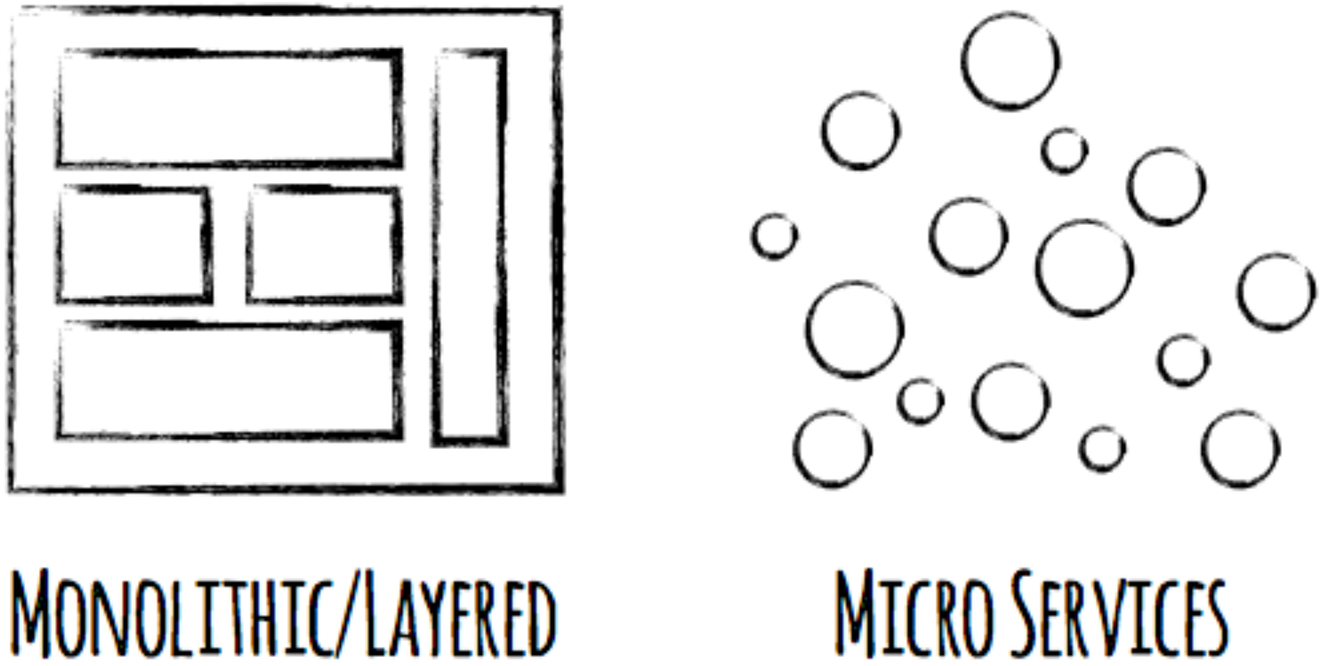
“Fixed the bug that would add something to everything. Turn to not add something to everything”

Manage containers

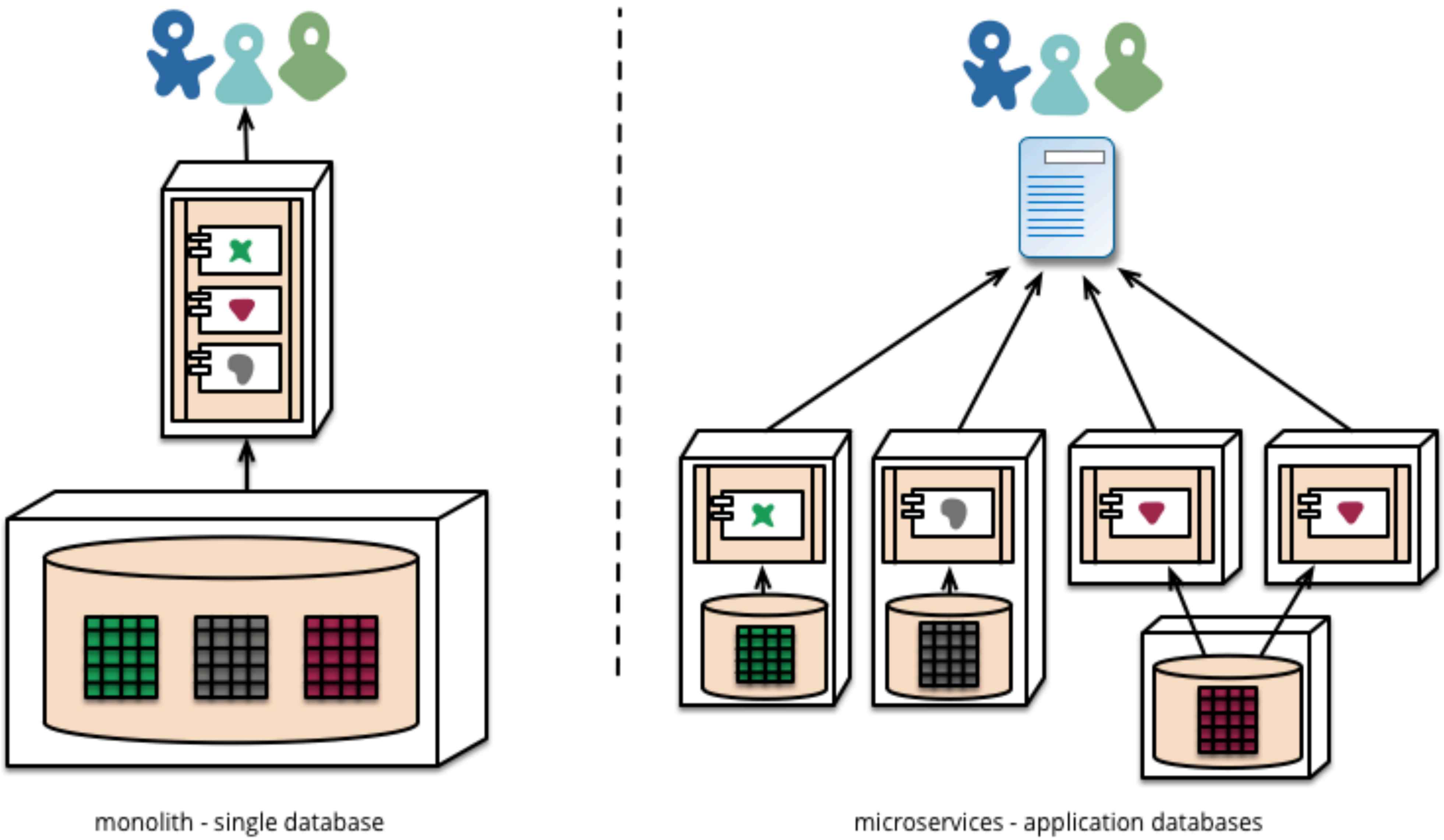
docker

Why docker ?

Software industry has changed !!



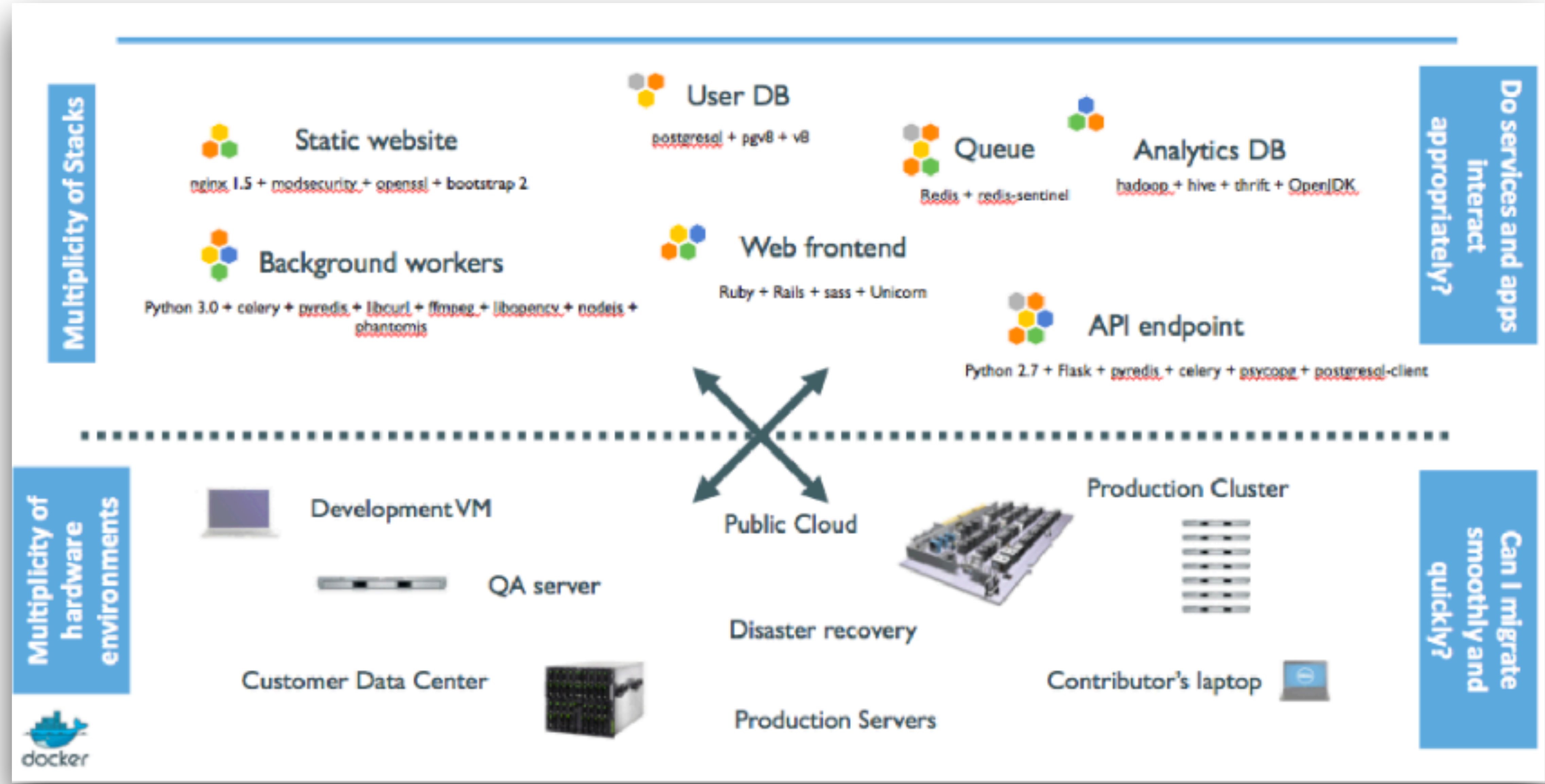
Microservice architecture



https://martinfowler.com/articles/microservices.html

Why docker ?

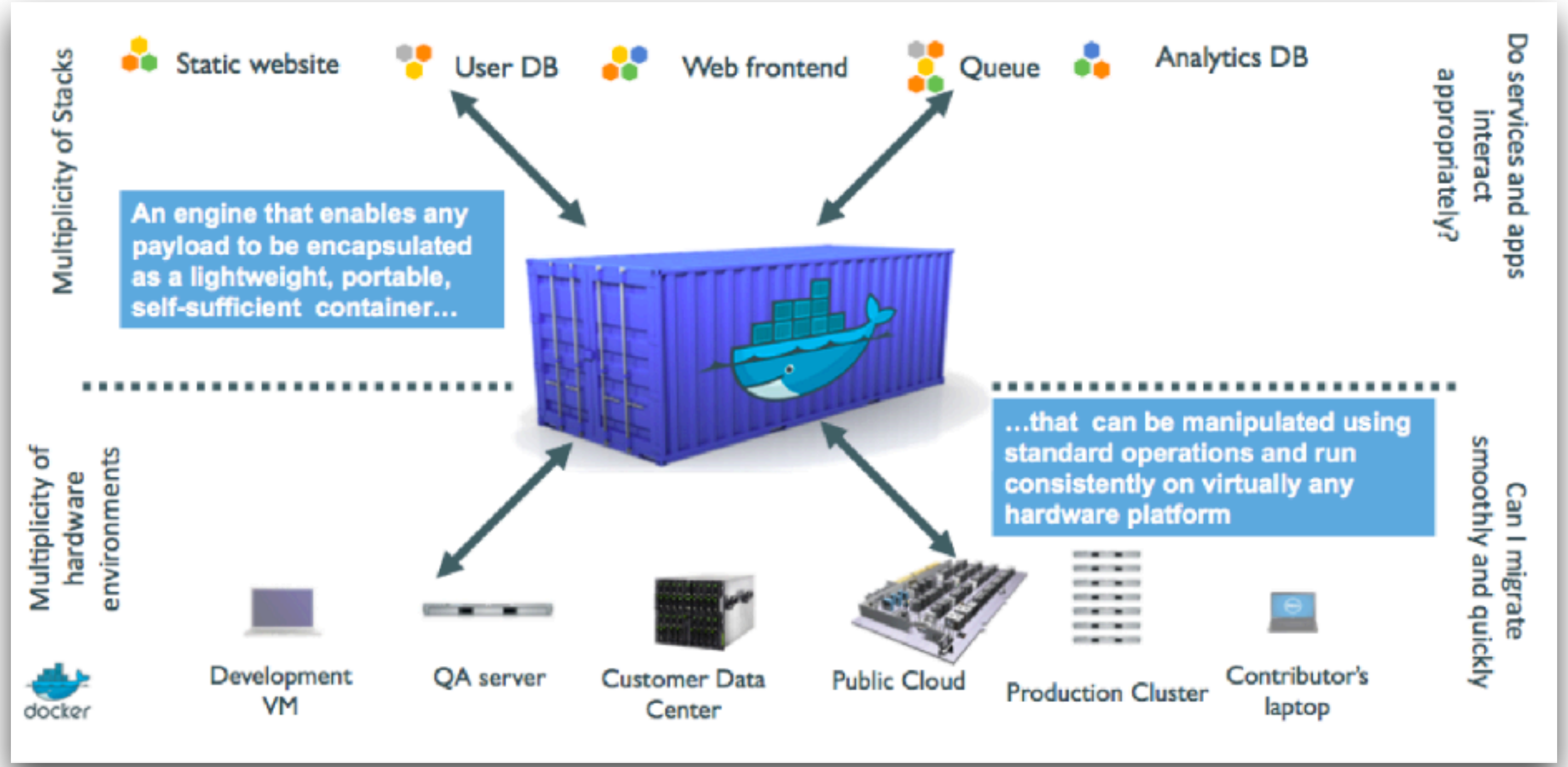
We have problem !!



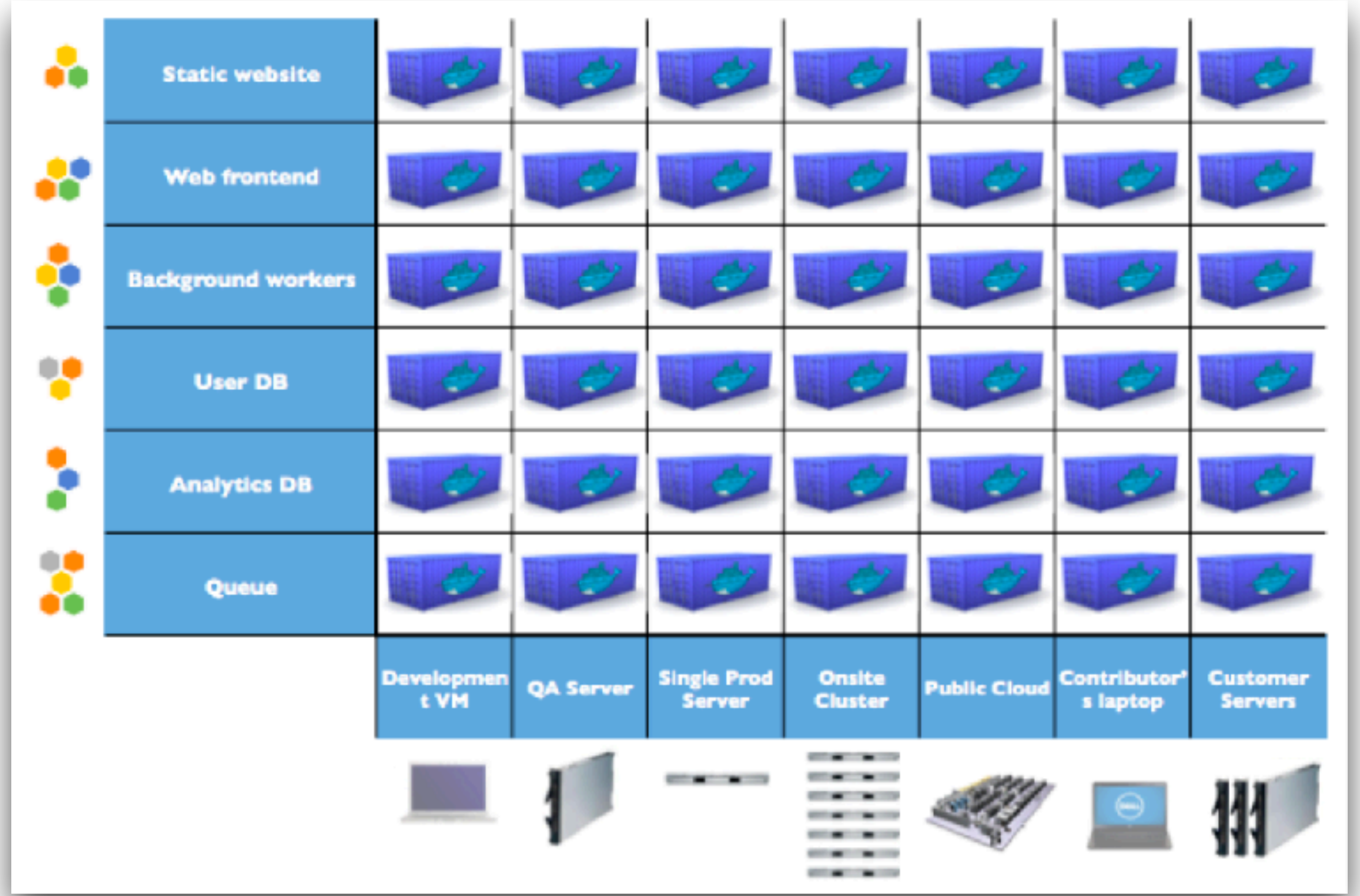
Problem ?

	Development VM	QA Server	Single Prod Server	Onsite Cluster	Public Cloud	Contributor's laptop	Customer Servers
Static website	?	?	?	?	?	?	?
Web frontend	?	?	?	?	?	?	?
Background workers	?	?	?	?	?	?	?
User DB	?	?	?	?	?	?	?
Analytics DB	?	?	?	?	?	?	?
Queue	?	?	?	?	?	?	?

Solution ?

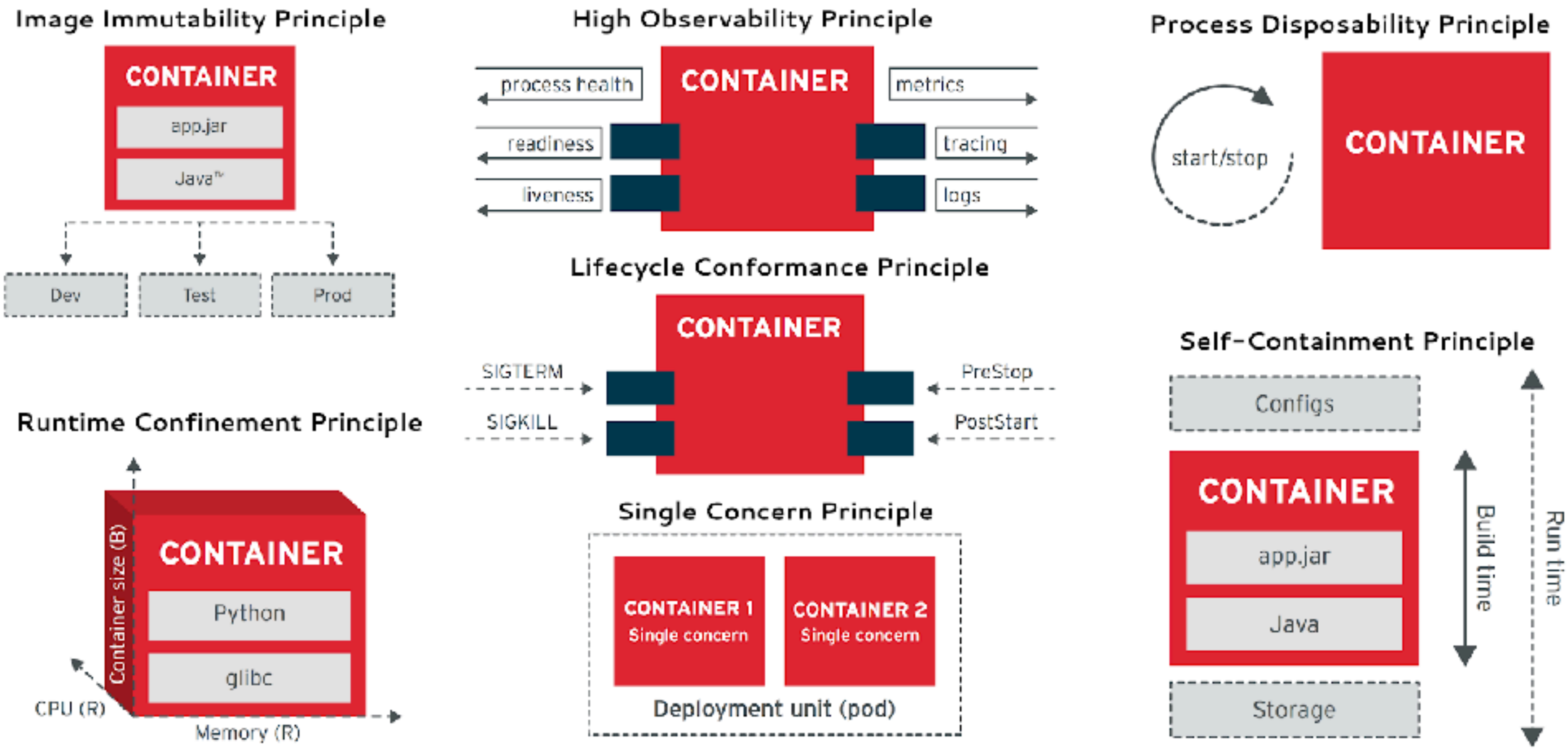

Multiplicity of Stacks
Static website
User DB
Web frontend
Queue
Analytics DB
Do services and apps interact appropriately?
An engine that enables any payload to be encapsulated as a lightweight, portable, self-sufficient container...
...that can be manipulated using standard operations and run consistently on virtually any hardware platform
Multiplicity of hardware environments
Can I migrate smoothly and quickly
docker
Development VM
QA server
Customer Data Center
Public Cloud
Production Cluster
Contributor's laptop

Solution ?

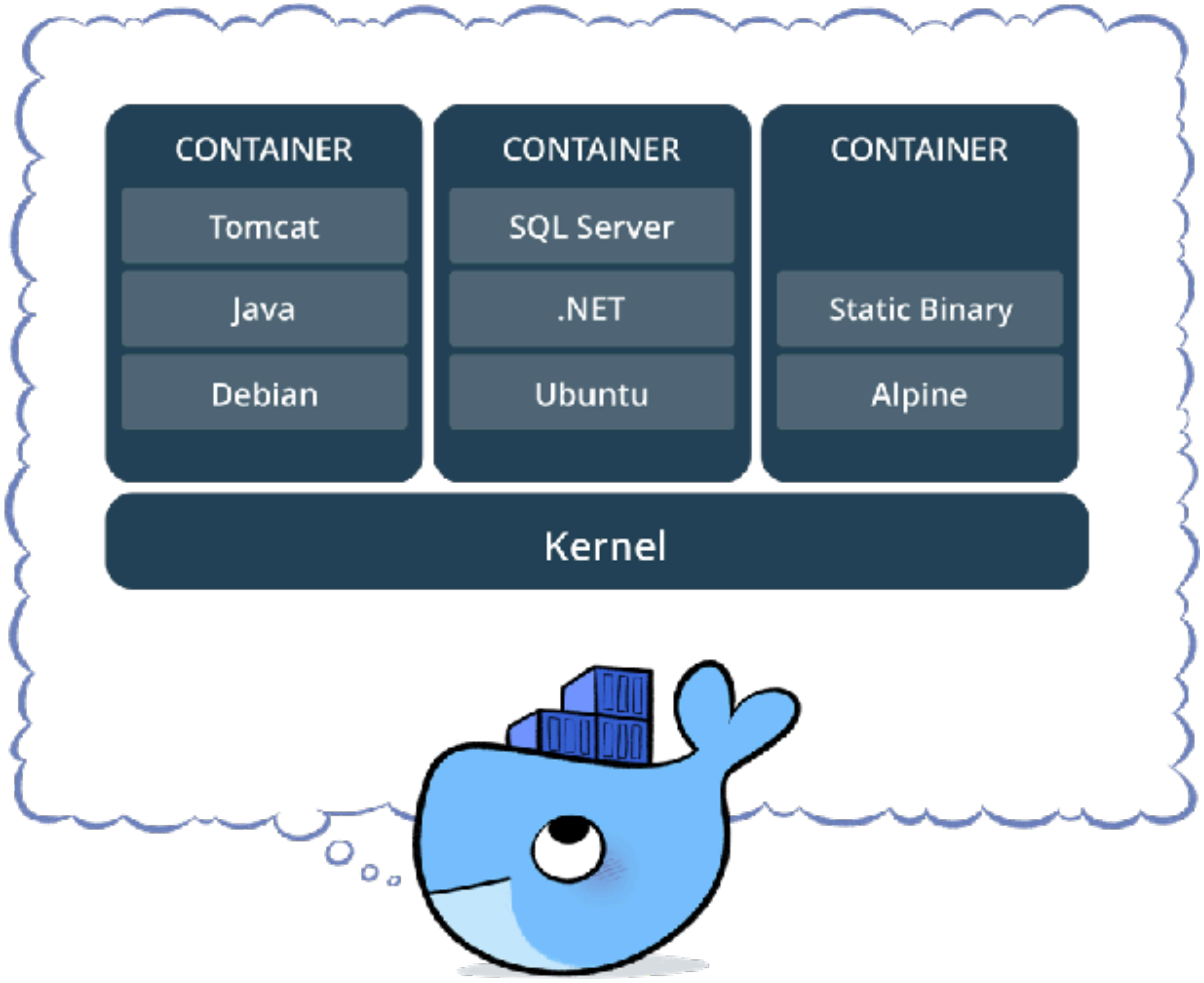


Static website
Web frontend
Background workers
User DB
Analytics DB
Queue
Development VM
QA Server
Single Prod Server
Onsite Cluster
Public Cloud
Contributor's laptop
Customer Servers

Container Design Principles



Container with Docker



Basic of Docker

Image
Container
Dockerfile
Registry
Volume and network

Image vs Container

Image like template/blueprint
Containers are created from image

Docker image

Collection of files and some meta data

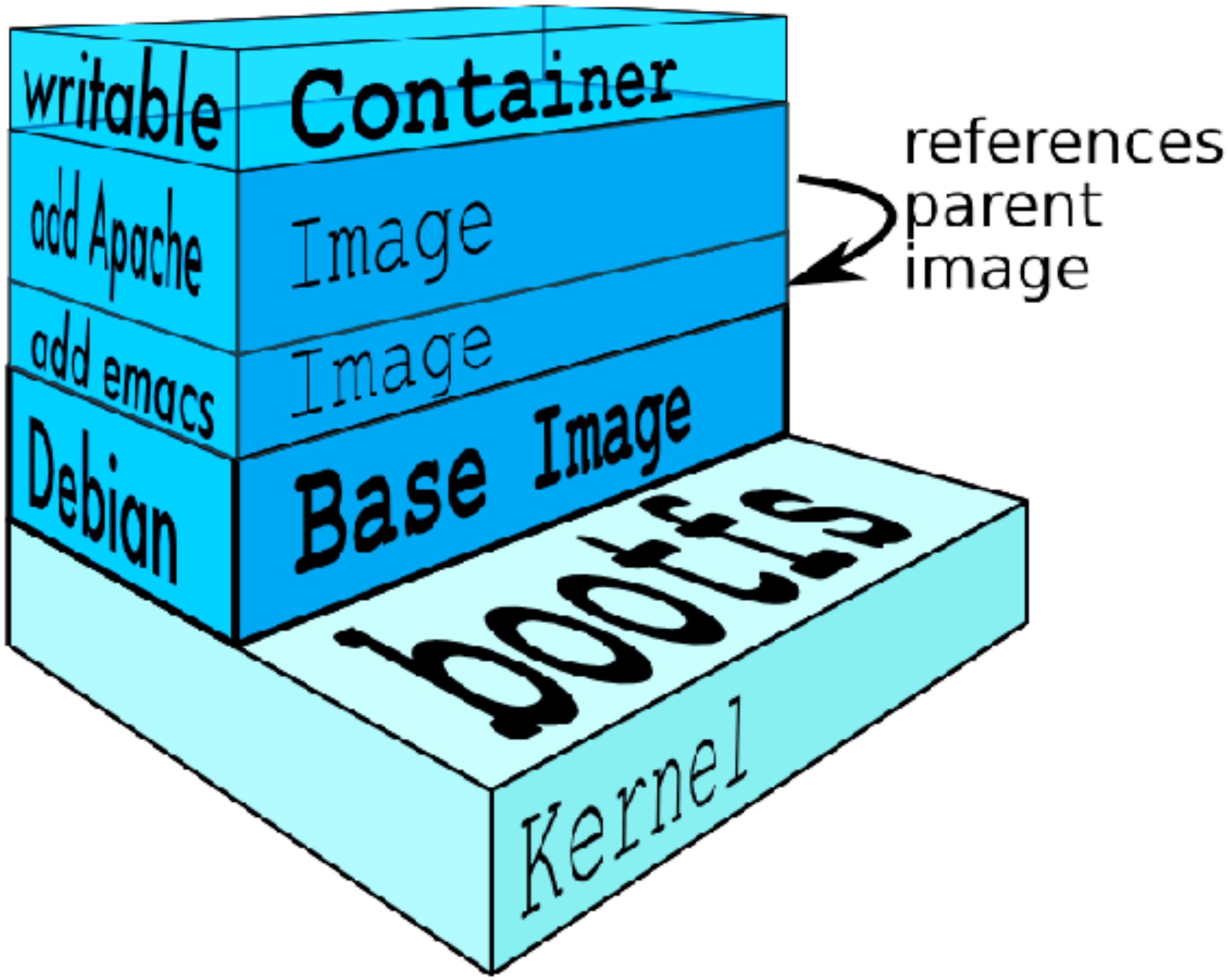
Made of **layers**

Each layer can add/change/remove files

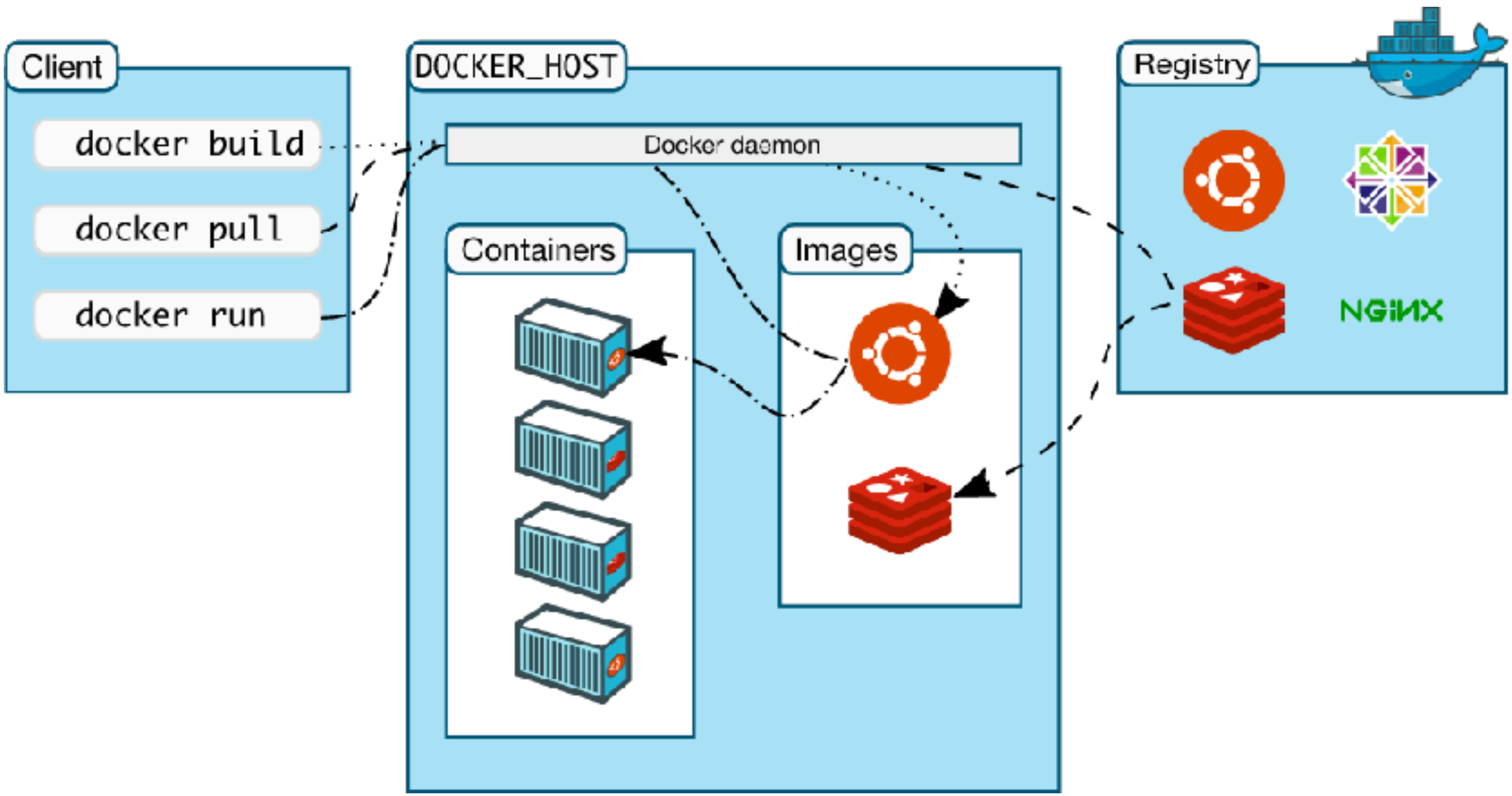
Image can share layers

Read-only file system

Docker image layer

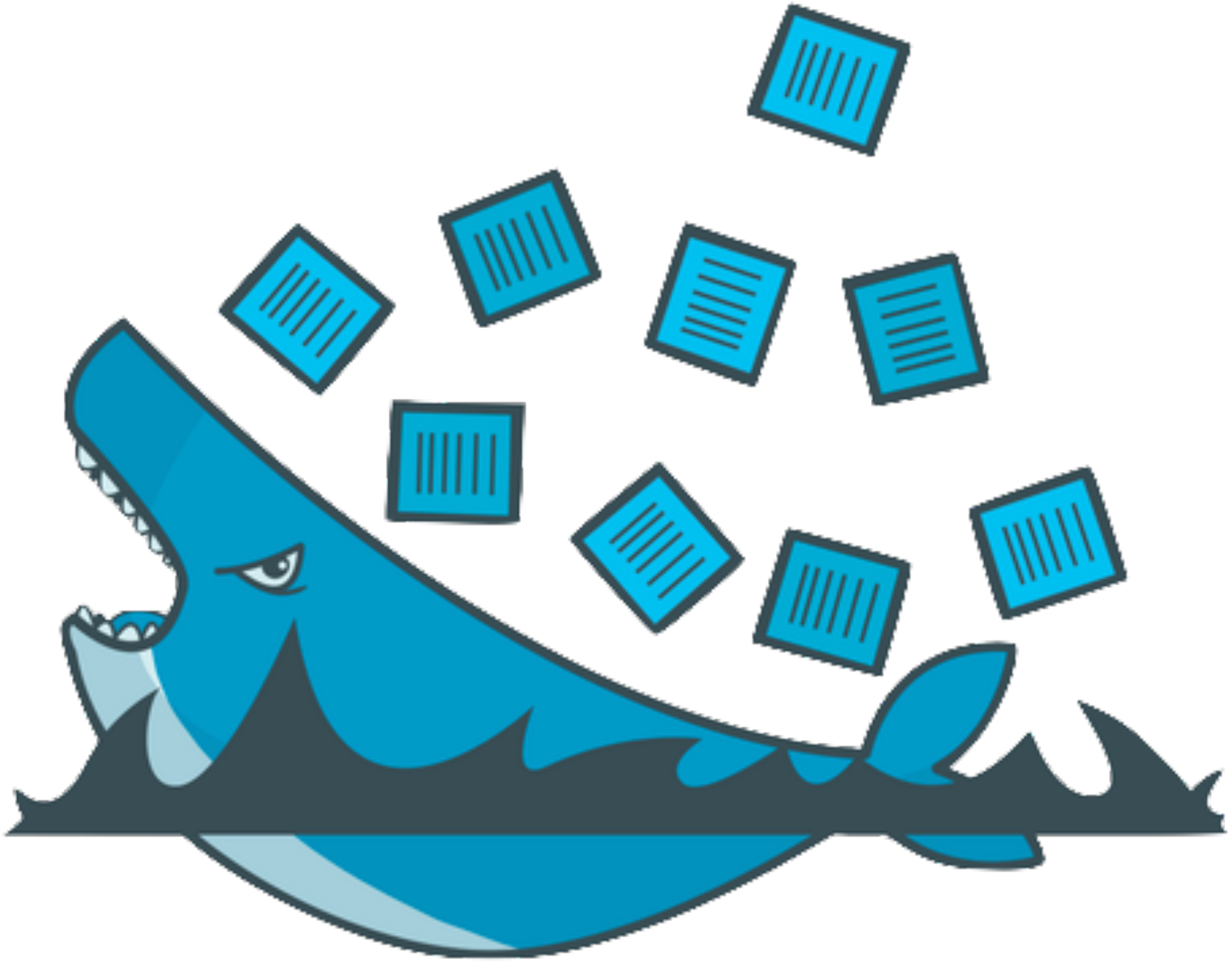


How docker works ?



https://docs.docker.com/get-started/overview/

Workshop



Working with Docker compose

https://docs.docker.com/compose/

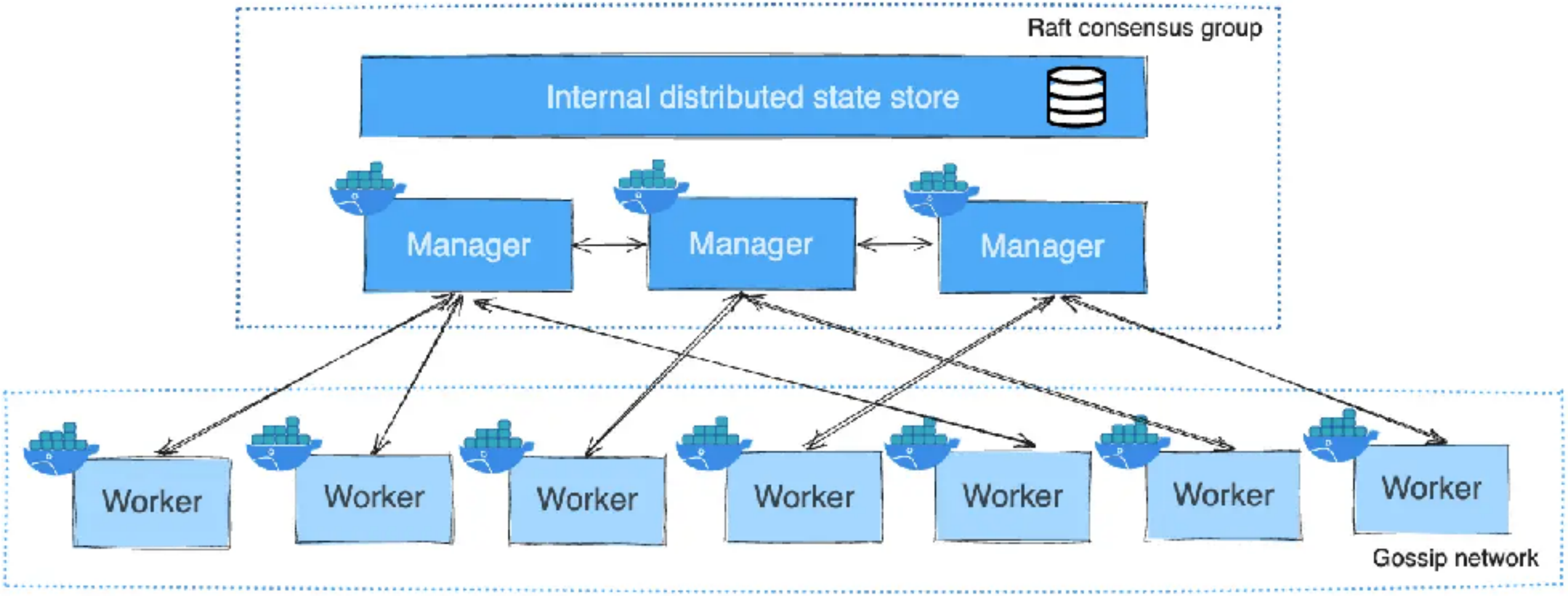
Multilple containers app!!

Difficult to create and manage !!

Mutilple containers app!!

Build images from Dockerfile
Pull images from Hub/private/cache
Configuration and create containers
Start and stop containers
Stream their logs

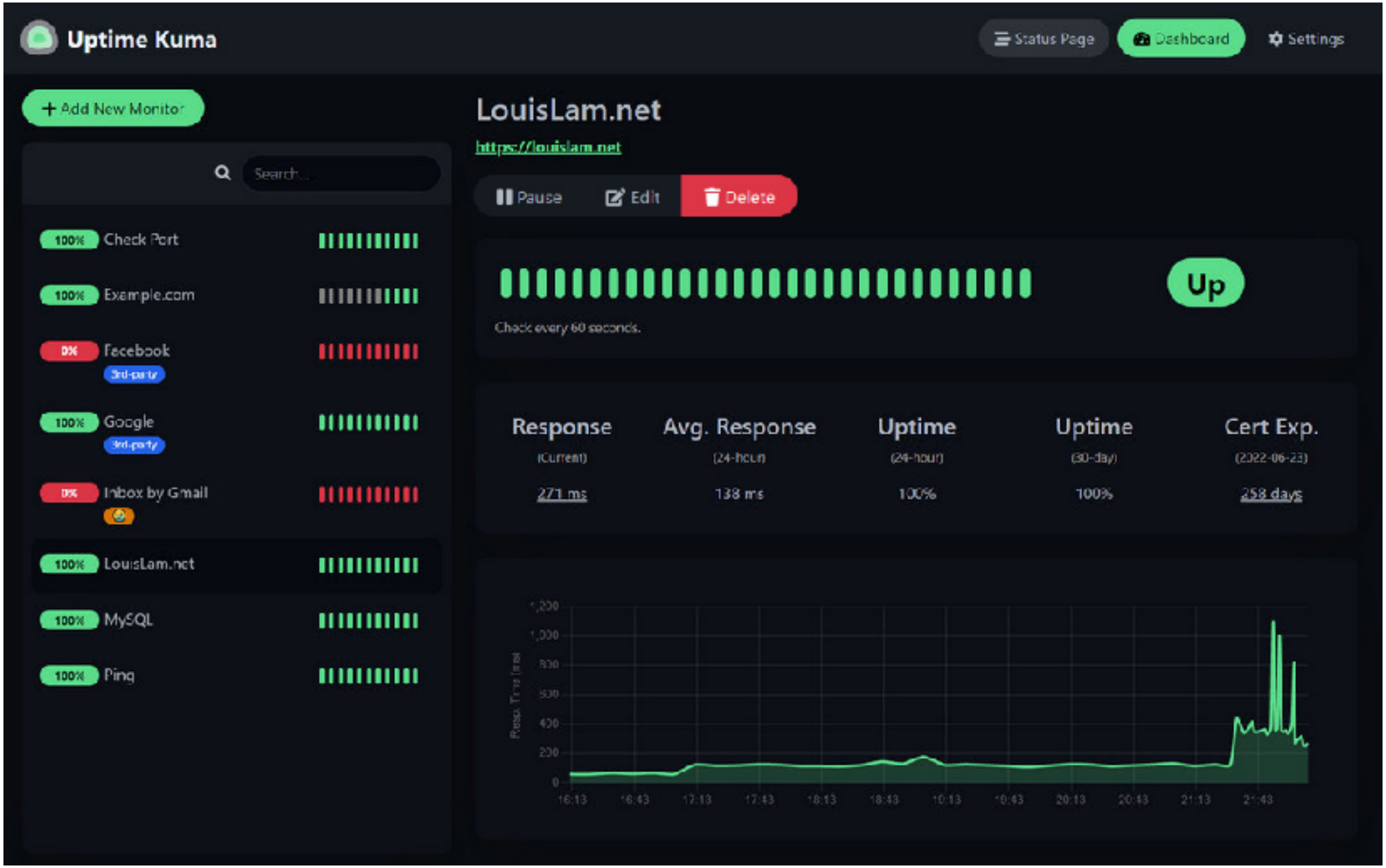
Docker Swarm



https://docs.docker.com/engine/swarm/how-swarm-mode-works/nodes/

Observability
Monitoring

Uptime Kuma

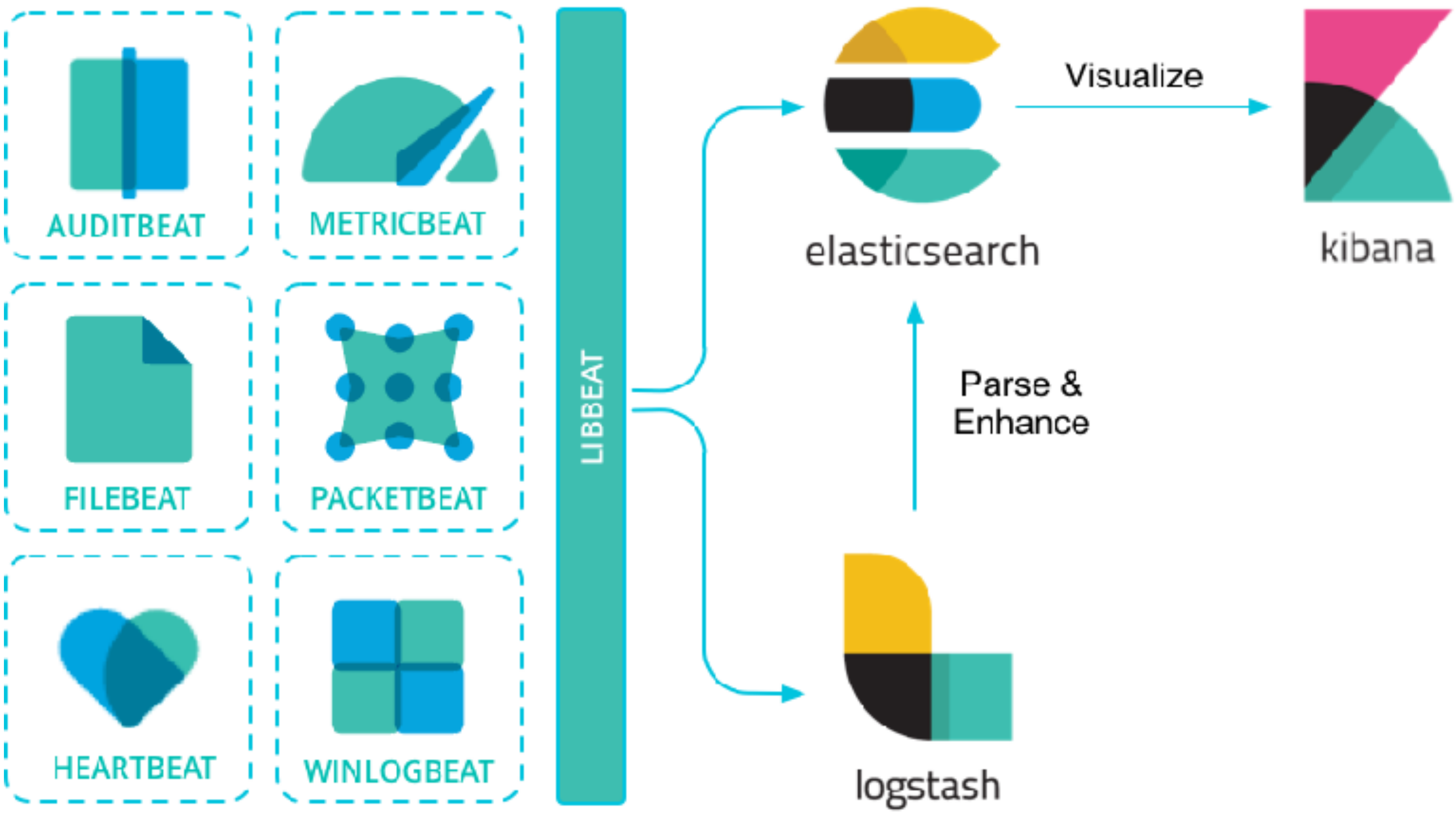


https://github.com/louislam/uptime-kuma

Gatus

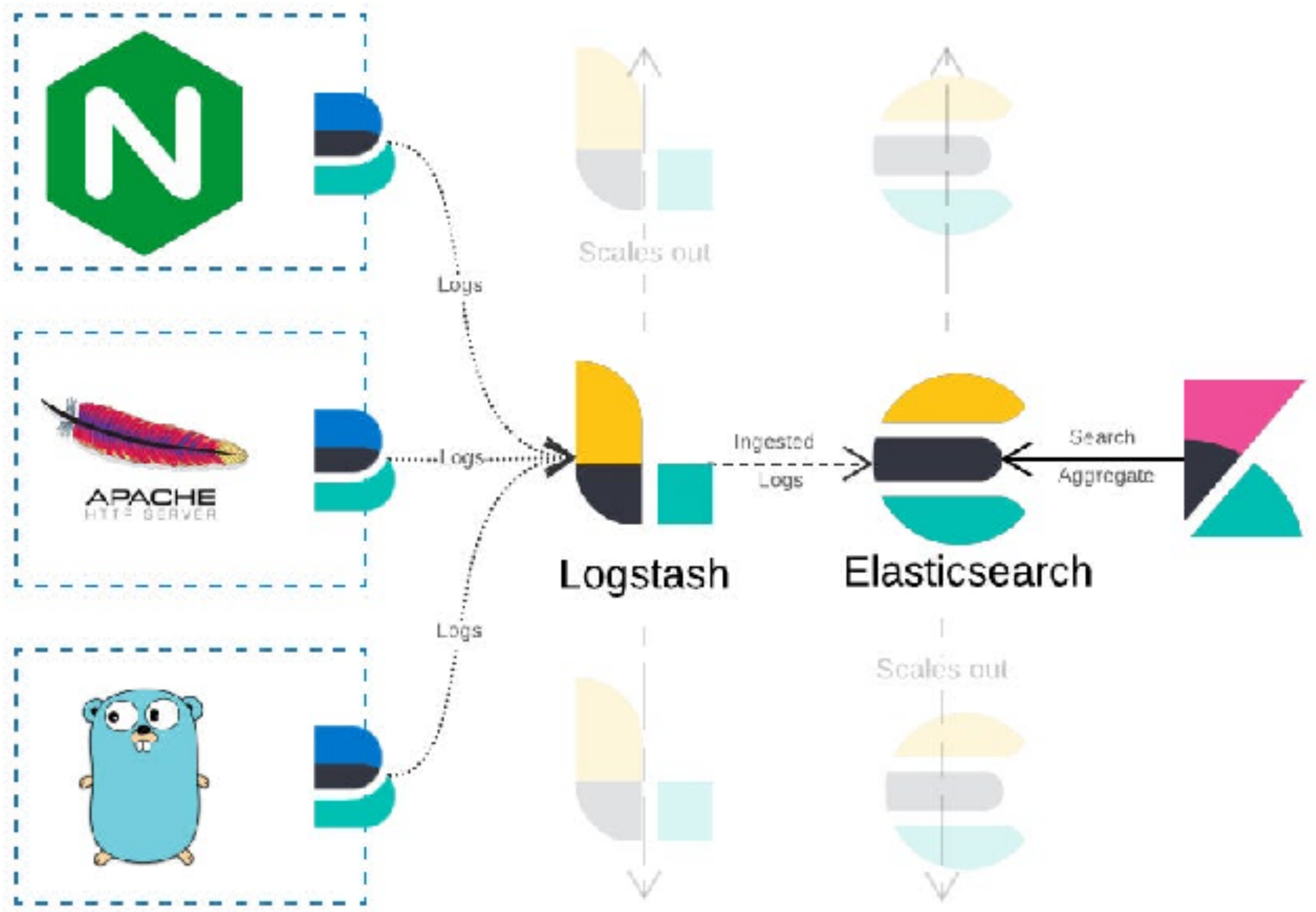
https://gatus.io/

ELK Stack



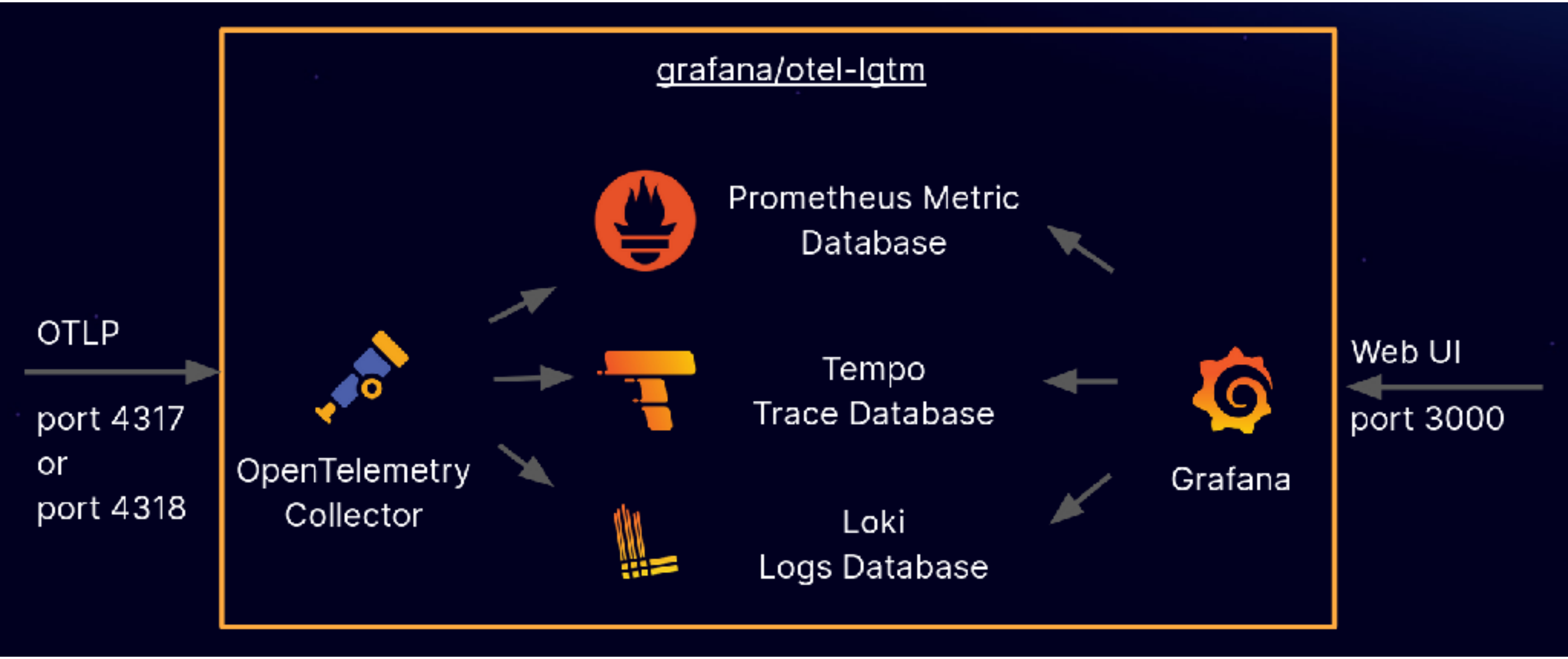
https://www.elastic.co/elastic-stack

ELK Stack



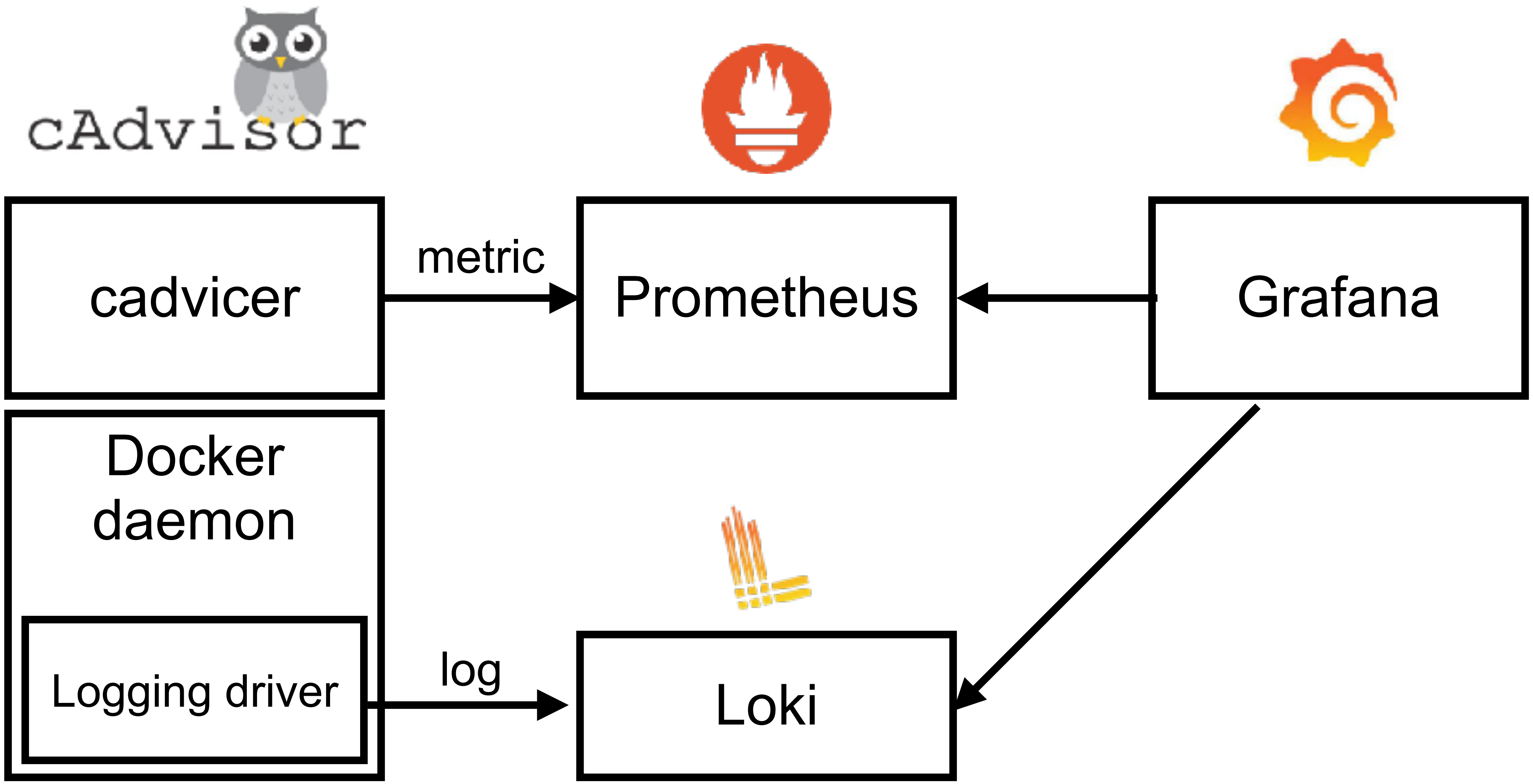
https://www.elastic.co/elastic-stack

LGTM Stack



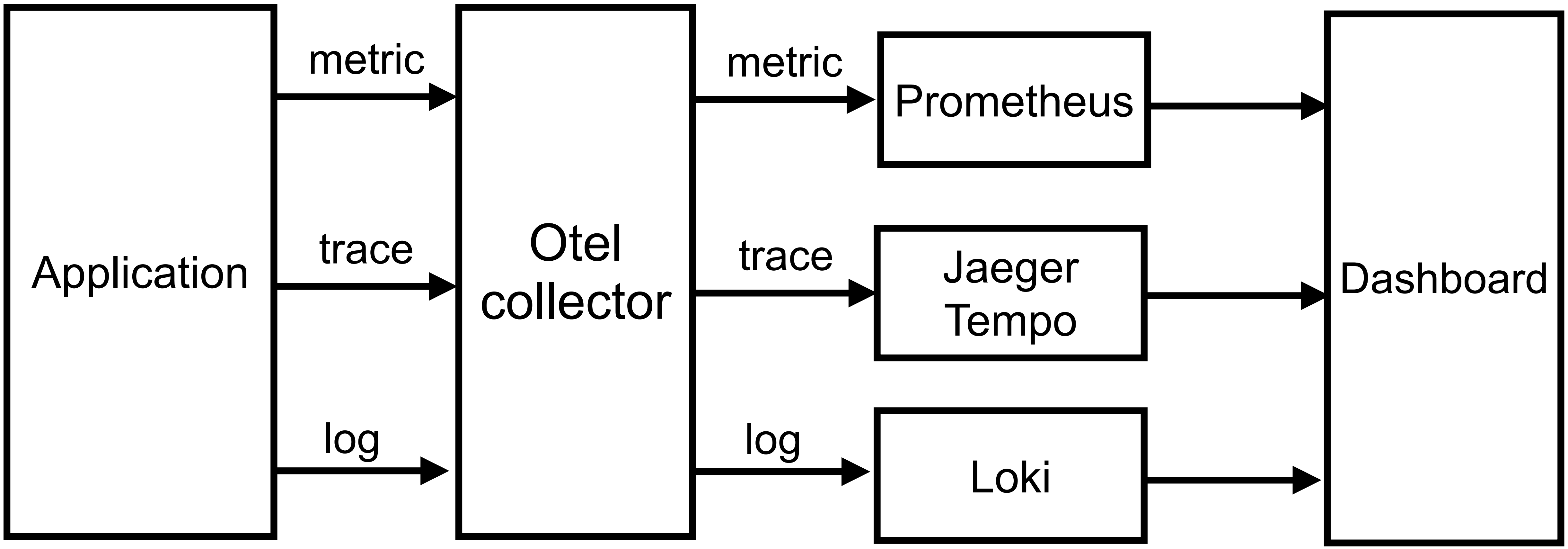
https://github.com/grafana/docker-otel-lgtm/

Docker monitoring



https://github.com/up1/workshop-docker-monitoring

OpenTelemetry

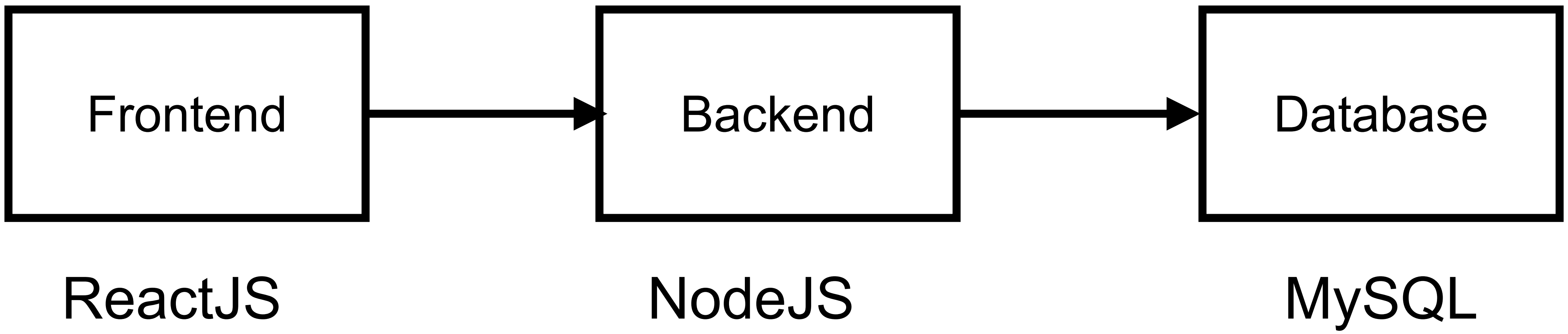


https://github.com/up1/workshop-lgtm-go

Observability Workshop

Workshop

Simple Architecture



Delivery process

Working with Docker

https://github.com/up1/workshop-ci-nodejs-web-api